

HND



Unit-29: Network Security

Designing a Secure Network for Myanmar Data Services



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LO1: Examine Network Security principles, protocols and standards.

Introduction

Myanmar Data Services (MDS) consists of two offices; the Head Office located in Yangon and the Branch Office in Mandalay. The company relies on the secure exchange of data in its day-to-day operations of five departments, namely; Finance, Sales and Marketing, HR, Customer Support, and IT. But in the absence of network security policies, MDS may also be vulnerable to cyber-attacks such as hacking or information leakage. As a Network Security Engineer, I would have to create a secure network system, install firewalls and VPNs, apply authentication and encryption systems, and locate and resolve flaws within the network. Another task I will perform is developing security rules and documents that will protect the company data and enhance the overall cybersecurity of MDS.

P1: Discuss the different types of network security devices.

Network security principles

Network security relies on a number of key principles that can be used to safeguard information and systems against attacks or abuse. These principles are confidentiality, privacy whereby the information is not shared with other users and integrity whereby information is correct and not modified by other people without permission; and availability whereby it is ready to be used when the need arises. Authentication is used to identify who the users are, and non-repudiation is used to ensure that nobody can deny any act he or she has performed on the network. The least privilege principle only grants the user the access that they require, minimizing the risks and defense in depth employs numerous layers of security providing additional protection in case any layer does not work.

Network security devices

Network security devices refer to special hardware or software that is utilized to guard against unauthorized access, data loss, and other forms of cyberattacks to computer networks. Such devices assist organizations to ensure the safety, confidentiality and integrity of their data whether being transferred through networks.

They are used to monitor and control flow of information between computers, servers and other external networks such as the internet. Network security devices provide levels of security by blocking unauthorized users, detecting suspicious activities, and screening harmful traffic, which (in addition) decreases the danger of cyber threats through multiple layers of protection. Concisely, they are important in ensuring that only safe and certified data is sent or received in a network and this is used to ensure the security of both personal and organizational information. (restorepoint, n.d.)

Firewall

One of the most significant network security tools is a firewall. It provides a shield between trusted internal network and untrusted external network like the internet. The firewall will scan the incoming and outgoing traffic and block or allow certain traffic depending on a set security rule. It may be hardware based, software based or both. Firewalls do this by marking packets of data and inspecting the source, destination and content of the packet. In case the information is suspicious or does not comply with the security regulations, it is blocked. This will prevent network attacks by hackers, malware and unauthorized access.

The advanced features are also provided in modern firewalls making them more effective. URL filtering enables the firewall to disallow users access to malicious or inappropriate websites and prevent phishing or malwares. Access control regulates the people with access to specific resources or services ensuring that only authorized individuals or machines gain access. Application restriction is used to block certain applications that can be considered risky like file-sharing or social media applications to avoid data leaks or distraction. These characteristics combined provide organizations with enhanced authority when it comes to network protection and user actions. (fortinet, n.d.)

Router

A router is a machine that links the various networks and routes the data to the appropriate destination. Routers are very vital in network security since they control the flow of information between the local network and the internet. These may have in-built security solutions, including firewalls, encryption and access control list (ACL) to block unwanted traffic. The internal network addresses can also be hidden by routers

using Network Address Translation (NAT) and this makes it difficult to discover the devices within the network and attack them.

Routing protocols like RIP (Routing Information Protocol), OSPF (Open Shortest Path First), or EIGRP (Enhanced Interior Gateway Routing Protocol) are also used by routers in exchanging routing information with other routers. Such protocols enable routers to constantly refresh their routing tables as they become more dependable and flexible when paths are altered or disconnected.

Routers also have various features in security that assist in securing the network. They will be able to filter the packets and block the unwanted traffic in the network, implement Access Control Lists (ACLs) to restrict access to specific resources, and Network Address Translation (NAT) to conceal the internal IP addresses among the outsiders. VPN connections via IPSec (Internet Protocol Security) are also supported by many routers and this encrypts data that is transmitted via the internet. This enables remote user or branch offices to connect to the main network safely as though they were right in the same position. (cloudflare, n.d.)

Switch

A switch is used to provide connections among various devices of the same local area network (LAN) which include computers, printers, and servers. It transmits data to the exact device that requires it and not to all the devices. This enhances performance as well as security by minimizing the possibilities of data interception. Most switches today are compatible with VLANs (Virtual Local Area Networks) which are useful to segregate network traffic into various groups to enhance security as well as facilitates easier management.

Switches do possess a number of significant security and reliability features. Switchport security prevents network security by restricting the device that can access a given port. It is able to block unknown devices or disconnect a port in case some unknown device is detected. MAC filtering is an addition of protection as it permits only the use of specific equipment with accepted MAC addresses to transmit and/or receive data on the switch.

Smooth performance is guaranteed by collision and congestion control features of switches, including full-duplex communication and buffering that ensure that there are no data collisions and that heavy traffic will be better managed. Further switches are equipped with redundancy and reliability i.e. the Spanning Tree Protocol (STP) to prevent network loops and maintain connection even when a single link goes offline. Combined, these features contribute to stable, safe and efficient network environments of the switches. (cisco, n.d.)

Intrusion Detection System (IDS)

An Intrusion Detection System (IDS) is a type of software that is used to monitor network traffic to identify instances of an activity that looks suspicious or dangerous. It also checks current network activity and matches it with established attack patterns or suspicious activity, which may be a cyberattack. Once the IDS notices something suspicious, it alerts the administrators but does not explicitly act. This is an early alert that enables the network security teams to act promptly in a possible attack such as hacking, malware or data break.

The IDS detection methods consist of two primary types of detectors, the anomaly-based and signature-based detectors.

The principle with anomaly-based detection is that it learns what normal network behavior should look like, and then, it detects any unexpected or unusual behavior that is not in the normal behavior pattern. This technique is useful in the detection of new or unfamiliar attacks that have never been detected previously, but may also generate false alarms.

Signature based detection is an alternative method which operates by matching network traffic to a database of known attack signatures as with an antivirus program. When an IDS finds a match then the package sends an alert. This approach is highly precise with already known attacks but might fail to detect new or unknown attacks until the corresponding signatures are updated.

Combination of the two detection techniques assists the IDS systems to offer enhanced and improved coverage against various cyber threats to networks (techtargert, n.d.)

Intrusion Prevention System (IPS)

An Intrusion Prevention System (IPS) is more like an IDS but it is a step further. It does not just identify the threats, but it goes ahead to prevent or stop the threats. The IPS may drop the malicious data packets or block traffic on a malicious source or reset the connections. It is directly positioned in the stream of the traffic on the network and it is actively safeguarding systems in real time. This renders IPS an effective means of averting attacks like denial-of-service (DoS), exploits and unauthorized access. (checkpoint, n.d.)

Virtual Private Network (VPN) Gateway

A VPN gateway enables its users to access a private network in the internet using a secure connection. It also encrypts all the information that is passing through the user device and the internal network making sure that the hackers will not be able to read or intercept data. VPNs are particularly beneficial to the remote employees who have to be able to reach the company resources in the most secure way in the external places. The VPN gateway is able to preserve the confidentiality of data and prevent eavesdropping by providing a very safe way to conduct communications, called a tunnel. (ninjaone, n.d.)

Proxy Server

A proxy server is a server that acts between the computer and the internet of the user. In the event that a user makes a request to gain access to a web site, a proxy server receives the request, verifies its safety and thereafter forwards the request to the destination. This process conceals the true IP address of the user thus offering privacy and anonymity. Proxies also have the ability to block out the application of the websites that are harmful or restricted, store regularly used web pages so that they can be accessed faster and also filter the content to avoid accessing the websites that are harmful. By so doing, they increase network performance and security. (vmware, n.d.)

A network design of Myanmar Data Services includes several important devices and technologies that are applied to ensure communication between the Head Office and the Branch Office. VPN that uses IPSec is adopted to form an encrypted tunnel connecting the two places, where all information sent via the internet is kept confidential and free of interception. Switches are set to Switchport security to avoid

connection of unauthorized devices to the network whereas the Intrusion Prevention System (IPS) is involved to monitor traffic within the network in real time and blocking potential attacks.

The two sites are also connected by a GRE tunnel that enables internal communication to be done safely through the public network. Authentication, Authorisation, and accounting server is used to control access of users and record an account history of user activities to ensure accountability. The network management is more efficient, as the DHCP server automatically provides network devices with an IP address, and it minimizes chances of configuration errors. FTP server provide safe sharing of files amongst departments.

All these elements combined offer a powerful, tiered security framework that secures the data of the company, provides a secure communication channel as well as facilitates effective and efficient network operations on behalf of all departments of Myanmar Data Services.

P2: Examine network security protocols and the use of different cryptographic types in network security.

Network security protocols refer to special guidelines and procedures that assist in safeguarding data that is travelling over a network. They ensure that the information is not stolen, modified, or accessed by the wrong individuals. These protocols apply encryption, authentication and data integrity checks techniques to ensure the security of communication between the devices. Simply put, network security protocols can be compared to the role of digital bodyguards that keep the information secure when the online communication takes place; it makes sure that only the right individuals can receive the information. The network security protocols are-

- HTTPS (Hypertext Transfer Protocol Secure)
- SSL/TLS (Secure Sockets Layer / Transport Layer Security)
- IPSec (Internet Protocol Security)
- SSH (Secure Shell)
- SFTP (Secure File Transfer Protocol)
- Kerberos
- RADIUS (Remote Authentication Dial-In User Service)
- WPA2/WPA3 (Wi-Fi Protected Access)

Hypertext Transfer Protocol Secure (HTTPS)

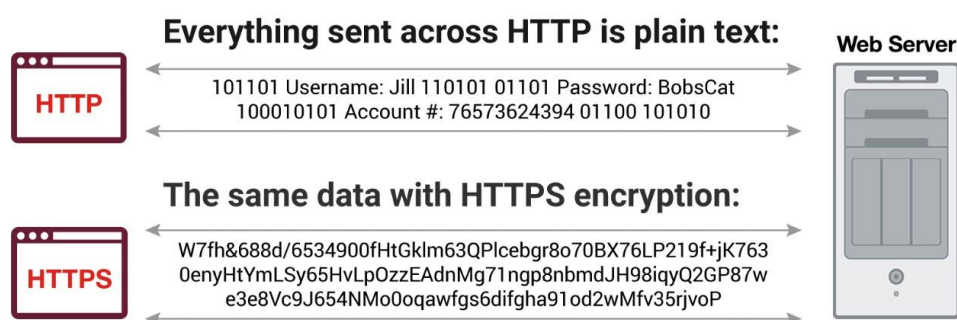


Figure (1.0) HTTP and HTTPS

The secure version of HTTP is known as HTTPS, and is applied in browsing websites. It provides an encryption layer which privacy and security of communication

between a browser of the user and the web server. When you access an Internet site with HTTPS, the data you give to it (i.e. log-ins, personal, credit cards, etc.) are turned into code that hackers will hardly be able to decipher. This will eliminate the possibility of data theft and the authenticity of the site. With banks, online stores, and social media, the use of HTTPS is common to secure the information of users and generate trust.

The use of HTTPS is combined with the use of SSL (Secure Sipp Layer) or a newer protocol TLS (Transport Layer Security) to secure online communication. The encryption that secures HTTPS is achieved with the help of SSL/TLS in which data is transformed into unreadable code during its transit between a web server and the browser of a user. Simply put, HTTPS is the secure form of HTTP and the technology that goes into the creation of the secure, encrypted connection behind HTTPS is called, simply, SSL/TLS. Lack of the use of the non-SSL/TLS would render the HTTPS incapable of providing a sense of privacy, authentication, and integrity of data over the web (bunny.net, n.d.)

SSL/TLS

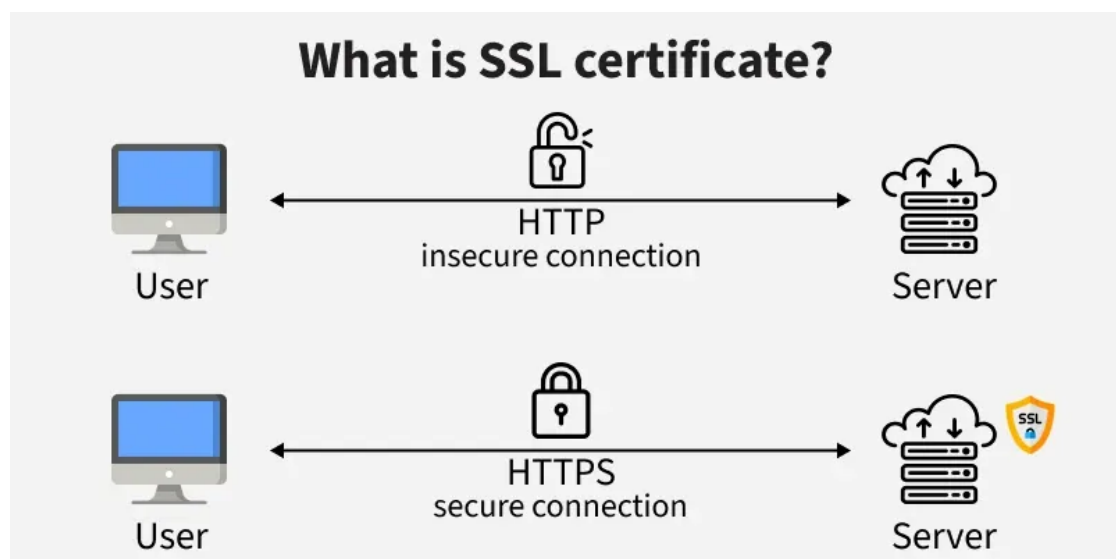


Figure (1.1) SSL Certificate

SSL and TLS are the protocols that provide security to the internet communication. They operate on the principle of establishing a secure link between two gadgets such as a computer and a web server so that no other individual can access

or alter the information under communication. TLS The new and more sophisticated version of the SSL is TLS that is more strongly encrypted and more performance-friendly. Such protocols find application in most web -based services including web browsing, email and instant messaging. With the help of the SSL/TLS, the organizations may make sure that the data of their users will be confidential and the communication will not be under the risk of cyberattacks. (f5.com, n.d.)

Internet Protocol Security (IPSec)

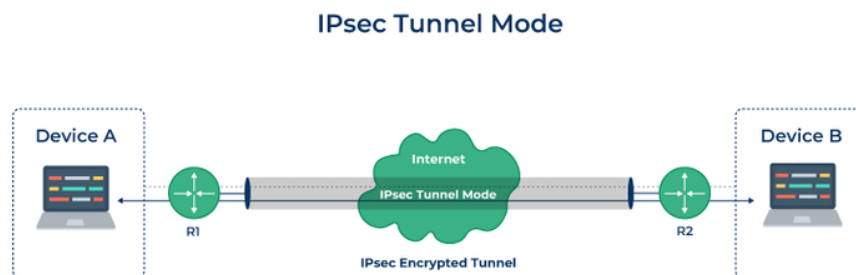


Figure (1.2) IPsec tunnel mode

IPSec is an effective protocol that is built with the aim of safeguarding data transfer among networks and particularly the internet. It operates on the network layer, encrypting and authenticating every packet of data in order to make it in-readable and in-writable. IPSec is usually applied in Virtual Private Networks (VPNs), which allow establishing a secure communication between remote users and office networks. As an example, an employee can be working at home and IPsec can assist the employee to securely log in to the internal systems of the company. It also checks the identity of every device which communicates with the network which is an added assurance of trust and security. (techtargget, n.d.)

SSH (Secure Shell)

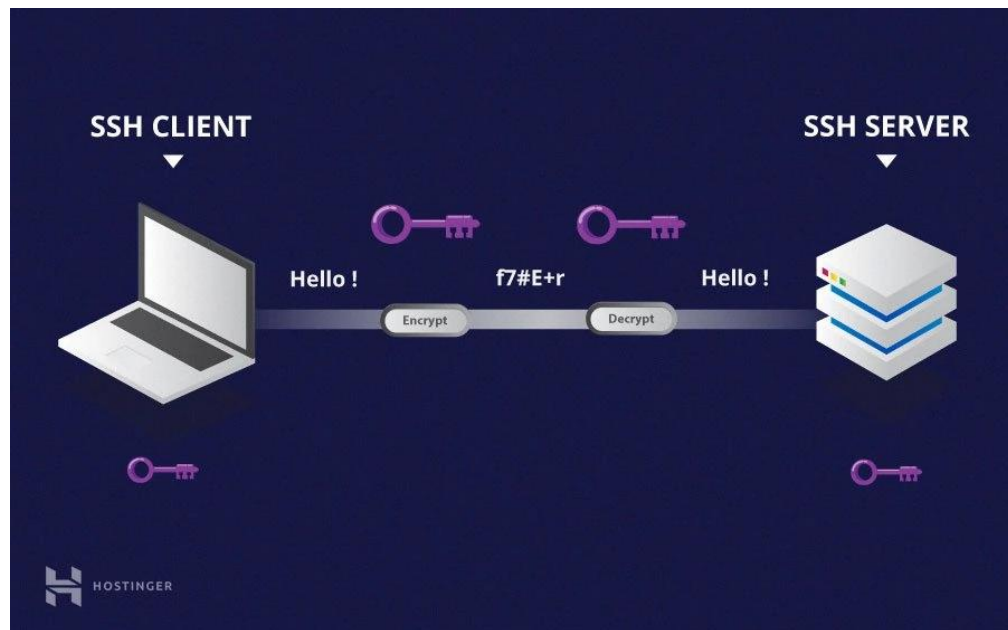


Figure (1.3) SSH working procedure

SSH is a network protocol of secure access and management of remote computers or servers. It will replace older and less secure communications such as Telnet and will offer the user encrypted communications between the user and the remote device. Network administrators and IT professionals frequently use SSH to configure, monitor and troubleshoot remote servers. It avoids the situation of interception of passwords or commands by outsiders and this makes it a consistent tool in ensuring secure management of the network. Under SSH, organizations are able to manage their systems securely without their data being leaked or accessed by unauthorized persons. (techtargt, n.d.)

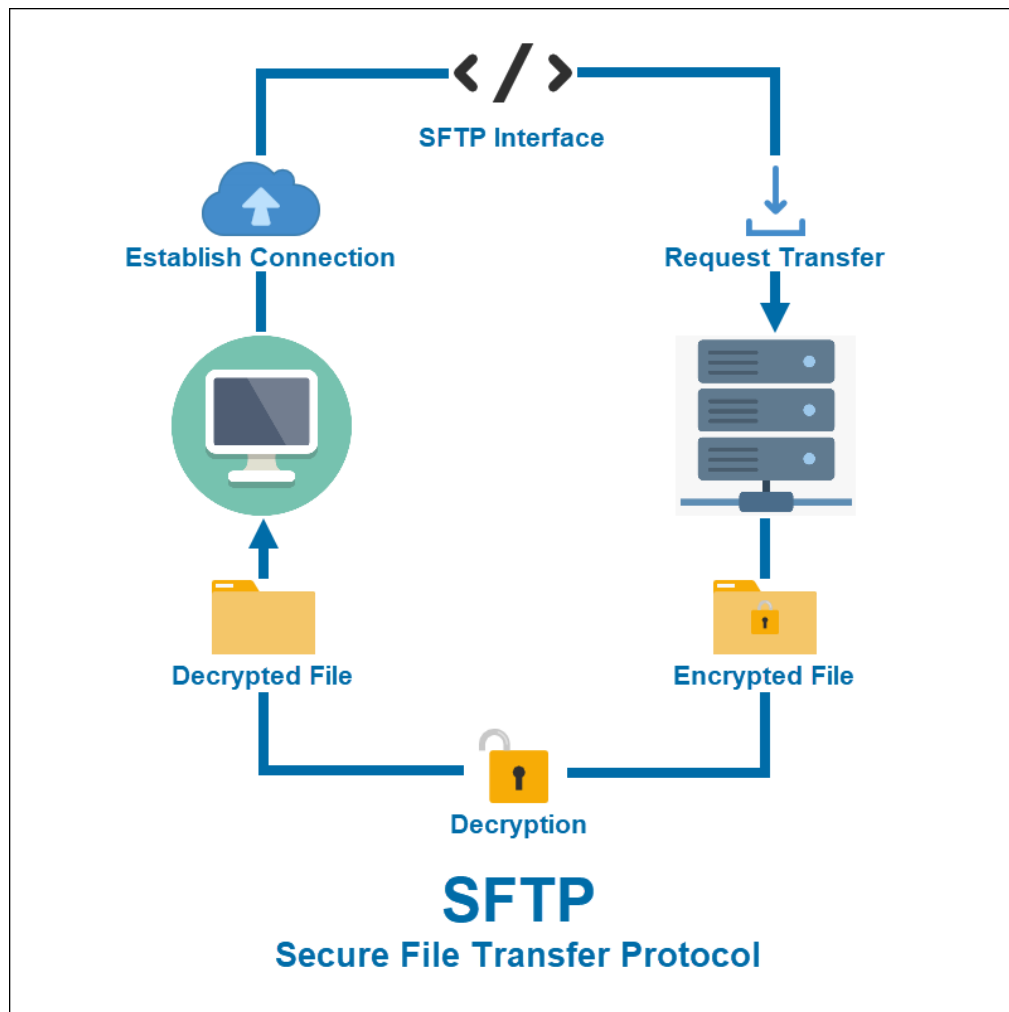
Secure File Transfer Protocol (SFTP)

Figure (1.4) SFTP

The SFTP is a secure way of transferring files between systems. It works on the basis of SSH protocol, i.e., all information exchanged in SFTP is encrypted. This ensures that files are not read or altered by hackers when they are being transmitted. Businesses will frequently utilize SFTP to transmit sensitive information, like financial documents, reports or customer data, between offices or to an outside partner. It is also compatible with secure authentication where only the authorized user can either upload or download files. The SFTP is a reliable and effective means of exchanging information in a safe manner in networks. (solarwinds, n.d.)

Kerberos

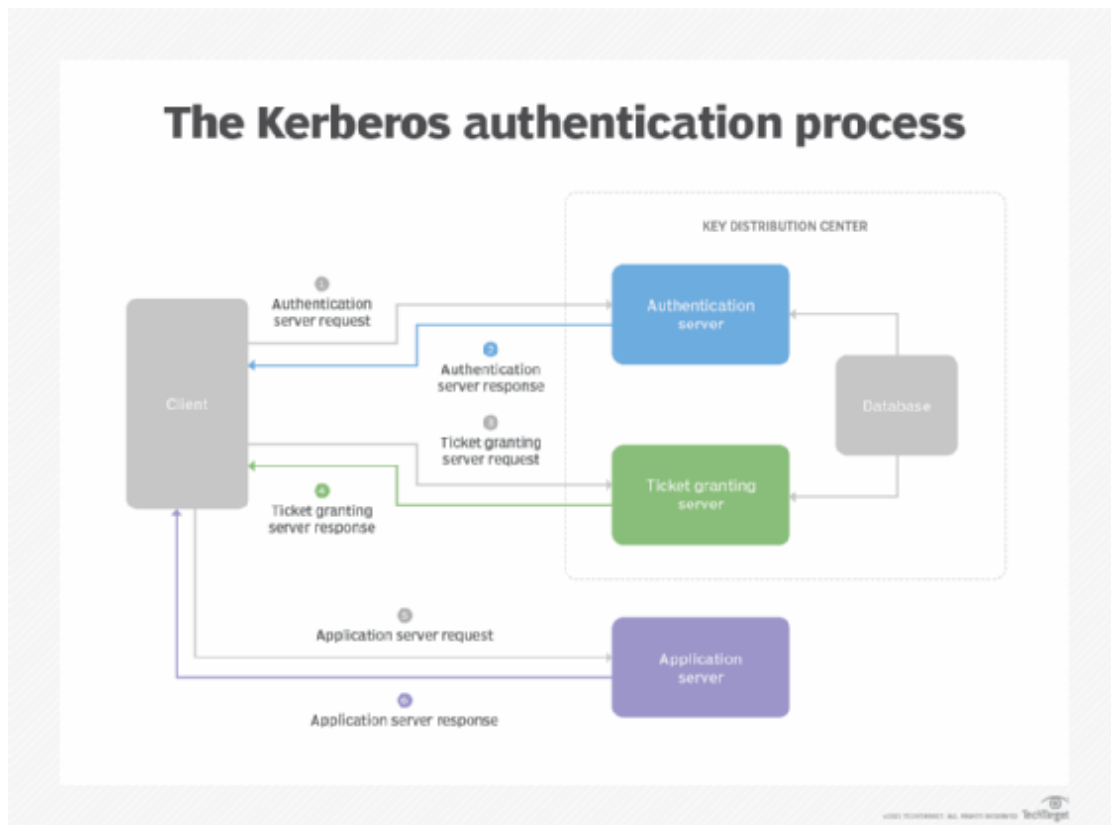


Figure (1.5) Kerberos

Kerberos is a network authentication protocol which operates with the help of secret keys to identify network users and devices, then authorize them to access network services. It is founded on a ticket system; users are given an electronic ticket by a central server to identify themselves. The ticket will enable them to get access to numerous services without the need to type in their password again. Kerberos is useful in preventing password theft, replay attacks, in which a hacker attempts to use previously used login information. It is popularly applied in big organizations and computer networks to provide efficient and secure user authentication.

RADIUS (Remote Authentication Dial-In User Service)

RADIUS is a protocol, which assists in regulating access to a network. It operates by transmitting the log-in details of a user (e.g., username and password) to a central server to conduct verification. When the information is a match, then the user is given access and otherwise the connection is denied. RADIUS is typically applied by

the internet services providers, companies and universities to control access to the network. It also logs user activities; this is useful in monitoring and controlling the network by the administrators. The centralization of authentication makes RADIUS permit only authorized users to enter the network and the network resources.

Cryptography

Cryptography is the art of safeguarding messages by encoding them into a secret message in order that only authorized individuals should be able to decipher it. It is applied to store data against hackers and unauthorised persons. The major concept of cryptography is to encode a readable message, which is referred to as plaintext, and to convert it into unreadable message, referred to as ciphertext, a process known as encryption. Once a data is given to the right individual, they are able to apply a secret key or password to transform it into readable format, and this is referred to as decryption.



Figure (1.6) Cryptography

Cryptography can be found in many daily forms of technology online banking, emails, and messaging applications among others to ensure that personal and financial data remains confidential. Cryptography has two broad categories; one is the symmetric where the same key is used in both encryption and decryption and the other is the asymmetric where two keys are used (public key and a private key). In modern crypto as well, the data integrity (enabling the information to remain untampered),

authentication (checking identity) and non-repudiation (so that the individual does not deny his actions) are guaranteed.

Symmetric Key Cryptography

This kind of encryption is known as symmetric key cryptography whereby the same key is used to encrypt (lock) and decrypt (unlock) the information. This implies that the sender and receiver should get access to the same secret key which they should keep secure and inaccessible to other people. It is also quick and effective, and thus it can be used to encrypt a lot of information including files or even a full network traffic. But its biggest weakness is key distribution - in case one intercepts the key he/she can easily decode or alter the encrypted information.

AES (Advanced encryption Standard)

AES has replaced DES as the modern one and it is currently of the most reliable encryption techniques in the world. It has a key size of 128, 192 or 256 bits and is therefore very difficult to crack by hackers. Another way that AES utilizes blocks of data is to make multiple rounds of replacement and conversion to transform plain text into ciphertext. It has many applications such as Wi-Fi security (WPA2/WPA3), internet banking and secure storage of files. AES is very secure, fast and flexible making it the existing standard in symmetric encryption.

DES (Data Encryption Standard)

One of the first commonly used symmetric encryption algorithms was DES. It encrypts the information with a key of 56 bits, that was a secure key when it was first developed in the 1970s. The mechanism of DES is that data is broken into small segments and undergoes several sophisticated mathematical operations on them so that the data becomes unreadable. Nonetheless, later with the enhanced computer power, DES was brute-force attackable, making it no longer regarded as securable in terms of modern application.

Asymmetric key cryptography

Otherwise called public key cryptography, asymmetric key cryptography is an encryption system that involves two keys, namely the public and the private, in order to safeguard information. As opposed to symmetric encryption where a single key is applied during the encryption and decryption process, in asymmetric encryption a specific key is used to encrypt and another is used to decrypt the information. Public key can be distributed publicly and the private key should be engaged in secret to its owner. This system works whereby regardless of whether one knows the public key, no one can decrypt the data unless they have the matching one the private key.

The RSA algorithm is the most common asymmetric encryption algorithm. RSA operates under complicated mathematical concepts of very large prime numbers. It generates two keys which are mathematically associated with each other, yet it is nearly impossible to determine the private key solely by holding the public one. The encrypted data can be decrypted only through the corresponding private key when it is encrypted with the use of the public key. This helps preserve the confidentiality, integrity and authenticity of the data.

RSA algorithm is primarily based on the key size of the algorithm. The popular key size in RSA is 1024-bit, 2048-bit and 4096-bit. An increased key size is more secure since it renders it significantly difficult to crack by brute force by a hacker. In the day and age of computer technology, 2048-bit keys are regarded as the standard of good security although 4096-bit keys are even stronger, but demand an extra amount of processing power. In the contemporary systems, 2048-bit RSA is mostly employed in most organizations since it offers decent security and performance.

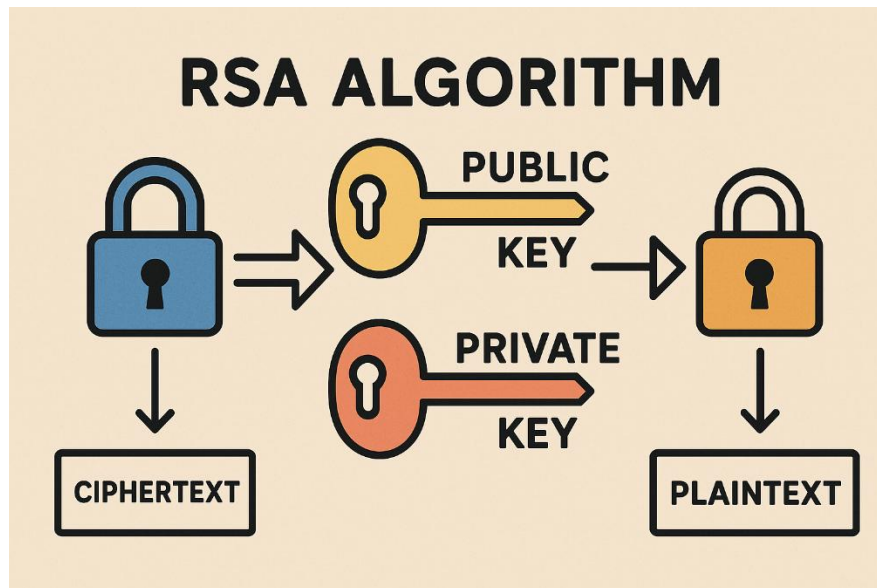


Figure (1.7) RSA Algorithm

RSA is also very efficient in guarding against unwarranted access and manipulation. It is said to be very secure since it would require huge computing power and time to intercept the encryption by guessing the secret code. Nonetheless, compared to the symmetric encryption, RSA is slower and is therefore commonly applied when it is necessary to encrypt smaller data segments such as encryption keys or authentication procedures instead of large files.

To sum up, asymmetric key cryptography, and the RSA in particular, contributes to the key aspects of the new cyber security as it is a secure and efficient method of information exchange, identity verification, and privacy protection in online communication.

Hash Function

A hash function is a mathematical operation (special process) in cryptography to transform data of arbitrary size into a fixed-size string of characters, also known as a hash value or a digest. A hash function is primarily aimed at maintaining the integrity of data, i.e. checking whether the information has not been distorted or modified in the process of transfer or storage. Any slight modification of the original data will result in a totally different hash value, which is easy to identify any unauthorized modification.

Hash functions are one-way functions, and this implies that once a data has been transformed to a hash, it cannot be transformed back to its original format. This renders them quite safe in data verification. They find various applications in digital signatures, password storage, as well as blockchain technology. To give an instance, in a site when you type a password the system does not store the password, but the hash of the password that you typed. The next time you log in, the hash of your password input is taken and compared with the stored hash to determine whether you enter the correct one or not.

Two hash functions that are the most extensively used are the MD5 (Message Digest 5) and the SHA (Secure Hash Algorithm). The MD5 hash function generates 128 bits in a hash, and is highly fast though it is no longer regarded as being completely secure due to the fact that there are times when two distinct inputs can produce the same hash (often referred to as a collision). Non-critical uses are however still done by it like in checking file checksums or data verification.

SHA family has been created by the National security agency of the United States (NSA) which has variants such as SHA-1, SHA-256 and SHA-512. These are more secure than MD5 as they have longer valued of hash and stronger algorithms to avoid collision. An example is SHA-256, which generates a 256-bit hash and is popular in the contemporary systems, such as digital signatures, cryptography via SSL/TLS, and blockchain. All in all, SHA algorithms can be recommended to the modern security applications as they are more resilient to attacks.

How Hashing Works

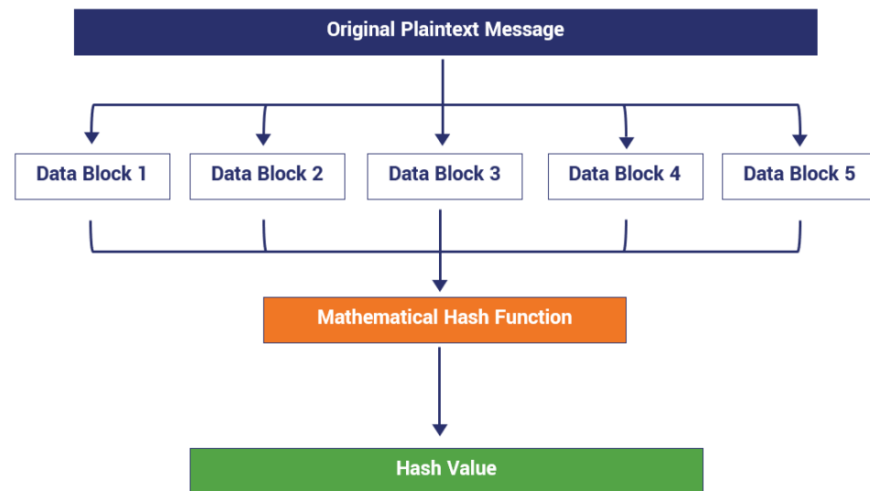


Figure (1.8) Hash Function

Other additional features of Cryptography

Cryptography is not simply the act of concealing information, but it also offers a number of valuable properties to secure communications and the storage of data, which are safe, reliable, and trustworthy. These characteristics will make information to be safe even in those open or untrusted networks.

Confidentiality

Confidentiality refers to the ability of withholding information to other parties who are not supposed to access it. The data can only be read by individuals with the right decryption key. This ensures that the privacy of information is not stolen or seen by hackers or outsiders during the transmission or storage process.

Integrity

Integrity is used to be sure that the data has not been modified, altered, or otherwise tampered with in its flow. Cryptography methods apply hashing algorithms in order to verify any unauthorized alteration. In case any alteration is made to the data, the receiver will be aware of that since the value will not align with the hash value.

Authentication

Authentication helps confirm the identity of the sender or the receiver and then communication will take place. It makes sure that the individual or machinery that you are conversing with is real and not a fake. Cryptography is commonly used to authenticate identities using digital certificates and digital signatures.

M1 Compare and contrast at least two major network security protocols.

Two major network security protocols

The internet security protocols are significant in securing communication on the internet. SSL/TLS (Secure Socket Layer / Transport Layer Security) and IPSec (Internet Protocol Security) are two protocols that are the most popular. As much as both offer data protection in terms of encryption and authentication, they vary in the way and location in which they address a network, and also in their general applications.

SSL/TLS is implemented on the application layer of the network. It is primarily used to protect information relayed between a web browser and a web server of a user. As a case study, when you access a secure site that employs HTTPS an encryption of the communication is provided via the use of the SSL/TLS which ensures that no one can read or modify the information being exchanged. It offers protection of online transactions, emails and web applications in encryption, authentication and integrity. SSL/TLS is simple to install on web-based services and provide secure browsing to users without any network modification.

SSL and TLS Versions The following are examples of SSL and TLS Versions.

SSL Versions:

Earlier version of the SSL is SSL 2.0 and 3.0, however, they are no longer in use as they are very vulnerable to security breaches. Such versions are not suggested or supported anymore in modern systems.

TLS Versions:

SSL was superseded by TLS that offers more protection and encryption. The main versions are:

TLS 1.0 and 1.1 - obsolete and deemed to be insecure.

TLS 1.2 - This is a commonly used and strong encryption and authentication.

TLS 1.3 - The newest and most secure version that is suggested in all the modern networks. It eliminates there outdated and weak algorithms and enhances the security and performance.

Recommended Versions

SSL Version Recommended None (SSL is outdated and insecure).

Recommended Version of TLS: TLS 1.3 - it supports better connections and better security features with perfect forward secrecy and higher hand shake encryption.

SSL/TLS operate on the basis of an encrypted connection between a browser and a web server used by a user. They employ digital certificates to confirm the identity of the server and encryption to secure the data to be transferred. This eliminates the possibility of attackers reading or manipulating, as well as impersonating secure sites. To the point, online communication is provided with confidentiality, integrity, and authentication with the help of SSL/TLS.

IPSec, in its turn, is an application of the network layer. It ensures that all information flowing through the IP networks is secured through encryption of each packet of information and authentication of their origin. IPSec is widely applied in Virtual private network (VPN) to provide secure networks between remote users and confidential networks. Contrary to the SS/TLS that secures selected applications, IPSec secures the entire network traffic irrespective of the nature of the application. It is more powerful, more system-wide protective yet may be more challenging to setup and maintain.

Transport Mode

IPSec also leaves the IP header alone but encrypts only the data segment (payload) of every IP packet in transport mode. This mode is primarily employed in secured end- to -end communication between two hosts e.g a client and a server within a single network.

Tunnel Mode

In tunnel mode, the IPSec encrypts the data and the IP header. It generates a routing outer IP header. It is a typical VPN mode that is used in connecting two networks (such as the Head Office and Branch Office). It is more secure as the original IP information is concealed.

AH (Authentication Header):

The AH protocol ensures a packet has authentication and integrity, that is it is not changed and that it is actually sent by a sender. Nonetheless, it lacks encryption, i.e. the content of the data remains readable.

ESP (Encapsulating Security Payload)

ESP offers authentication, integrity and encryption. It conceals the data content by encrypting it and it also has the capability of checking the origin of the packet. ESP is considered the most popular IPSec protocol since it provides not only confidentiality but also the protection of the data.

IKE (Internet Key Exchange)

During this stage, the two devices initially authenticate each other and also settle on security parameters like encryption algorithms and encryption keys. This forms a safe management channel, also referred to as the IKE Security Association (SA). It is then through this safe channel that keys are exchanged safely.

Comparing SSL/TLS and IPSec

Aspect	SSL/TLS (Secure Socket Layer / Transport Layer Security)	IPSec (Internet Protocol Security)
OSI Layer	Application Layer	Network Layer
Purpose	Secures communication between a web browser and a web server.	Secures data transmission between networks or hosts over IP.
Encryption Scope	Encrypts specific application data (e.g., web, email, etc.).	Encrypts all IP packets passing between two endpoints.

Configuration	Easier to implement — configured at the application or server level.	More complex — requires configuration on network devices (routers, gateways, firewalls).
Common Usage	HTTPS for websites, email encryption, instant messaging.	Virtual Private Networks (VPNs), site-to-site or remote access security.
Authentication	Uses digital certificates for verifying website identity.	Uses pre-shared keys or certificates to authenticate devices.
Encryption Type	Uses symmetric (AES) and asymmetric (RSA) cryptography.	Supports multiple encryption algorithms (AES, 3DES, etc.).
Performance	Lightweight; minimal overhead suitable for individual sessions.	Higher overhead; protects all traffic, which may reduce performance.
Visibility	Operates at user level — end users can see HTTPS padlock indicators.	Transparent to users — runs behind the scenes.
Ideal For	Securing web-based transactions and browser communications.	Securing network-to-network or site-to-site communications.

LO2: Design a secure network for a corporate environment.**P3: Investigate the purpose and requirements of a secure network according to a given scenario.**

The primary aim of the secure network of Myanmar Data Services (MDS) is to secure the valuable data and communication of the organization in its 2 main offices located in Yangon and Mandalay. As MDS is dealing with sensitive data that involves the Finance, HR, Sales, Customer Support, and IT departments, a secure network will ensure that only the authorized people can access certain data and systems. It assists in ensuring confidentiality through encryption of the information on the transfer, integrity through avoiding unauthorized alteration and availability by ensuring the network is dependable and operational on a daily basis.

Purpose of a Secure Network for MDS

The secure network also facilitates both internal and external communication of the company via VPN using IPSec and GRE tunnels that enables safe transfer of data between the branches using the internet. It prevents security risks like hacking, malwares and so on that may affect business operations or loss of reputation of the company. It is also useful in controlling the users access using AAA servers, automatic IP control using DHCP servers, and security measures such as switchport protection and IPS monitoring. In general, it is possible to say that the secure network enables MDS to work effectively, safely, and being sure that its data and resources are properly protected.

The company can also have a secure network that supports its internal and external communication by the use of VPN using IPSec and GRE tunnels that enable the transfer of data in the company between the branches safely over the internet. It prevents any form of security threats like hacking, malware and data breaches that might interfere with the functioning of business or bring a bad image to the company. It will also assist in control of access to users using AAA servers, automatic IP management using DHCP servers, and security measures such as switchport protection and IPS monitoring. The secure network in general enables MDS to run its operations effectively and without any fear that its data and resources are not effectively secured.

Requirements of a Secure Network for MDS

In order to create a safe and trustworthy network in Myanmar Data Services (MDS), there are various critical conditions that have to be met to save data and ensure the proper functioning of communication between the Yangon Head Office and the Mandalay Branch Office. The network should also contain powerful data protection and encryption with the use of technologies like IPSec VPN to make sure that the all information shared between the two sites will be confidential and not exposed to cyber threats. It is also important to provide user authentication and access control which can be achieved by using an AAA server to authenticate users, grant access privileges and maintain logs of user activities. A GRE tunnel needs to be installed to provide an encrypted and private internet-based connection between the two sites, which would allow secure and stable communication between them.

Also switchport security and MAC filtering should be enabled on switches to block unauthorized devices and internal breaches but VLAN segmentation allows separating departmental traffic to enable easier control and protection. The implementation of an Intrusion Prevention System (IPS) is expected to monitor traffic on the network in real time and prevent the occurrence of malicious activities before they can impact the system. A network will also need a DHCP server that will automatically allocate IP addresses and ease the management as well as a secure FTP server where the departments will share files safely. To make service availability continuous, redundancy must be incorporated in network links and devices such that in case of failure of one of the components, the service will still be available. Lastly, MDS should establish effective security policies and documents that stipulate user roles, password rules and implementation of incident management procedures. Addressing these needs will enable MDS to have a very secure, efficient and reliable network in both offices.

Designing a Network Prototype with a Security Perspective

A security-viewed design of a network prototype entails designing a structured and secure network layout that is capable of supporting communication requirements of the organization and also against cyber threats. In the case of Myanmar Data Services (MDS), the prototype design aims at linking the Yangon Head Office and Mandalay

Branch Office insecurely with a VPN of IPSec and GRE tunnel, which will offer encrypted and secret channels of communication over the internet. The network is subdivided into various VLANs of the departments like Finance, HR, IT, Sales and Customer Support to regulate traffic flow and minimize internal risks. MAC filtering and switchport security are used to prevent the unauthorized access of devices, user authentication and access control is done by AAA servers. An IPS (Intrusion Prevention System) is applied to prevent and identify threatful activities automatically and DHCP provides IP addresses automatically to a device with a view to managing it efficiently. Moreover, secure FTP server will enable safe transfer of files in the organization.

Hardware Requirements

In order to achieve a secure and efficient network to Myanmar Data Services (MDS), a number of hardware components are needed. To communicate encrypted and reliable between Yangon Head Office and Mandalay Branch Office, routers are required to link them by use of VPN using IPSec and GRE tunnel. VLAN and switchport security switches are applied in segregating departments and avoiding unauthorized connections. Some of the services that servers are needed to host include AAA server, which is used to perform authentication and DHCP server which is used to provide automatic assignment of IP addresses and FTP server which is used to transfer files in a secure way. All the departments must possess workstations or PCs where the employees will work as well as network interface cards (NICs) that will enable them to be connected. Furthermore, cabling and wireless access points will be required to offer the wired and wireless connections in the offices. It should also have a backup power supply (UPS) and redundant links in order to ensure that the network is available when there is a power outage or equipment failure.

Software Requirements

The software modules are critical in the process of network security and management. It must be based on a secure operating system, e.g. windows server or Linux (ubuntu Server) where services like DHCP, AAA and FTP are hosted. VPN software and IPsec protocols need to be installed on routers to develop safe tunnels between the two branches. A network monitoring system and Intrusion Prevention System (IPS) software is necessary to identify and prevent suspicious activities as they

occur. Simulation, testing, and configuration could be performed using network management tools, including cisco packet Tracer or gns3. Also, encrypting and authentication software should be provided to provide security to the remote users, antivirus and endpoint protection software is required to be installed on every computer to avoid malware infection. A combination of these software tools guarantees the confidentiality, integrity and availability of data in the MDS network.

Network Segmentation

Secure network design includes network segmentation which is a division of a large network by small sections that are easy to manage and are called segments or VLANs (Virtual Local Area Networks). In the case of Myanmar Data Services (MDS), segmentation is useful in enhancing security as well as performance since various departments like Finance, HR, IT, Sales and Customer Support are separated into individual VLANs. This will avoid cross-departmental access and minimizes the transmission of malware or network attacks. As an example, a user working in the Sales VLAN cannot directly access the Finance VLAN unless explicitly assigned some permission to do so. Segmentation is also useful in reducing network congestion through the restriction of unwarranted traffic across devices. Every segment has its security policies and access control policies as well as monitoring systems and it is more straightforward to identify and isolate threats. Through switchport security, VLANs and Access Control Lists (Access Control List) MDS can ensure that its data deals move safely, effectively and in an orderly arrangement through both the Yangon and Mandalay offices.

P4: Determine which network hardware and software to use in a secure network.

The choice of the hardware and software components to use to create a secure and efficient network is quite crucial. Each device and application has its particular role to play in the context of data protection, the control of the access of users, and the organization of the smooth communication within the enterprise. In the case of Myanmar Data Services (MDS) with its offices located in Yangon and Mandalay, it is essential to select well-known and proven hardware equipment (routers, switches, servers) and safe software tools (monitoring, encryption, and authentication) to address

business and security requirements. This combination of these elements is useful in safeguarding the data of the company, deterring unauthorized access, and maintaining a constant connection between the two sites.

Hardware requirements

The most appropriate hardware solutions will be required in ensuring the smooth flow of communication as well as security and sound performances between the Yangon Head Office and the Mandalay Branch Office to establish a safe and stable network in Myanmar Data Services (MDS). The hardware devices are important in ensuring the efficiency and security of the network infrastructure of the company as a whole.

ISP Router



Figure (2.0) Cisco 2911 router

ISP router is the key point of connection between the internal network of the organization and the Internet offered by the Internet Service Provider. It regulates the incoming and outgoing data in the network and should be capable of supporting VPN using IPSec in order to encrypt all the communications between the two offices. This router also controls the use of bandwidth, avoids unauthorized access and offers the initial level of protection against external threats.

Edge Router



Figure (2.1) 2911 router

The edge router is placed on the edge of the local network. It directs traffic between the internal LAN and the external world and implements such security functions as Access Control Lists (ACLs), GRE tunneling, and Network Address Translation (NAT). The functions assist in securing the internal systems against direct exposure with the internet. The edge router also assists in optimization by controlling the route of the packets as well as the data traffic along the various subnets.

Multilayer Switches Gateways



Figure (2.2) Multilayer switch

Multilayer switches work both as routers as well as switches that enable communication and routing between VLANs at high rates. They are programmed with switchport security, VLAN segmentation and MAC address filtering so that

unauthorized devices cannot gain access into the network. These switches assist in isolating traffic between departments like Finance, HR, IT and Sales and minimize congestion and enhance performance and security.

Layer 2 switches



Figure (2.3) Layer 2 switch

A Layer 2 access switch contains a way to reliably connect end devices in a LAN and the addition of redundancy will mean that connectivity will not be interrupted when a device or a link fails. The network is stable and efficient because protocols like Rapid Spanning Tree Protocol (RSTP) will be used in trying to avoid network loops and Link Aggregation Control Protocol (LACP) to bundle links together with increased bandwidth and failover. Case Study of Myanmar Data Services (MDS), installation of redundancy is achieved by having Layer 2 access switches which are interconnected with more than one multilayer switch to provide continuity in the communication between the Yangon Head Office and Mandalay Branch. Such an arrangement improves availability, load balancing and fault tolerance hence ensuring smooth and secure operations in all the departments.

Servers



Figure (2.4) Sever PT

The servers are the internal backbone of the network. This DHCP server assigns IP addresses automatically to equipment to facilitate management and avoid address conflicts. The AAA server is responsible for user authentication, authorization and accounting so that only valid users can access network resources. The FTP server gives a secure platform for the movement of files from department to department. Each server requires proper maintenance to ensure it is up to date and backed up in case of data loss or downtime.

End Devices

End devices consist of all user machines, such as computers, laptops and printers that connect to the network. These devices allow employees to perform their daily tasks of getting files, sending e-mail and facilities of use. All end devices should be connected by secure VLANs and protected by virus checks, password and user access methodology to avoid data loss and breaches or unauthorized activity.

Software Requirements

Software components are equally important as regard network security, management and performance. The right type of software enables MDS to keep data secure, discover threats early and gain and maintain and sort out stable communications between the two offices.

Gateway antivirus

A Gateway antivirus scans all data which either enters or leaves the network, and will block malicious material before that information reaches sensitive internal systems. This is different from normal endpoint antivirus, which checks for all the individual computers but which works at the entry point of the network. This assists in stopping the spread of malware, phishing and virus attacks over the individual servers and machines in the network which, in turn, keeps the network clean and secure.

NetFlow and SNMP

The tools used in network monitoring such as NetFlow and SNMP are vital in ensuring the visibility, performance and security in the network of Myanmar Data Services (MDS). NetFlow is a traffic monitoring tool of Cisco designed to capture the flow of traffic between routers and switches to enable the administrators to analyze

bandwidth consumption, monitor traffic patterns, identify anomalies, and achieve optimal network performance between Yangon Head Office and Mandalay Branch. It assists in identifying high users, applications consuming too much bandwidth or any other suspicious data transfers that can be a sign of a security breach. In the meantime, SNMP allows network devices (routers, switches, and servers) to be centrally monitored and managed. It offers an up-to-date information on device status, interface usage and fault conditions and notifies about this state (trap) to respond promptly to failures or performance problems. NetFlow and SNMP used together provide a fully-developed monitoring system that provides better visibility, higher troubleshooting, and increases the overall reliability and security of the network infrastructure of MDS.

Network Operating Systems (NOS)

Network Operating Systems (NOS) such as Cisco IOS, Windows Server, and Linux Server are essential for configuring and administering network devices such as routers and switches. They provide a platform for routing, switching, authenticating and sharing resources. NOS also contains tools for monitor traffic, error detection and troubleshooting connectivity problems, both of which are essential for security and performance management

VPN and IPSec Configuration Software

This VPN and IPSec configuration software allows for a secure, encrypted connection between the Yangon and Mandalay branches. It gives assurance that any information transmitted is confidential and not a subject to being intercepted or changed by outsiders. This software additionally permits the employees who work away from the office to safely obtain access to data available via the internet network and available computer programs from various locations, but without having to sacrifice security for flexibility.

Intrusion Protection System (IPS)

An Intrusion Protection System (IPS) scans all of the network traffic to determine if there are any unusual or dangerous actions taking place. It is capable of automatically identifying, alerting and blocking various types of threats such as denial of service (DoS) attacks, intrusions from various types of malwares, and unauthorized

entry attempts. IPS is useful in minimizing security problems and maintaining the network in a stable, rapidly responsive and well protected condition.

Network Management and Simulation Tools

Various network management and simulation tools, such as Cisco Packet Tracer and GNS3, are employed to design, test and manage the network before actual utilization. They permit engineers to simulate the operation of the network, verify the operation of the design, as well as locate potential and recognized problems prior to actual application. They are essential for education, debugging, and for an efficient and secure network structure.

The role of Hardware and Software in improving Network Security

Hardware as well as software is a crucial component in enhancing the general security of a network. The physical layer of defense contains hardware devices like routers, switches, servers, etc., which regulate the flow of data and its accessibility. Such capabilities as VLAN segmentation, switchport security, and Access Control Lists (ACLs) assist department isolation and exclude unauthorized connections in the network. Routers that are programmed with VPN and IPSec will make the communications between sites encrypted to safeguard sensitive information against hackers. AAA and DHCP are servers that can be used to secure user access and network resources.

Conversely, software solutions can serve to improve protection by actively monitoring, encrypting and identifying cyber threats. Antivirus such as Gateway Antivirus and Intrusion Prevention Systems (IPS) are used to scan the network traffic to block harmful activities that may cause damage before they can. Network operating systems (NOS) are systems that control devices and guarantee safe settings and VPN applications and encryption programs keep privacy in online communication. The combination of hardware and software gives the basis of safe connectivity and constant observation, regulation, and security against threats, forming a multi-layered security system that keeps Myanmar Data Services (MDS) network safe, efficient, and reliable.

M2 Create a design of a secure network according to a given scenario.

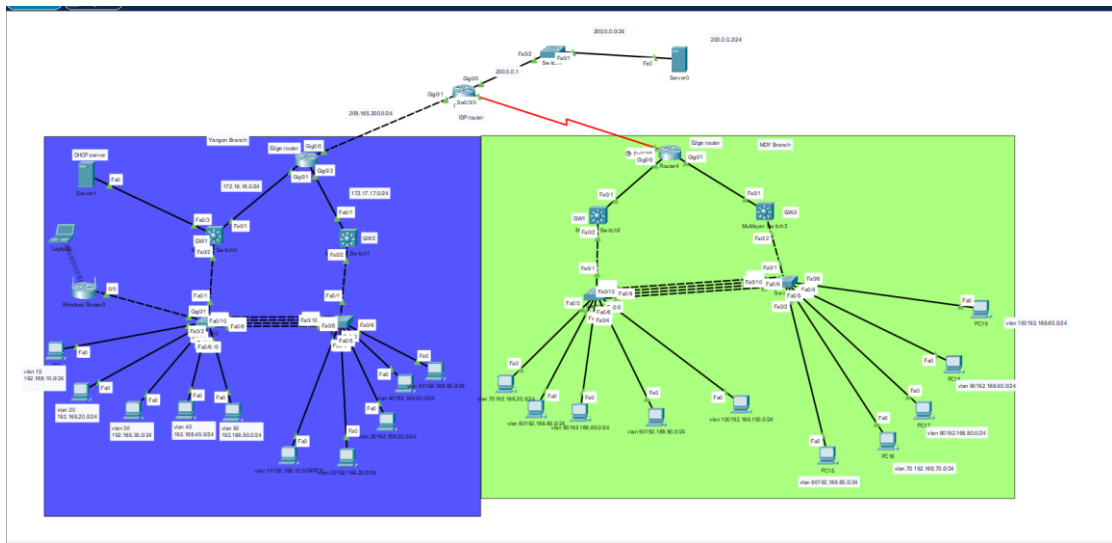


Figure (2.5) Secure Network design

Introduction to MDS Secure Network Design

Myanmar Data Services (MDS) secure network design is created to integrate both the Head Office located in Yangon and the Branch Office located in Mandalay and also to provide the confidentiality, integrity and availability of data. The key objective of this design is to have a safe, efficient and scalable structure of communication among the departments of the company which comprise of Finance, HR, Sales, IT and Customer Support. The design incorporates a mix of technologies, which include but not limited to VPN with IPsec, GRE tunnels, VLAN segmentation, switchport security, AAA authentication, Intrusion Prevention Systems (IPS) and DHCP services to ensure that both sites are fully secured.

IP addressing table

Device	Interface Ports	IP address	Subnet Mask	Default Gateway
Internet Server	-	200.0.0.2	255.255.255.0	200.0.0.1

ISP Router	Gig0/0	200.0.0.1	255.255.255.0	
	Gig0/1	209.165.200.1	255.255.255.0	
	Se0/3/0	209.165.100.1	255.255.255.0	
Yangon Edge Router	Gig0/0	209.165.200.2	255.255.255.0	
	Gig0/1	172.16.16.1	255.255.255.0	
	Gig0/2	172.17.17.1	255.255.255.0	
Yangon Gateway 1	VLAN 1	172.16.16.2	255.255.255.0	
Finance	VLAN 10	192.168.10.1	255.255.255.0	
HR	VLAN 20	192.168.20.1	255.255.255.0	
Sales and Marketing	VLAN 30	192.168.30.1	255.255.255.0	
Customer Support	VLAN 40	192.168.40.1	255.255.255.0	
IT	VLAN 50	192.168.50.1	255.255.255.0	
Yangon Gateway 2	VLAN 1	172.17.17.1	255.255.255.0	
Finance	VLAN 10	192.168.10.2	255.255.255.0	
HR	VLAN 20	192.168.20.2	255.255.255.0	
Sales and Marketing	VLAN 30	192.168.30.2	255.255.255.0	
Customer Support	VLAN 40	192.168.40.2	255.255.255.0	
IT	VLAN 50	192.168.50.2	255.255.255.0	

Mandalay Edge Router	Se0/3/0	209.165.100.2	255.255.255.0	
	Gig0/0	172.18.18.1	255.255.255.0	
	Gig0/1	172.19.19.1	255.255.255.0	
Mandalay Gateway 1	VLAN 1	172.18.18.2	255.255.255.0	
Finance	VLAN 60	192.168.60.1	255.255.255.0	
HR	VLAN 70	192.168.70.1	255.255.255.0	
Sales and Marketing	VLAN 80	192.168.80.1	255.255.255.0	
Customer Support	VLAN 90	192.168.90.1	255.255.255.0	
IT	VLAN 100	192.168.100.1	255.255.255.0	
Mandalay Gateway 2	VLAN 1	172.19.19.2	255.255.255.0	
Finance	VLAN 60	192.168.60.2	255.255.255.0	
HR	VLAN 70	192.168.70.2	255.255.255.0	
Sales and Marketing	VLAN 80	192.168.80.2	255.255.255.0	
Customer Support	VLAN 90	192.168.90.2	255.255.255.0	
IT	VLAN 100	192.168.100.2	255.255.255.0	
DHCP Server	-	192.168.10.253	255.255.255.0	172.16.16.1

GRE Tunnel	Gig0/0	10.0.0.1	255.0.0.0	
	Se0/3/0	10.0.0.2	255.0.0.0	
HSRP Gateway	-	192.168.10.254	255.255.255.0	
		192.168.20.254	255.255.255.0	
		192.168.30.254	255.255.255.0	
		192.168.40.254	255.255.255.0	
		192.168.50.254	255.255.255.0	
		192.168.60.254	255.255.255.0	
		192.168.70.254	255.255.255.0	
		192.168.80.254	255.255.255.0	
		192.168.90.254	255.255.255.0	
		192.168.100.254	255.255.255.0	

VPN and GRE Tunnel Implementation for Secure Communication

The network employs IPsec VPN, GRE tunnel between the office of Yangon and Mandalay to provide security to all information traversed through the internet by encrypting it. This keeps away the confidential company data and also makes sure that the internal systems are only accessed by the authorized users and devices. To divide the departments and to regulate traffic within a particular office, VLANs (Virtual Local Area Networks) are adopted. This kind of segmentation will avoid cross-departmental unauthorized access and can control the resource of the network. Switchport security and MAC address filters are also set in order to stop any unknown or untrusted device to connect to the network, which is also another level of protection against internal threats.

Role of Multilayer Switches and Network Services

Multilayer switches are used in both locations as gateways performing inter-VLAN routing and also implementing Access Control Lists (ACLs) to block or permit traffic. The AAA server offers user authentication, authorization as well as accounting in a centralized system where only authenticated users are able to access sensitive network resources. To prevent the entry of malicious activities like denial-of-service attacks or unauthorized access, an IPS continuously scans traffic in real time and prevents these harmful activities. The DHCP server will automatically assign IP addresses to the devices under each VLAN and will eliminate errors in manual configuration and enhance network efficiency.

Importance of Layer 2 Switches in MDS Network

The Layer 2 switches are essential in the Myanmar Data Services (MDS) secure network design as it is necessary in the access layer which links all the end-user devices like computers, printers, and IP phones across each department. These switches work on the data link layer (Layer 2) of the OSI model, and they use MAC addresses to queue data, hence, they can facilitate quick and efficient communication within a single VLAN. This structure assists the various departments such as the Finance, HR, and Sales to share information within the organization without routing it to other routers, which lead to network congestion and performance decline.

Layer 2 Switch Security Features and Benefits

Security wise, the Layer 2 switches have a number of built-in protection mechanisms. Switchport security helps to limit the number of devices that can access a port and automatically shuts down the port in case an unknown device attempts to connect. MAC address filtering can provide an extra control as only registered devices can be supported to use the network and prevents spoofing and unauthorized connections. All VLANs are separated and that is why a system of a department could be compromised but it will not be able to impact the other sections of the network easily. This segmentation enhances the overall network safety and it is easy to monitor and control traffic in each department.

Inter-VLAN Routing and Overall Network Performance

These Layer 2 switches are attached to multilayer switches, which do inter-VLAN routing and impose on the traffic between departments and sites an access Control List (ACL) to filter the traffic. This stratified system makes the communication within the company safe, reliable, and properly structured. The Layer 2 switches are very useful by offering a combination of security measures and performance to the security and effectiveness of the MDS network in the Yangon Head Office and the Mandalay Branch.

D1 Evaluate the importance of network security to an organisation.

Network security is a necessity to all organisations that depend on digital systems and communication networks to carry out its business activities. It gives the basis of securing worthwhile data, preserving trust and ensuring that day to day operations proceed without hitches. A network security, in the example of Myanmar Data Services (MDS) with numerous branches and a vast amount of sensitive data about clients, is crucial to preventing the threats to the digital infrastructure of the company, both internal and external. The increase in the complexity of cyberattacks, including ransomware, phishing, and denial-of-service (DoS) attacks, has placed the need to establish a strong network security level that can help organisations avoid data loss, service downtime, and monetary harm.

Network security is an important factor due to one of its core reasons to ensure confidentiality, integrity, and availability (the CIA triad) which are the three primary tenets of information security. Confidentiality is an assurance that sensitive information is available to only the authorised users which include details of the employees, customer records, and financial information. Integrity ensures that information will be correct and will not be altered or changed by hackers or other unauthorized people. The availability guarantees that network resources and services are available to legitimate users at the right time. When any of these factors are affected, the organisation can suffer drastic operation and reputation wise setbacks. To give an example, in the event

that the financial information of MDS was leaked or manipulated, it will result in loss of customer confidence, fines as well as severe setback to the business.

Cyber threats protection is another significant issue of network security. The attacks that are experienced by organisations today have a high variety comprising malware, spyware, ransomware and brute-force attacks that can penetrate an unprotected network. MDS can use the technologies of VPN with IPSec, Intrusion Prevention Systems (IPS) and the switchport security to ensure that its data transmissions between offices in Yangon and Mandalay are protected and that the network is not exposed to other devices. Access Control Lists (ACLs) and VLAN segmentation also aids in restricting interdepartmental access so that the confidential information in the HR or Finance department cannot be accessed by other users. Moreover, AAA will be used to enhance the verification of users, since only trusted persons will have access to the network and the activities of the users would be logged to make them accountable.

In case of improper configuration of VPN, this may pose grave threats to an organization such as Myanmar Data Services (MDS). An incorrectly configured VPN cannot encrypt data, so hackers can intercept confidential data in the form of client information or financial data. Lack or poor authentication settings may also be used to allow unauthorized persons access to the network resulting in data breaches. Lack of proper routing or proper configuration of a tunnel may result in some traffic going outside the secure VPN tunnel and the internal communication is exposed to the open internet. Not only do they affect the confidentiality of data, but these problems may also interfere with the communication process between the Yangon and Mandalay offices. Hence, it is essential that MDS implements its VPN with a lot of attention, strong encryption, and authentication, and does some routine monitoring of the connections to ensure the safety and reliability of communication.

Access Control Lists (ACLs) and Virtual Local Area Networks (VLANs) are effective measures of mitigating insider threat and data breach in an organisation. VLANs separate a big network into smaller, isolated groupings of every department including Finance, HR, and Sales, so that users simply have access to those resources that are pertinent to the job that they do. This isolation makes it difficult to get the

sensitive information of other departments. Instead, ACLs are adopted to direct traffic between these VLANs by establishing certain rules that either permit or forbid access according to IP addresses, protocols or ports. In the example, ACLs can prevent employees of the general departments to access the financial servers or HR databases. Combined, VLANs and ACLs provide an opportunity of preventing unjustified disclosures of data, the internal propagation of attacks, and granting only authorized users access to vital information, enhancing the overall level of network security.

Network security plays an important role in preserving business continuity and reliability. In case of a cyberattack, it will lead to significant downtime, which interferes with internal processes and customer service. In the case of a data-driven company such as MDS, loss of productivity and money due to few hours of downtime is possible. The adoption of redundancy systems, secure VPN tunnel and monitoring of IPS can help to minimize the chances of downtime and also ensures that the company is able to run smoothly even when at its knees. It is not only better operational efficiency, but also enhances customer trust because customers can trust the system that their data and services are always secure.

The Port Aggregation Protocol (PAgP) is employed in the secure network design of Myanmar Data Services (MDS) to set up EtherChannel between switches in order to offer redundancy and enhance network reliability. PAgP enables a combination of several physical links amongst switches into one logical link and also provides greater bandwidth as well as failover. This implies that failure by one physical connection will not disrupt network connectivity as rest of the connections maintain network connections. Consequently, this has ensured that communication among departments and the Yangon Head Office and the Mandalay Branch does not take a hit.

PAgP would also achieve network performance balancing the data traffic in all the links of the EtherChannel, which reduces congestion and increases the overall throughput. Moreover, it assists in sustaining the loop-free topology through collaboration with Spanning Tree Protocol (STP) which avoids network loops but still provides the backup pathways. This redundancy will guarantee that the network is not vulnerable to the failure of a single point, which poses a threat to the overall business processes of MDS, which rely on the active availability of common resources and

services. Through adoption of PAgP-based EtherChannel, MDS will enjoy stability, resilience, and uptime of the network enabling individual departments to continue their productivity in the event of hardware failures, or even hardware maintenance.

Moreover, network security is also necessary due to regulatory compliance and protection under the law. Data protection laws and data protection standards (the GDPR, ISO 27001, or local data privacy laws) regulate many industries, which compel organisations to ensure they protect their data. An effective network security system would assist MDS to adhere to such regulations, which would save it the possible fines, litigation, or business licensing. There is also the benefit of compliance in making the company appear more professional to customers and partners in that the company cares about data protection.

The fact that network security reduces expenses and controls risks is yet another significant advantage. Investing in security technologies and training may seem expensive at first, but over time, the costs of handling data breaches, system recovery, and damaged reputation due to security intrusions are greatly decreased. A single cyberattack could cost a company millions of dollars in lost revenue and recovery expenses. Therefore, a proactive approach to network security—that is, frequent updates, ongoing intrusion monitoring, and user training—will save money and avoid larger-scale expenses down the road. Lastly, network security supports the organization's overall expansion and competitiveness.

It makes it possible for departments and branches to work together seamlessly, enhancing productivity and creativity while preserving control over data access and flow.

In summary, network security is a strategic element of contemporary business management as well as a technical requirement. Sensitive data is safeguarded, cyber threats are avoided, legal compliance is supported, and uninterrupted operations are guaranteed. Strong network security features like IPSec VPNs, AAA authentication, VLAN segmentation, and intrusion prevention systems are essential for preserving trust, safeguarding company assets, and attaining long-term sustainability for a company like Myanmar Data Services, which primarily relies on digital communication

and data exchange. A successful, robust, and future-ready organization is built on a secure network.

LO3: Configure Network Security measures for the corporate environment.

P5: Configure network security for a network.

WAN configuration

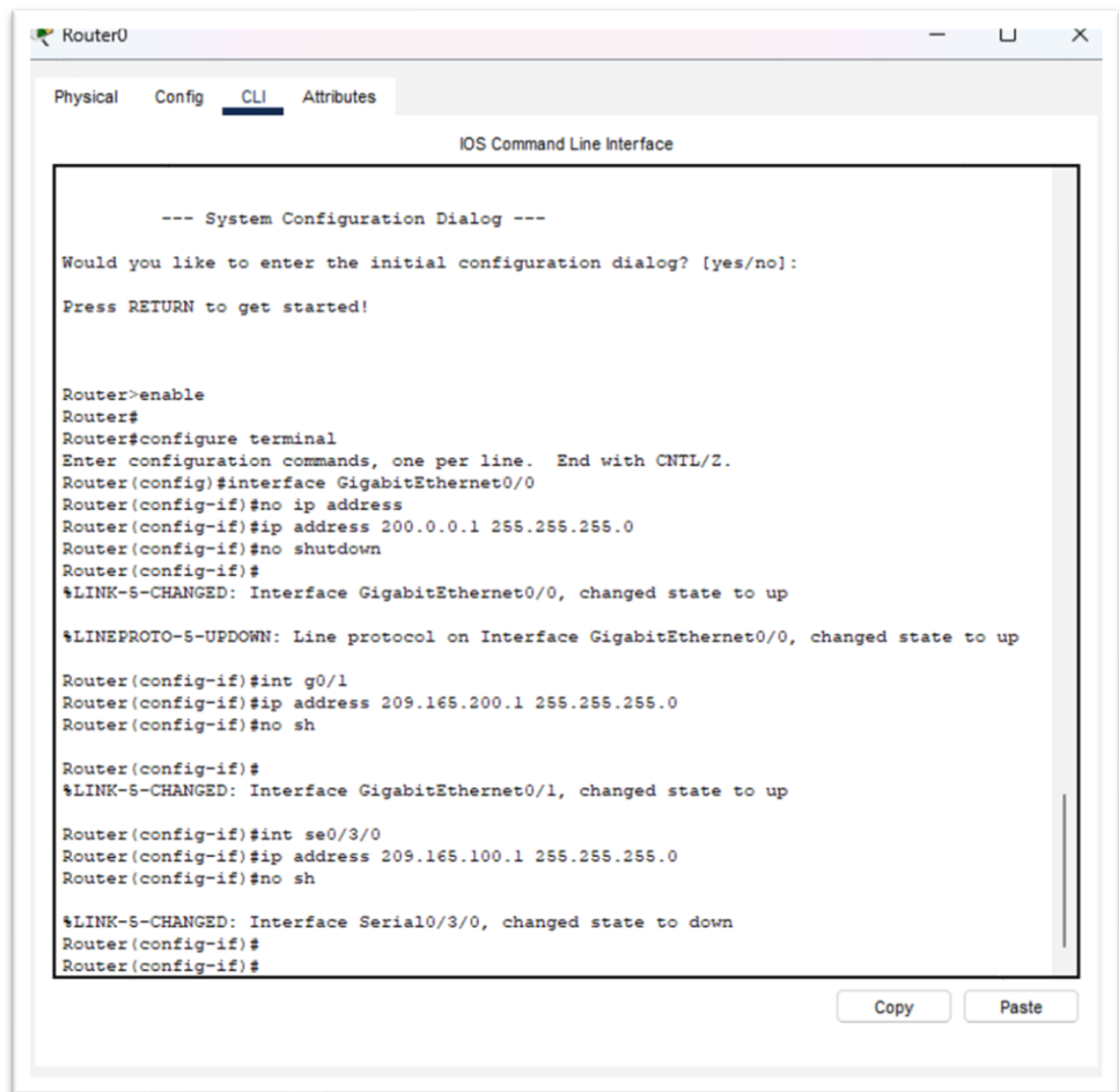


Figure (3.0) Ip addressing for ISP router

These commands bring the router's interfaces online and configure them using their assigned IP addresses. A unique port, such as GigabitEthernet0/0, GigabitEthernet0/1, or Serial0/3/0, can be chosen using the `interface` command. Each interface's IP address and subnet mask are assigned by the `Ip address` command; for instance, 200.0.0.1/24 joins to the Internet Server, 209.165.200.1/24 linked to the Yangon Edge Router, and 209.165.100.1/24 is linked to the Mandalay router. Each interface is activated by the `no shutdown` command, which changes the status from administratively down to up, enabling data to flow through. In order to connect the Internet server, edge routers, and ISP router in the MDS network, these procedures are necessary.

The screenshot shows the configuration interface for a device named 'Server0'. The 'Desktop' tab is selected. The 'IP Configuration' section is active, showing the following settings:

Configuration Type	Static
IP Configuration	☒ Static
IPv4 Address	200.0.0.2
Subnet Mask	255.255.255.0
Default Gateway	200.0.0.1
DNS Server	0.0.0.0

The 'IPv6 Configuration' section is also visible, showing the following settings:

Configuration Type	Static
IPv6 Configuration	☒ Static
IPv6 Address	
Link Local Address	FE80::206:2AFF:FE34:32ED
Default Gateway	
DNS Server	

The '802.1X' section is also visible, showing the following settings:

802.1X	
Use 802.1X Security	☐
Authentication	MDS
Username	
Password	

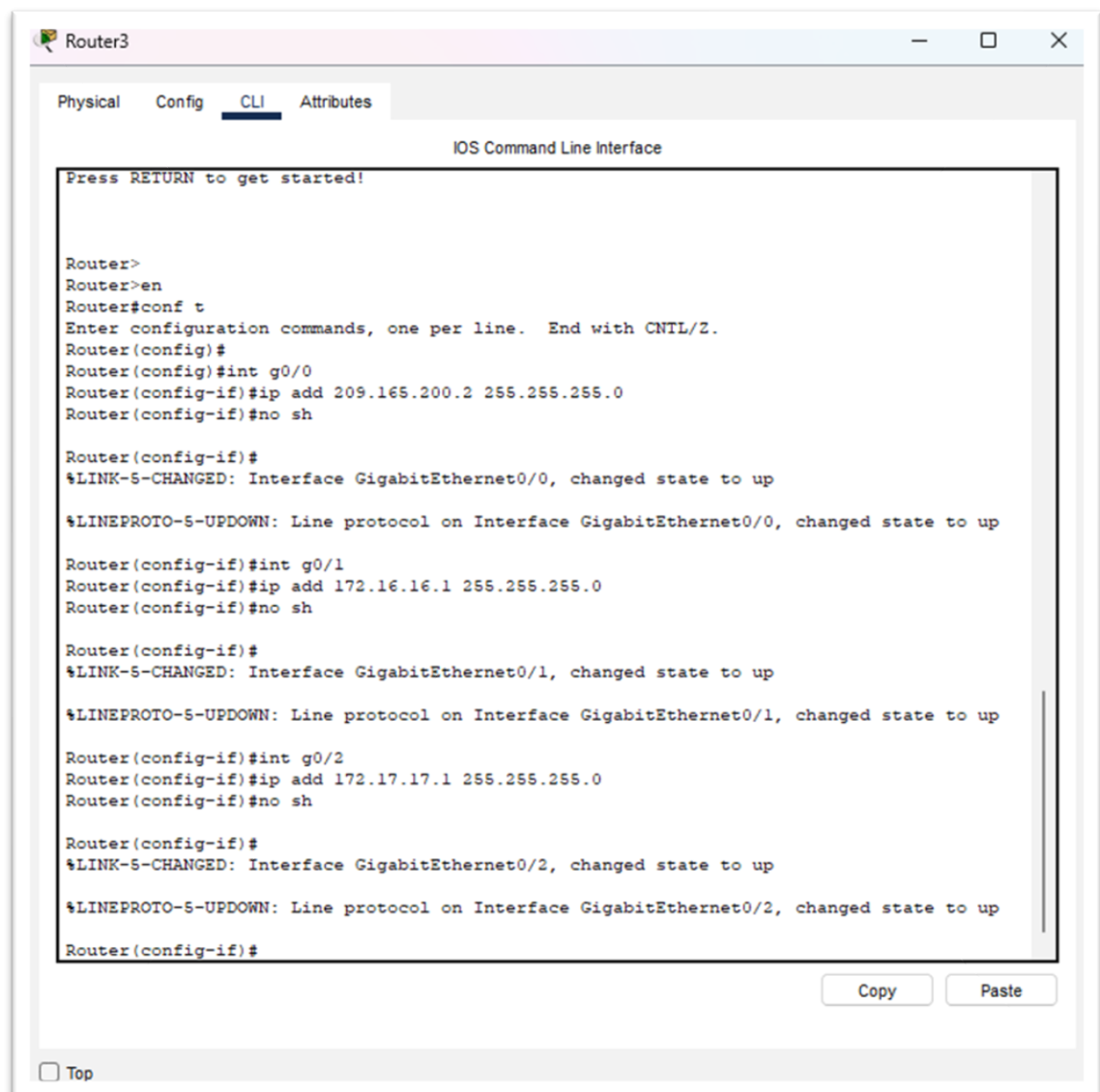
At the bottom left, there is a 'Top' button.

Figure (3.1) Server IP addressing

The server has been given an IPv4 address of 200.0.0.2 and a subnet mask of 255.255.255.0 by hand. This puts it on the same network as the ISP router interface 200.0.0.1. The default gateway is set to 200.0.0.1, which means that all traffic going to networks outside of the ISP will be sent to the ISP router. Because DNS is not being used in this setup, the DNS server is set to 0.0.0.0. This static setup makes sure that the server always has the same IP address, which makes it possible to reliably talk to the ISP router and other devices on the MDS network.

Yangon Main Branch

Edge router configuration on Yangon main branch



```
Router3
Physical Config CLI Attributes
IOS Command Line Interface
Press RETURN to get started!

Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#int g0/0
Router(config-if)#ip add 209.165.200.2 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#int g0/1
Router(config-if)#ip add 172.16.16.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#int g0/2
Router(config-if)#ip add 172.17.17.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
Router(config-if)#
```

Copy Paste

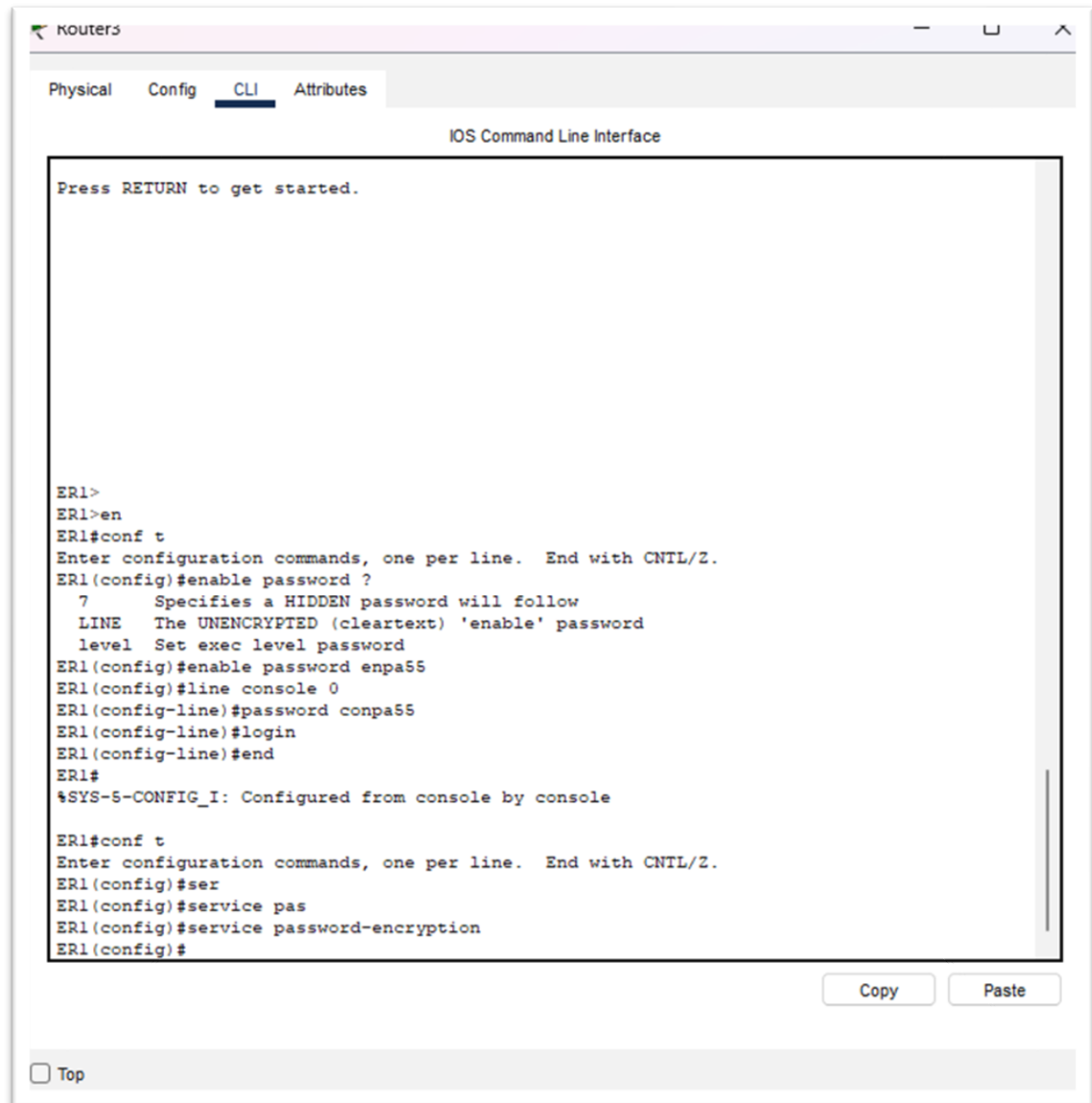
☐ Top

Figure (3.2) IP addressing for Edge router

These commands are preparing the GigabitEthernet0/0, Giga-bitEthernet0/1, and GigabitEthernet0/2 interfaces on the Yangon Edge Router. The interface G0/0 is getting the IP address 209.165.200.2/24, which connects to the ISP router. This is also the public-facing interface and the one used for internet and VPN connectivity. The G0/1 interface is configured with 172.16.16.1/24 and connects the router to Gateway 1. within the Yangon network. The G0/2 interface uses the address space of 172.17.17.1/24 for its connectivity to Gateway 2. The command no shutdown enables each interface bringing it up and ready for the transmission of data. It is these configurations that are setting the routing base that will allow the internal VLAN's in the Yangon network to communicate with the ISP, and ultimately forming an IPSec VPN with the Mandalay branch of Ying Huang, thus facilitating secure and reliable inter-branch connectivity.

```
outer(config)#  
outer#  
SYS-5-CONFIG_I: Configured from console by console  
  
outer#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
outer(config)#hostname ER1  
D1(config)#
```

Figure (3.3) Give edge router hostname for ER1



```
Press RETURN to get started.

ER1>
ER1>en
ER1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
ER1(config)#enable password ?
    7      Specifies a HIDDEN password will follow
    LINE   The UNENCRYPTED (cleartext) 'enable' password
    level  Set exec level password
ER1(config)#enable password enpa55
ER1(config)#line console 0
ER1(config-line)#password conpa55
ER1(config-line)#login
ER1(config-line)#end
ER1#
%SYS-5-CONFIG_I: Configured from console by console

ER1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
ER1(config)#ser
ER1(config)#service pas
ER1(config)#service password-encryption
ER1(config)#
```

Figure (3.4) Giving console, enable, service password-encryption configuration

The router is configured with password security. While the console is protected by the line console 0 function with password conpa55 and login, privileged EXEC mode is secured by the enable password enpa55. For the purpose of to prevent unauthorized users from reading the configuration file's plain-text passwords, the service password-encryption command encrypts them all. The router's basic access security is improved by this configuration.



```
ER1(config)#ip route 0.0.0.0 0.0.0.0 209.165.200.1
ER1(config)#
```

Figure (3.5) Default route

In order to enable internet or external network access, the line `ip route 0.0.0.0 0.0.0.0 209.165.200.1` instructs the router to forward all traffic intended for unidentified networks to the ISP router, which is the next-hop address.

```
R1(config)#access-list 1 per
R1(config)#access-list 1 permit any
R1(config)#ip nat in
R1(config)#ip nat inside source list 1
  Incomplete command.
R1(config)#ip nat inside source list ?
  <1-199> Access list number for local addresses
  WORD    Access list name for local addresses
R1(config)#ip nat inside source list 1 ?
  interface Specify interface for global address
  pool      Name pool of global addresses
R1(config)#ip nat inside source list 1 int
R1(config)#ip nat inside source list 1 int g0/0 overload
R1(config)#int g0/0
R1(config-if)#ip nat outside
R1(config-if)#int range g0/1-2
R1(config-if-range)#ip nat inside
D1(config-if-range)#
```

Figure (3.6) NAT overload

All internal traffic can be translated using the command `access-list 1`. PAT (Port Address Translation) is then enabled by `ip nat inside source list 1 interface g0/0 overload`, allowing several internal devices to share a single public IP address on interface GigabitEthernet0/0. Internal networks are represented by the inside interfaces (g0/1–2), while the outside interface that is linked to the ISP is denoted by g0/0.

```
R1(config)#router rip
R1(config-router)#ver
R1(config-router)#version 2
R1(config-router)#ne
R1(config-router)#network 172.16.16.0
R1(config-router)#network 172.17.17.0
R1(config-router)#de
R1(config-router)#default-information or
R1(config-router)#default-information originate
D1(config-router)#
```

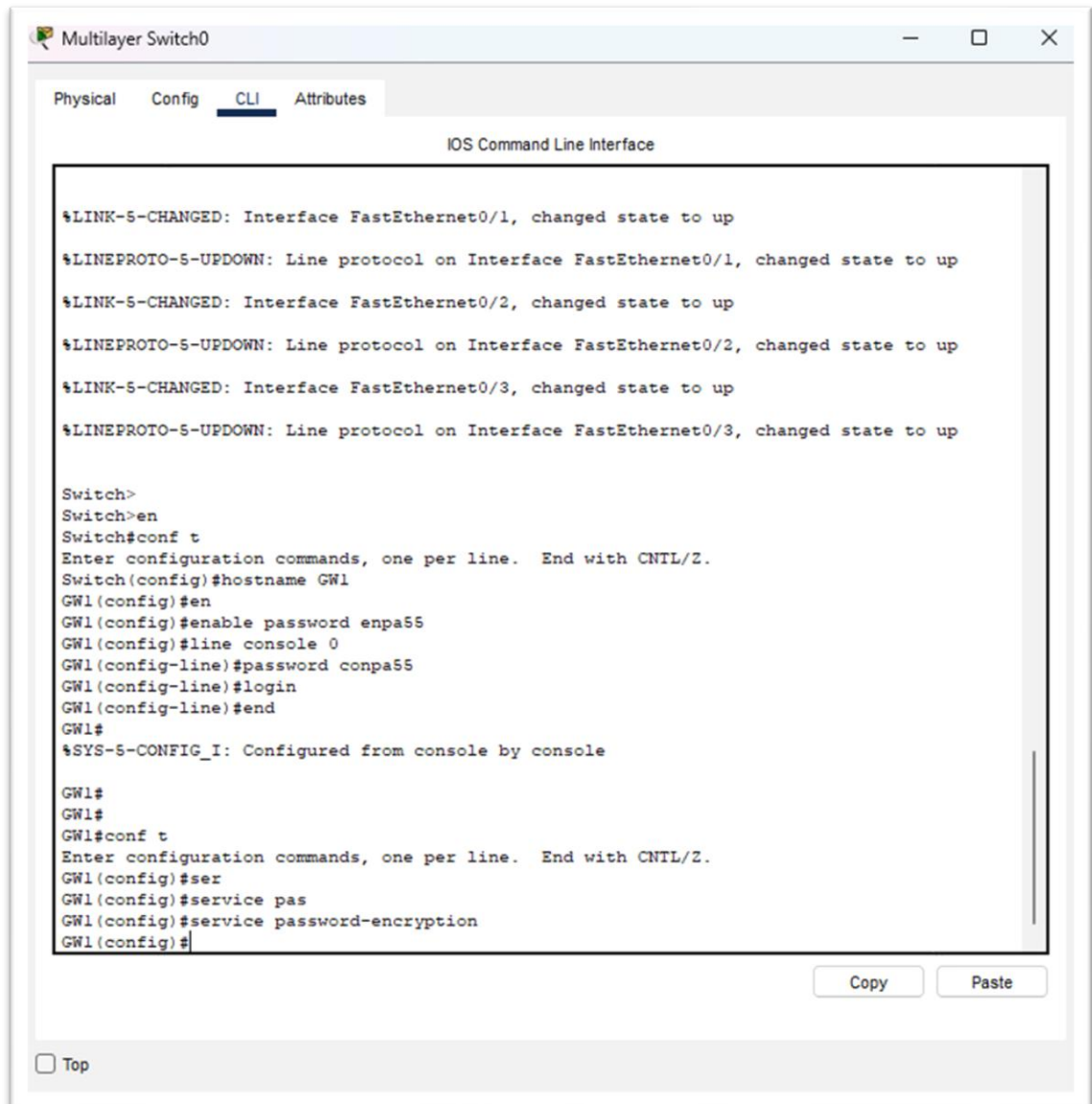
Figure (3.7) RIP routing

RIP version 2 is set up for dynamic routing with this configuration. Those routes can be readily shared by routers through to the network 172.16.16.0 and 172.17.17.0 commands, that suggest local networks. By distributing the router's default

route to other routers, the default-information originate command makes sure that every device is aware of how to connect to external networks using this router.

YGN GW1 configuration

Password configuration for security purpose



The screenshot shows the CLI interface of a Multilayer Switch0. The 'CLI' tab is selected. The interface displays the following commands and their outputs:

```
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname GW1
GW1(config)#en
GW1(config)#enable password enpa55
GW1(config)#line console 0
GW1(config-line)#password conpa55
GW1(config-line)#login
GW1(config-line)#end
GW1#
%SYS-5-CONFIG_I: Configured from console by console

GW1#
GW1#
GW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
GW1(config)#ser
GW1(config)#service pas
GW1(config)#service password-encryption
GW1(config)#
```

At the bottom of the window, there are 'Copy' and 'Paste' buttons, and a 'Top' button with a checkbox.

Figure (3.8) giving hostname, enable password and console password for GW1

By configuring the hostname, turning on console and host passwords, and employing service password-encryption to conceal passwords, this configuration secures the multilayer switch and prevents unauthorized access.

Giving Ip address for connection

```
GW1(config)#int vlan 1
GW1(config-if)#no sh

GW1(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

GW1(config-if)#
GW1(config-if)#ip add 172.16.16.2 255.255.255.0
GW1(config-if)#
```

Figure (3.9) Enable default and giving ip address VLAN 1 for GW 1

This setup gives the multilayer switch the IP address 172.16.16.2/24 and activates VLAN 1. The interface is brought up by the `no shutdown` command, enabling network administration and communication with other devices within the same VLAN.

Assigning Vlan name

```
GW1(config-if)#
GW1(config-if)#ip add 172.16.16.2 255.255.255.0
GW1(config-if)#vlan 10
GW1(config-vlan)#name finance
GW1(config-vlan)#vlan 20
GW1(config-vlan)#name salemart
GW1(config-vlan)#vlan 30
GW1(config-vlan)#name HR
GW1(config-vlan)#name 40
GW1(config-vlan)#name HR
GW1(config-vlan)#vlan 40
GW1(config-vlan)#name Customersp
GW1(config-vlan)#vlan 50
GW1(config-vlan)#name IT
```

Figure (3.10) Assigning department name to each vlan

This configuration produces specific VLANs on the multilayer switch that split the network by department. VLAN 10, 20, 30, 40, and 50 are called Finance , Salesmart, HR, Customer Support, and IT, respectively.

Assigning vlan ip address for connection

```
W1(config)#int vlan 10
W1(config-if)#
LINK-S-CHANGED: Interface Vlan10, changed state to up

W1(config-if)#ip add 192.168.10.1 255.255.255.0
W1(config-if)#int vlan 20
W1(config-if)#
LINK-S-CHANGED: Interface Vlan20, changed state to up

W1(config-if)#ip add 192.168.20.1 255.255.255.0
W1(config-if)#int vlan 30
W1(config-if)#
LINK-S-CHANGED: Interface Vlan30, changed state to up

W1(config-if)#ip add 192.168.30.1 255.255.255.0
W1(config-if)#int vlan 40
W1(config-if)#
LINK-S-CHANGED: Interface Vlan40, changed state to up

W1(config-if)#ip add 192.168.40.1 255.255.255.0
W1(config-if)#int vlan 50
W1(config-if)#
LINK-S-CHANGED: Interface Vlan50, changed state to up

W1(config-if)#ip add 192.168.50.1 255.255.255.0
W1(config-if)#
```

Figure (3.11) Assigning VLAN IP address to each VLAN

It enables inter-VLAN routing on the multilayer switch by assigning IP addresses to VLAN interfaces. each VLAN (10–50) is assigned a distinct subnet, such as `192.168.10.1/24` for VLAN 10 and `192.168.50.1/24` for VLAN 50, and is activated with `no shutdown`. This keeps network traffic segmentation while enabling safe communication between devices in different organizations by using the switch.

Ip routing to get connection

```
Enter configuration commands, one per line. End with CNTL/Z.
GW1(config)#ip routing
GW1(config)#router rip
GW1(config-router)#version 2
GW1(config-router)#network 172.16.16.0
GW1(config-router)#network 192.168.10.0
GW1(config-router)#network 192.168.20.0
GW1(config-router)#network 192.168.30.0
GW1(config-router)#network 192.168.40.0
GW1(config-router)#network 192.168.50.0
GW1(config-router)#
```

Copy

Paste

Figure (3.12) IP routing on GW1

In order to exchange routes between VLANs and the router, this configuration allows IP routing and configures RIP version 2 on the multilayer switch. Devices in various VLANs and subnets can communicate with each other without any problems thanks to the `network` commands, which advertise all VLAN networks (192.168.10.0–50.0) and the connection to the router (172.16.16.0).

Switchport mode trunk for vlan traffic

```
FW1(config-router)#int range f0/1-2
FW1(config-if-range)#sw
FW1(config-if-range)#switchport tru
FW1(config-if-range)#switchport trunk en
FW1(config-if-range)#switchport trunk encapsulation do
FW1(config-if-range)#switchport trunk encapsulation dot1q
FW1(config-if-range)#sw
FW1(config-if-range)#switchport mo
FW1(config-if-range)#switchport mode tr
FW1(config-if-range)#switchport mode trunk

FW1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan40, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan50, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
```

Figure (3.13) Changing switchport mode trunk on GW1

Multiple VLANs can move within a single link between switches or routers because to this configuration, which uses FastEthernet0/1–2 interfaces in trunk mode. IEEE 802.1Q tagging is specified by the switchport trunk encapsulation dot1q command, which permits VLAN traffic separation across the trunk.

Redundancy HSRP configuration

```
% Invalid input detected at '^' marker.

GW1(config-if)#standby 10 track f0/1
GW1(config-if)#int vlan 20
GW1(config-if)#standby 20 ip 192.168.20.254
GW1(config-if)#standby 20 preempt
GW1(config-if)#standby 20 priority 105
GW1(config-if)#standby 20 track f0/1
GW1(config-if)#
%HSRP-6-STATECHANGE: Vlan20 Grp 20 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan20 Grp 20 state Standby -> Active

GW1(config-if)#int vlan 30
GW1(config-if)#standby 30 ip 192.168.30.254
GW1(config-if)#standby 30 preempt
GW1(config-if)#standby 30 priority 105
GW1(config-if)#standby 30 track f0/1
GW1(config-if)#
%HSRP-6-STATECHANGE: Vlan30 Grp 30 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan30 Grp 30 state Standby -> Active

GW1(config-if)#
GW1(config-if)#int vlan 50
GW1(config-if)#standby 50 ip 192.168.50.254
GW1(config-if)#standby 50 preempt
GW1(config-if)#standby 50 priority 105
GW1(config-if)#standby 50 track f0/1
GW1(config-if)#
%HSRP-6-STATECHANGE: Vlan50 Grp 50 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan50 Grp 50 state Standby -> Active
|
```

Figure (3.14) HSRP configuration on GW1

It activates in HSRP in VLANs 20, 30, and 50 to give gateway redundancy. Each VLAN's default gateway is a virtual IP address, like 192.168.20.254. These priority, preemptive, and track commands make sure that the standby switch takes over instantly in case that the current switch or downstream link fails, offering constant network connectivity.

SSH configuration for remote management

```
GW1(config)#username admin password telpa55
^
: Invalid input detected at '^' marker.

GW1(config)#username?
username
GW1(config)#username admin ?
password  Specify the password for the user
privilege Set user privilege level
secret    Specify the secret for the user
<cr>
GW1(config)#username admin ?
password  Specify the password for the user
privilege Set user privilege level
secret    Specify the secret for the user
<cr>
GW1(config)#username admin pas
GW1(config)#username admin password ?
0         Specifies an UNENCRYPTED password will follow
7         Specifies a HIDDEN password will follow
LINE The UNENCRYPTED (cleartext) user password
GW1(config)#username admin password telpa55
GW1(config)#cr
GW1(config)#crypto key
GW1(config)#crypto key generate rsa
The name for the keys will be: GW1.info.com
Choose the size of the key modulus in the range of 360 to 2048 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
: Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

GW1 /root#
```

Figure (3.15) SSH configuration on GW1

A 1024-bit RSA key is created for SSH, and a user account called admin is set up with the password telpa55. By offering encrypted remote access, this guarantees safe communication and stops unwanted logins.

Yangon GW2

Password configuration for security purpose

```
Compiled Thu 05-Jul-07 22:22 by pt_team

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname GW2
GW2(config)#enable password enpa55
GW2(config)#line console 0
GW2(config-line)#password compa55
GW2(config-line)#login
GW2(config-line)#
GW2(config-line)#end
GW2#
%SYS-5-CONFIG_I: Configured from console by console

GW2#
GW2#ser
GW2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
GW2(config)#ser
GW2(config)#service pas
GW2(config)#service password-encryption
GW2(config)#
```

Figure (3.16) giving hostname, enable password and console password for GW2

GW2 is the switch's name, and it is secured by console and enable passwords. By hiding these passwords, the service password-encryption command improves device security in general.

Giving Ip address for connection

```
FW2(config)#int vlan 1
FW2(config-if)#no sh

FW2(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

FW2(config-if)#ip add 172.17.17.2 255.255.255.0
FW2(config-if)#
```

Figure (3.17) enable default VLAN and giving ip address

With the above configuration, VLAN 1 is activated and given the IP address 172.17.17.2/24 . The VLAN interface is set up by the `no shutdown` command, allowing network administration and device communication.

VLAN naming

```
GW2(config-if)#ip add 172.17.17.2 255.255.255.0
GW2(config-if)#vlan 10
GW2(config-vlan)#name finance
GW2(config-vlan)#vlan 20
GW2(config-vlan)#name salemart
GW2(config-vlan)#vlan 30
GW2(config-vlan)#name HR
GW2(config-vlan)#vlan 40
GW2(config-vlan)#name customersp
GW2(config-vlan)#vlan 50
GW2(config-vlan)#name IT
GW2(config-vlan)#
```

Figure (3.18) Giving VLAN name to each vlan

The switch's VLANs 10, 20, 30, 40, and 50 are created in this setting, and names like Finance , Salesmart , HR , CustomerSP , and IT are assigned. By splitting departmental traffic, these VLANs enhance network performance, security, and organization.

Assigning Vlan ip address for connection

```
W2(config-vlan)#int vlan 10
W2(config-if)#
LINK-5-CHANGED: Interface Vlan10, changed state to up

W2(config-if)#ip add 192.168.10.2 255.255.255.0
W2(config-if)#int vlan 20
W2(config-if)#
LINK-5-CHANGED: Interface Vlan20, changed state to up

W2(config-if)#ip add 192.168.20.2 255.255.255.0
W2(config-if)#int vlan 30
W2(config-if)#
LINK-5-CHANGED: Interface Vlan30, changed state to up

W2(config-if)#ip add 192.168.30.2 255.255.255.0
W2(config-if)#int vlan 40
W2(config-if)#
LINK-5-CHANGED: Interface Vlan40, changed state to up

W2(config-if)#ip add 192.168.40.2 255.255.255.0
W2(config-if)#int vlan 50
W2(config-if)#
LINK-5-CHANGED: Interface Vlan50, changed state to up

W2(config-if)#ip add 192.168.50.2 255.255.255.0
W2(config-if)#
```

Figure (3.19) Giving ip address to each vlan

This configuration configures each VLAN interface and assigns IP addresses to VLAN interfaces 10–50, such as `192.168.10.2` and `192.168.50.2`. In doing this, inter-VLAN communication becomes possible, allowing different groups to securely link through the switch while maintaining traffic separation.

IP routing configuration for the connection

```
GW2(config)#ip routing
GW2(config)#router rip
GW2(config-router)#ver
GW2(config-router)#version 2
GW2(config-router)#network 172.17.17.0
GW2(config-router)#network 192.168.10.0
GW2(config-router)#network 192.168.20.0
GW2(config-router)#network 192.168.30.0
GW2(config-router)#network 192.168.40.0
GW2(config-router)#network 192.168.50.0
GW2(config-router)#
```

Figure (3.20) Ip routing on GW 2

It configures RIP version 2 on the switch and sets up IP routing. The networks on the list provide automatic route exchange between VLANs and the router, to ensure smooth communication between all of the organization's subnets.

Switchport mode trunk for vlan traffic

```
GW2(config-if-range)#int range 10/1-2
GW2(config-if-range)#sw
GW2(config-if-range)#switchport tr
GW2(config-if-range)#switchport trunk en
GW2(config-if-range)#switchport trunk encapsulation do
GW2(config-if-range)#switchport trunk encapsulation dot1q
GW2(config-if-range)#sw
GW2(config-if-range)#switchport mo
GW2(config-if-range)#switchport mode tr
GW2(config-if-range)#switchport mode trunk

GW2(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan40, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan50, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
```

Figure (3.21) Changing switchport to trunk mode

Using the dot1q encapsulation technique, interfaces FastEthernet0/1–2 is set up as trunk ports. This enables a single physical connection between switches to be shared by several VLANs.

HSRP for redundancy configuration

```
W2(config-if-range)#int vlan 10
W2(config-if)#standby 10 ip 192.168.10.254
W2(config-if)#standby 10 preempt
W2(config-if)#standby 10 priority 100
W2(config-if)#standby 10 track f0/1
W2(config-if)#int vlan 20
W2(config-if)#standby 20 ip 192.168.20.254
W2(config-if)#standby 20 preempt
W2(config-if)#standby 20 priority 100
W2(config-if)#standby 20 track f0/1
W2(config-if)#int vlan 30
W2(config-if)#standby 30 ip 192.168.30.254
W2(config-if)#standby 30 preempt
W2(config-if)#standby 30 priority 100
W2(config-if)#standby 30 track f0/1
W2(config-if)#int vlan 40
W2(config-if)#standby 40 ip 192.168.40.254
W2(config-if)#standby 40 preempt
W2(config-if)#standby 40 priority 100
W2(config-if)#standby 40 track f0/1
W2(config-if)#int vlan 50
W2(config-if)#standby 50 ip 192.168.50.254
W2(config-if)#standby 50 preempt
W2(config-if)#standby 50 priority 100
W2(config-if)#standby 50 track f0/1
W2(config-if)#
:USBD-6-STATE/CHANGE: Vlan20 Gm 30 state Speak -> Standby
```

Figure (3.22) HSRP on GW 2

To be able to provide gateway redundancy, this configuration sets HSRP on VLANs 10, 20, 30, 40, and 50. A shared virtual IP (such as 192.168.10.254) acts as the default gateway for every VLAN.

Remote management configuration SSH

```
GW2(config-if)#
GW2(config-if)#exir
^
% Invalid input detected at '^' marker.

GW2(config-if)#
GW2(config-if)#exit
GW2(config)#ip dom
GW2(config)#ip domain-name info.com
GW2(config)#username admin password telpa55
GW2(config)#cr
GW2(config)#crypto key
GW2(config)#crypto key ge
GW2(config)#crypto key generate rs
GW2(config)#crypto key generate rsa
The name for the keys will be: GW2.info.com
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

GW2(config)#
```

Figure (3.23) SSH configuration on GW 2

For authentication, a user account admin is created with a username and password telpa55 and a domain name info.com was set. After that, an RSA key with a 1024-bit modulus is created, allowing SSH to be used for encrypted communication with the switch and safe remote access.

Switch 1 configuration

Password configuration for security

```
switch>en
switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
switch(config)#hostname AS1
AS1(config)#enable password enpa55
AS1(config)#line console 0
AS1(config-line)#pas
AS1(config-line)#password conpa55
AS1(config-line)#login
AS1(config-line)#exit
AS1(config)#
AS1(config)#ser
AS1(config)#service pas
AS1(config)#service password-encryption
```

Figure (3.24) giving hostname, enable password and console password for AS1

To control access, security settings are configured using enable and console passwords, and the switch is renamed AS1. To enhance security against unwanted access, all stored passwords are encrypted using the service password-encryption command.

Assigning the VLAN

```
S1(config)#vlan 10
S1(config-vlan)#name finance
S1(config-vlan)#vlan 20
S1(config-vlan)#name salemart
S1(config-vlan)#vlan 30
S1(config-vlan)#name HR
S1(config-vlan)#vlan 40
S1(config-vlan)#name customersp
S1(config-vlan)#vlan 50
S1(config-vlan)#
S1(config-vlan)#name IT
S1(config-vlan)#
```

Figure (3.25) naming each vlan

The configuration and names of VLANs 10, 20, 30, 40, and 50 are Finance, Salesmart , HR , CustomerSP , and IT . By separating departmental traffic, each VLAN contributes to better security, less congestion, and more effective network management.

Change Switchport mode trunk for etherchannel

```
AS1(config)#int E0/1
AS1(config-if)#sw
AS1(config-if)#switchport mo
AS1(config-if)#switchport mode tr
AS1(config-if)#switchport mode trunk
AS1(config-if)#int range f0/7-10
AS1(config-if-range)#chan
AS1(config-if-range)#channel-protocol pagp
AS1(config-if-range)#channel-group 1 mode auto
AS1(config-if-range)#
Creating a port-channel interface Port-channel 1

!LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to down
!LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up
!LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to down
!LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up
!LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to down
!LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
!LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to down
!LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
```

Figure (3.26) Changing switchport mode trunk and configuring EtherChannel

Port-Channel 1 is created by connecting FastEthernet0/7–10 interfaces using PAgP EtherChannel. Higher bandwidth and redundancy are provided by combining several links into a single logical connection. Traffic flows uninterrupted through the remaining active links in the event of a link failure.

Changing switchport mode access for traffic

```
AS1(config-if-range)#int f0/2
AS1(config-if)#sw
AS1(config-if)#switchport mode access
AS1(config-if)#sw
AS1(config-if)#switchport access vlan 10
AS1(config-if)#int f0/3
AS1(config-if)#sw
AS1(config-if)#switchport mode access
AS1(config-if)#sw
AS1(config-if)#switchport ac
AS1(config-if)#switchport access vlan 20
AS1(config-if)#int f0/4
AS1(config-if)#switchport mode access
AS1(config-if)#switchport access vlan 30
AS1(config-if)#int f0/5
AS1(config-if)#switchport mode access
AS1(config-if)#switchport access vlan 40
AS1(config-if)#int f0/6
AS1(config-if)#switchport mode access
AS1(config-if)#switchport access vlan 50
AS1(config-if)#
```

Figure (3.27) Changing switchport mode access for end devices interface

Every interface (FastEthernet0/2 to 0/6) is set up as a access port and given to a particular VLAN, such as VLAN 10 for Finance, VLAN 20 for Sales, VLAN 30 for HR, VLAN 40 for Customer Support, and VLAN 50 for IT. By separating departmental traffic, this configuration improves network security, performance, and organization.

Switch 2 configuration

Password configuration for security

```
AS2(config)#enable password enable
AS2(config)#line console 0
AS2(config-line)#password conpa55
AS2(config-line)#login
AS2(config-line)#exit
AS2(config)#
AS2(config)#service password
AS2(config)#service password-encryption
AS2(config)#
```

Figure (3.28) giving hostname, enable password and console password for AS2

To secure device access, enable console passwords. The service password-encryption command encrypts these passwords in the configuration file.

Assigning VLAN name

```
AS2(config)#vlan 10
AS2(config-vlan)#name finance
AS2(config-vlan)#vlan 20
AS2(config-vlan)#name salemart
AS2(config-vlan)#vlan 30
AS2(config-vlan)#name HR
AS2(config-vlan)#vlan 40
AS2(config-vlan)#name customersp
AS2(config-vlan)#vlan 50
AS2(config-vlan)#name IT
AS2(config-vlan)#
```

Figure (3.29) naming the each VLAN

This setup creates and names several VLANs (10–50) according to departments like IT, Finance, Sales, HR, and Customer Support.

Changing switchport mode trunk for EtherChannel

```
AS2(config)#int f0/1
AS2(config-if)#sw
AS2(config-if)#switchport mode trunk
AS2(config-if)#int range f0/7-10
AS2(config-if-range)#channel-protocol pagp
AS2(config-if-range)#channel-group 1 mode desirable
AS2(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up
```

Figure (3.30) Changing trunk mode and configuring etherchannel

By combining several FastEthernet links into Port-channel 1 , EtherChannel with PAgP ensures redundancy and boosts bandwidth. Multiple VLANs can pass through in trunk mode, increasing network efficiency and dependability.

Changing switchport mode access for VLAN traffic

```
AS2(config-if-range)#int f0/2
AS2(config-if)#sw
AS2(config-if)#switchport mode access
AS2(config-if)#sw
AS2(config-if)#switchport access vlan 10
AS2(config-if)#int f0/3
AS2(config-if)#switchport mode access
AS2(config-if)#switchport access vlan 20
AS2(config-if)#int f0/4
AS2(config-if)#switchport mode access
AS2(config-if)#switchport access vlan 30
AS2(config-if)#int f0/5
AS2(config-if)#switchport mode access
AS2(config-if)#switchport access vlan 40
AS2(config-if)#int f0/6
AS2(config-if)#switchport mode access
AS2(config-if)#switchport access vlan 50
AS2(config-if)#
```

Figure (3.31) Changing switchport mode access for end devices interface

Switch ports are assigned to various VLANs in this configuration, such as 10, 20, 30, 40, and 50, each of which stands for a distinct department.

Mandalay side

Mandalay Edge router configuration

Giving IP address for Connection

```
Router(config)#int se0/3/0
Router(config-if)#ip add 209.165.100.2 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-S-CHANGED: Interface Serial0/3/0, changed state to up
Router(config-if)#
```

Figure (3.32) Giving IP address on se0/3/0

This configuration puts the Mandalay Edge Router's Serial0/3/0 interface in an active state for WAN connectivity by assigning it the IP address 209.165.100.2 with subnet mask 255.255.255.0 and enabling it with the no shutdown command.

Password configuration for the security

```
Router(config)#hostname ER2
ER2(config)#enable secret enpa55
ER2(config)#line console 0
ER2(config-line)#password compa55
ER2(config-line)#login
ER2(config-line)#exit
ER2(config)#
ER2(config)#service
ER2(config)#service-pas
ER2(config)#service ?
    dhcp          Enable DHCP server and relay agent
    nagle          Enable Nagle's congestion control algorithm
    password-encryption  Encrypt system passwords
    timestamps     Timestamp debug/log messages
ER2(config)#service password-en
ER2(config)#service password-encryption
ER2(config)#
```

Figure (3.33) Changing hostname, giving enable password and console password

The router hostname in this configuration is ER2 , and enable secret and console passwords are used to create secure access. All plain-text passwords are encrypted using the command service password-encryption, which improves router security and stops unwanted access.

IP route for traffic

```
ER2(config)#ip route 0.0.0.0 0.0.0.0 209.165.100.1
ER2(config)#ac
ER2(config)#access-list 1 permit any
ER2(config)#
```

Figure (3.34) IP route on Mandalay side

All outgoing traffic is routed to the ISP gateway via IP route 0.0.0.0 0.0.0.0 209.165.100.1. Furthermore, an access control list (ACL 1) is set up to permit any, which permits all IP traffic to flow through without any limitations.

NAT service

```
ER2(config)#ip nat inside source list 1
ER2(config)#ip nat inside source list 1 int se0/3/0
ER2(config)#ip nat inside source list 1 int se0/3/0 overload
ER2(config)#int se0/3/0
ER2(config-if)#ip nat out
ER2(config-if)#ip nat outside
ER2(config-if)#int range g0/1-2
ER2(config-if-range)#ip nat inside
ER2(config-if-range)#
```

Figure (3.35) IP nat service

By enabling on NAT with overload (PAT), this configuration allows several internal devices to share a single public IP address. Gigabit0/1–2 is configured as inside interfaces for internal network traffic translation, while Serial0/3/0 is set as the outside NAT interface.

RIP routing default route

```
ER2(config)#router rip
ER2(config-router)#ver
ER2(config-router)#version 2
ER2(config-router)#network 172.18.18.0
ER2(config-router)#network 172.19.19.0
ER2(config-router)#de
ER2(config-router)#default-information ori
ER2(config-router)#default-information originate
ER2(config-router)#
ER2(config-router)#
```

Figure (3.36) Rip routing

With this configuration, networks 172.18.18.0 and 172.19.19.0 enable RIP version 2, enabling dynamic route sharing and promoting a default route for Internet connectivity.

Giving IP address for connection

```
ER2(config-if-range)#int g0/0
ER2(config-if)#ip add 172.18.18.1 255.255.255.0
ER2(config-if)#int g0/1
ER2(config-if)#ip add 172.19.19.1 255.255.255.0
ER2(config-if)#int g0/0
ER2(config-if)#no sh
```

Figure (3.37) Giving IP address on each interface

In order to enable network connectivity, this configuration gives interfaces GigabitEthernet0/0 and GigabitEthernet0/1 the IP addresses 172.18.18.1 and 172.19.19.1, respectively, and uses the no shutdown command to activate the interfaces.

GW3 Configuration

Giving password for security

```
Switch(config)#hostname GW3
GW3(config)#enable secret enpa55
GW3(config)#line console 0
^
% Invalid input detected at '^' marker.

GW3(config)#line console 0
GW3(config-line)#password conpa55
GW3(config-line)#login
GW3(config-line)#exit
GW3(config)#
GW3(config)#service-pass
GW3(config)#service pas
GW3(config)#service password-encryption
GW3(config)#
```

Figure (3.38) Changing hostname, giving enable password and console password

By creating a hostname and turning on both privileged and console passwords, this setup secures the switch GW3. To prevent unwanted access, the `service password-encryption` command makes sure that all stored passwords are encrypted.

Building VLAN 1 for IP address

```
GW3(config)#int vlan 1
GW3(config-if)#no sh

GW3(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

GW3(config-if)#ip add 172.18.18.2 255.255.255.0
GW3(config-if)#
```

Figure (3.39) Building default VLAN 1

This configuration gives switch GW3 the IP address 172.18.18.2/24 and activates VLAN 1. Connection within the Mandalay branch network is made possible by the `no shutdown` command, which guarantees that the VLAN interface is up and running.

Assigning VLAN name

```
GW3(config)#vlan 60
GW3(config-vlan)#name finance
GW3(config-vlan)#vlan 70
GW3(config-vlan)#name salemart
GW3(config-vlan)#vlan 80
GW3(config-vlan)#name HR
GW3(config-vlan)#vlan 90
GW3(config-vlan)#name customersp
GW3(config-vlan)#vlan 100
GW3(config-vlan)#name IT
GW3(config-vlan)#
```

Figure (3.40) Giving VLAN name to each VLAN

On switch GW3, this setting builds VLANs 60 to 100 with names such as Finance, Salemart, HR, Customersp, and IT. By segmenting the Mandalay branch network into groups of logic, these VLANs improve security and traffic management.

Assign IP to each VLAN

```
GW3(config-if)#ip add 192.168.60.1 255.255.255.0
GW3(config-if)#int vlan 70
GW3(config-if)#
%LINK-5-CHANGED: Interface Vlan70, changed state to up

GW3(config-if)#ip add 192.168.70.1 255.255.255.0
GW3(config-if)#int vlan 80
GW3(config-if)#
%LINK-5-CHANGED: Interface Vlan80, changed state to up

GW3(config-if)#ip add 192.168.80.1 255.255.255.0
GW3(config-if)#int vlan 90
GW3(config-if)#
%LINK-5-CHANGED: Interface Vlan90, changed state to up

GW3(config-if)#ip add 192.168.90.1 255.255.255.0
GW3(config-if)#int vlan 100
GW3(config-if)#
%LINK-5-CHANGED: Interface Vlan100, changed state to up

GW3(config-if)#ip add 192.168.100.1 255.255.255.0
GW3(config-if)#
```

Figure (3.41) Giving IP address to each VLAN

It enables departments such as Finance, HR, and IT to communicate across VLANs by allocating Internet Protocol (IP) addresses to VLAN interfaces 60–100 on switch GW3. In order to ensure appropriate network segmentation and effective routing within the Mandalay branch, each VLAN is assigned a distinct subnet.

RIP routing

```
GW3(config)#ip routing
GW3(config)#router rip
GW3(config-router)#version 2
GW3(config-router)#network 172.18.18.0
GW3(config-router)#network 192.168.60.0
GW3(config-router)#network 192.168.70.0
GW3(config-router)#network 192.168.80.0
GW3(config-router)#network 192.168.90.0
GW3(config-router)#network 192.168.100.0
GW3(config-router)#
```

Figure (3.42) Rip routing

On switch GW3, this configuration activates RIP version 2 routing, promoting the networks 172.18.18.0, 192.168.60.0, 192.168.70.0, 192.168.80.0, 192.168.90.0, and 192.168.100.0. In order to guarantee appropriate route exchange throughout the Mandalay branch network, it permits communication between VLANs and other linked routers.

Changing switchport mode trunk for EtherChannel

```
GW3(config)#int range f0/1-2
GW3(config-if-range)#sw
GW3(config-if-range)#switchport tr
GW3(config-if-range)#switchport trunk en
GW3(config-if-range)#switchport trunk encapsulation do
GW3(config-if-range)#switchport trunk encapsulation dot1q
GW3(config-if-range)#sw
GW3(config-if-range)#switchport mode trunk

GW3(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan60, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan70, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan80, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan90, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan100, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
```

Figure (3.43) Changing switchport into trunk mode

It uses dot1q encapsulation to configure FastEthernet0/1–0/2 on switch GW3 as trunk ports. These trunks ensure inter-VLAN communication between switches and routers by allowing multiple VLANs (60–100) to transmit over one link.

HSRP For redundancy

```
GW3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
GW3(config)#
GW3(config)#int vlan 60
GW3(config-if)#standby 60 ip 192.168.60.254
GW3(config-if)#standby 60 preempt
GW3(config-if)#standby 60 priority 105
GW3(config-if)#standby 60 track f0/1
GW3(config-if)#int vlan 70
GW3(config-if)#standby 70 ip 192.168.70.254
GW3(config-if)#standby 70 preempt
GW3(config-if)#standby 70 priority 105
GW3(config-if)#standby 70 track f0/1
GW3(config-if)#int vlan 80
GW3(config-if)#standby 80 ip 192.168.80.254
GW3(config-if)#standby 80 preempt
GW3(config-if)#standby 80 priority 105
GW3(config-if)#standby 80 track f0/1
GW3(config-if)#int vlan 90
GW3(config-if)#standby 90 ip 192.168.90.254
GW3(config-if)#standby 90 preempt
GW3(config-if)#standby 90 priority 105
GW3(config-if)#standby 90 track f0/1
GW3(config-if)#int vlan 100
GW3(config-if)#standby 100 ip 192.168.100.254
GW3(config-if)#standby 100 preempt
GW3(config-if)#standby 100 priority 105
GW3(config-if)#standby 100 track f0/1
GW3(config-if)#
%HSRP-6-STATECHANGE: Vlan100 Grp 100 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan100 Grp 100 state Standby -> Active

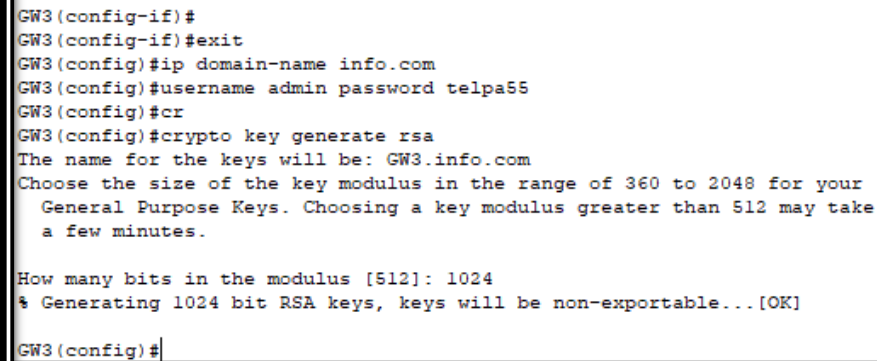
%HSRP-6-STATECHANGE: Vlan90 Grp 90 state Speak -> Standby

%HSRP-6-STATECHANGE: Vlan90 Grp 90 state Standby -> Active
```

Figure (3.44) HSRP configuration

It gives gateway redundancy and failover support by enabling HSRP on VLANs 60–100 in GW3. In the instance that the primary gateway fails, each VLAN has a virtual IP with a fixed priority and interface tracking to guarantee ongoing network availability.

SSH for remote management



```
GW3(config-if)#
GW3(config-if)#exit
GW3(config)#ip domain-name info.com
GW3(config)#username admin password telpa55
GW3(config)#cr
GW3(config)#crypto key generate rsa
The name for the keys will be: GW3.info.com
Choose the size of the key modulus in the range of 360 to 2048 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

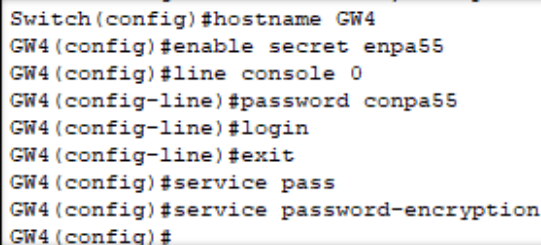
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
GW3(config)#
```

Figure (3.45) SSH configuration on GW3

This configuration creates a local admin user with a password, defines a domain name, and generates a 1024-bit RSA key to enable SSH security on switch GW3. This improves the Mandalay network's device management security and guarantees encrypted remote access.

GW4 Configuration

Giving password for the security purpose

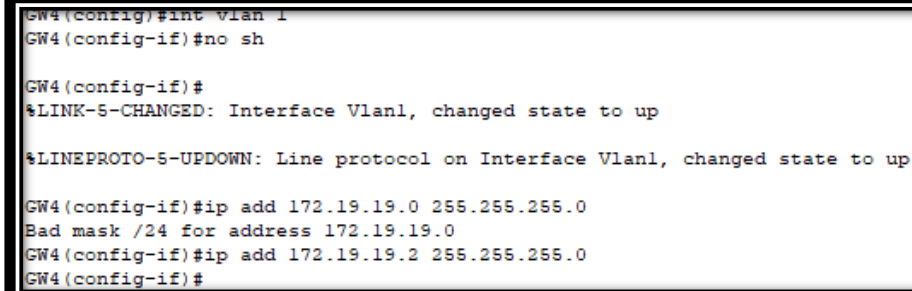


```
Switch(config)#hostname GW4
GW4(config)#enable secret enpa55
GW4(config)#line console 0
GW4(config-line)#password conpa55
GW4(config-line)#login
GW4(config-line)#exit
GW4(config)#service pass
GW4(config)#service password-encryption
GW4(config)#
```

Figure (3.46) Changing hostname, giving enable password and console password

When creating a enable secret password (`enpa55`) and a console password (`conpa55`) for login, this configuration on switch GW4 secures device access.

Building VLAN 1



```
GW4(Config)#int vlan 1
GW4(config-if)#no sh

GW4(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

GW4(config-if)#ip add 172.19.19.0 255.255.255.0
Bad mask /24 for address 172.19.19.0
GW4(config-if)#ip add 172.19.19.2 255.255.255.0
GW4(config-if)#
```

Figure (3.47) Enable the default VLAN 1 giving IP address

VLAN 1 is set with `no shutdown`, and then a management IP is assigned. The first try to employ `ip address 172.19.19.0 255.255.255.0` was unsuccessful because .0 is the address of the network and not a host. For the 172.19.19.0/24 subnet, the correct host IP is `172.19.19.2 255.255.255.0`.

Assign VLAN name

```
GW4(config)#vlan 60
GW4(config-vlan)#name finance
GW4(config-vlan)#vlan 70
GW4(config-vlan)#name salemart
GW4(config-vlan)#vlan 80
GW4(config-vlan)#name HR
GW4(config-vlan)#vlan 90
GW4(config-vlan)#name customersp
GW4(config-vlan)#vlan 100
GW4(config-vlan)#name IT
GW4(config-vlan)#
```

Figure (3.48) Naming the each VLAN

VLANs 60–100 is made and named for departments such as IT, HR, and finance on switch GW4. By segmenting the network, this configuration improves security and effectively handles traffic.

Assigning VLAN IP address to each VLAN

```
GW4(config-vlan)#int vlan 60
GW4(config-if)#
%LINK-5-CHANGED: Interface Vlan60, changed state to up

GW4(config-if)#ip add 192.168.60.2 255.255.255.0
GW4(config-if)#int vlan 70
GW4(config-if)#
%LINK-5-CHANGED: Interface Vlan70, changed state to up

GW4(config-if)#ip add 192.168.70.2 255.255.255.0
GW4(config-if)#int vlan 80
GW4(config-if)#
%LINK-5-CHANGED: Interface Vlan80, changed state to up

GW4(config-if)#ip add 192.168.80.2 255.255.255.0
GW4(config-if)#int vlan 90
GW4(config-if)#
%LINK-5-CHANGED: Interface Vlan90, changed state to up

GW4(config-if)#ip add 192.168.90.2 255.255.255.0
GW4(config-if)#int vlan 100
GW4(config-if)#
%LINK-5-CHANGED: Interface Vlan100, changed state to up

GW4(config-if)#ip add 192.168.100.2 255.255.255.0
GW4(config-if)#
```

Figure (3.49) Giving ip address to each VLAN in GW4

In this configuration, a switch or router's VLAN interfaces (60, 70, 80, 90, and 100) are created, assigned IP addresses, and activated. Through inter-VLAN routing, these IPs serve as gateways for devices within each VLAN, enabling communication between them.

RIP Routing for connection

```
GW4(config)#ip routing
GW4(config)#router rip
GW4(config-router)#version 2
GW4(config-router)#network 172.19.19.0
GW4(config-router)#network 192.168.60.0
GW4(config-router)#network 192.168.70.0
GW4(config-router)#network 192.168.80.0
GW4(config-router)#network 192.168.90.0
GW4(config-router)#network 192.168.100.0
GW4(config-router)#
```

Figure (3.50) Rip routing

Communication between all VLANs and linked networks is made possible by this configuration, which activates routing and uses RIP v2 to automatically exchange route information between networks such as 172.19.19.0 and 192.168.60.0 to 192.168.100.0.

Changing trunk mode for communication

```
GW4(config-router)#int range f0/1-2
GW4(config-if-range)#sw
GW4(config-if-range)#switchport trunk en
GW4(config-if-range)#switchport trunk encapsulation do
GW4(config-if-range)#switchport trunk encapsulation dot1q
GW4(config-if-range)#s
GW4(config-if-range)#sw
GW4(config-if-range)#switchport mode trunk

GW4(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan60, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan70, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan80, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan90, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan100, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
```

Figure (3.51) changing trunk mode

By turning ports F0/1 and F0/2 into trunk ports, all VLANs (60, 70, 80, 90, and 100) can use them to send data. It allows inter-VLAN routing and enables communication between switches or between a switch and a router.

HSRP for standby

```
GW4(config-if)#standby 70 track f0/1
GW4(config-if)#int vlan 80
GW4(config-if)#standby 80 ip 192.168.80.254
GW4(config-if)#standby 80 preempt
GW4(config-if)#standby 80 priority 105
GW4(config-if)#standby 80 track f0/1
GW4(config-if)#int vlan 90
GW4(config-if)#standby 90 ip 192.168.90.254
GW4(config-if)#standby 90 preempt
GW4(config-if)#standby 90 priority 105
GW4(config-if)#standby 90 track f0/1
GW4(config-if)#int vlan 100
GW4(config-if)#standby 100 ip 192.168.100.254
GW4(config-if)#standby 100 preempt
GW4(config-if)#standby 100 priority 105
GW4(config-if)#standby 100 track f0/1
GW4(config-if)#
%HSRP-6-STATECHANGE: Vlan60 Grp 60 state Speak -> Standby
%HSRP-6-STATECHANGE: Vlan60 Grp 60 state Standby -> Active
%HSRP-6-STATECHANGE: Vlan90 Grp 90 state Speak -> Standby
%HSRP-6-STATECHANGE: Vlan90 Grp 90 state Standby -> Active
%HSRP-6-STATECHANGE: Vlan70 Grp 70 state Speak -> Standby
%HSRP-6-STATECHANGE: Vlan70 Grp 70 state Standby -> Active
%HSRP-6-STATECHANGE: Vlan100 Grp 100 state Speak -> Standby
%HSRP-6-STATECHANGE: Vlan100 Grp 100 state Standby -> Active
%HSRP-6-STATECHANGE: Vlan80 Grp 80 state Speak -> Standby
%HSRP-6-STATECHANGE: Vlan80 Grp 80 state Standby -> Active
```

Figure (3.52) HSRP configuration on GW4

HSRP is used in this kind of configuration to improve network reliability. The two routers share a virtual IP address for each of the VLANs (60, 70, 80, 90, and 100). Another router takes over immediately in the case of a router failure, ensuring uninterrupted network service.

SSH remote management

```
GW4(config-if)#
GW4(config-if)#exit
GW4(config)#
GW4(config)#ip doma
GW4(config)#ip domain-name info.com
GW4(config)#username admin password telpa55
GW4(config)#crypto key generate rsa
The name for the keys will be: GW4.info.com
Choose the size of the key modulus in the range of 360 to 2048 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
GW4(config)#
```

Figure (3.53) SSH configuration on GW4

By assigning a domain name, user account, and encryption key, this configuration allows safe SSH login to the router. It provides secure and encrypted remote access to the router.

AS3 Configuration

Giving Enable and console password

```
Switch(config)#hostname AS3
AS3(config)#enable secret enpa55
AS3(config)#line console 0
AS3(config-line)#password conpa55
AS3(config-line)#login
AS3(config-line)#exit
AS3(config)#service pass
AS3(config)#service password-encryption
AS3(config)#
```

Figure (3.54) Changing hostname, giving enable password and console password

With this configuration, the switch receives the name AS3 and both console and administrator access are password protected.

Naming each VLAN

```
AS3(config-vlan)#name finance
AS3(config-vlan)#vlan 70
AS3(config-vlan)#name salemart
AS3(config-vlan)#vlan 80
AS3(config-vlan)#name HR
AS3(config-vlan)#vlan 90
AS3(config-vlan)#name customersp
AS3(config-vlan)#vlan 100
AS3(config-vlan)#name IT
AS3(config-vlan)#
```

Figure (3.55) naming the each VLAN on AS3

Five VLANs are created on the switch with this configuration: Finance (60), Salemart (70), HR(80),Customersp (90), and IT (100).By splitting network connections by department, each VLAN improves network management and security.

Switchport trunk for traffic

```
AS3(config-vlan)#int f0/1
AS3(config-if)#sw
AS3(config-if)#switchport mode trunk
AS3(config-if)#int range f0/7-10
AS3(config-if-range)#channel-prop
AS3(config-if-range)#channel-pro
AS3(config-if-range)#channel-protocol pagp
AS3(config-if-range)#channel-group 1 mode auto
AS3(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
```

Figure (3.56) Changing switchport mode trunk and configuration EtherChannel on AS3

By using PAgP to group ports F0/7 to F0/10 into Port-Channel 1, this configuration builds an EtherChannel on switch AS3. For increased speed and

dependability, it merges several links into one, enabling VLAN traffic via a trunk connection.

Switchport mode access for traffic

```
AS3(config-if-range)#int f0/2
AS3(config-if)#sw
AS3(config-if)#switchport mode access
AS3(config-if)#sw
AS3(config-if)#switchport access vlan 60
AS3(config-if)#int f0/3
AS3(config-if)#sw
AS3(config-if)#switchport mode access
AS3(config-if)#switchport access vlan 70
AS3(config-if)#int f0/4
AS3(config-if)#switchport mode access
AS3(config-if)#switchport access vlan 80
AS3(config-if)#int f0/5
AS3(config-if)#switchport mode access
AS3(config-if)#switchport access vlan 90
AS3(config-if)#int f0/6
AS3(config-if)#switchport mode access
AS3(config-if)#switchport access vlan 100
AS3(config-if)#
```

Figure (3.57) Changing switchport mode access for end devices

Every switch port is assigned to a unique VLAN in this configuration. Access mode is enabled on ports F0/2 through F0/6, which are linked to VLANs 60, 70, 80, 90, and 100. As a result, any device connected to those ports will automatically join the department-specific VLAN network.

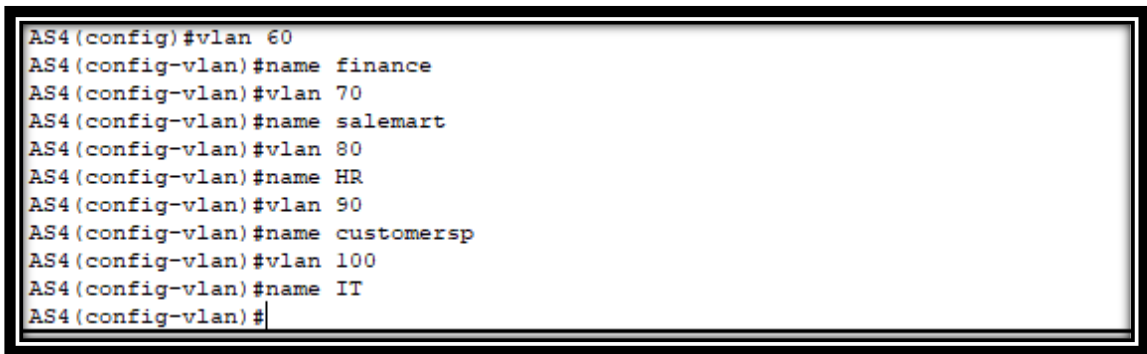
AS4 Configuration

Giving Enable and console password for security

```
Switch(config)#hostname AS4
AS4(config)#enable secret enpa55
AS4(config)#line console 0
AS4(config-line)#password conpa55
AS4(config-line)#login
AS4(config-line)#exit
AS4(config)#service pass
AS4(config)#service password-encryption
AS4(config)#
```

Figure (3.58) Changing hostname, giving enable password and console password

This configuration configures a switch called AS4 with basic security. It generates a console password (conpa55) for console login and a secure password (enpa55) for privileged access. To keep all passwords secure, the command service password-encryption conceals them in the configuration file.



```
AS4(config)#vlan 60
AS4(config-vlan)#name finance
AS4(config-vlan)#vlan 70
AS4(config-vlan)#name salemart
AS4(config-vlan)#vlan 80
AS4(config-vlan)#name HR
AS4(config-vlan)#vlan 90
AS4(config-vlan)#name customersp
AS4(config-vlan)#vlan 100
AS4(config-vlan)#name IT
AS4(config-vlan)#
```

Figure (3.59) naming each VLAN

VLANs are created and named on the switch AS4 with this configuration. Finance is VLAN 60, Salemart is VLAN 70, HR is VLAN 80, CustomersP is VLAN 90, and IT is VLAN 100.

EtherChannel for dependency

```
S4(config-if)#switchport mode trunk
S4(config-if)#int range f0/7-10
S4(config-if-range)#channel-proto
S4(config-if-range)#channel-protocol pagp
S4(config-if-range)#channel-group 1 mode des
S4(config-if-range)#channel-group 1 mode desirable
S4(config-if-range)#
reating a port-channel interface Port-channel 1

LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to down
LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up
LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to down
LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up
LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to down
LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to down
LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
LINK-5-CHANGED: Interface Port-channell, changed state to up
LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channell, changed state to up
```

Figure (3.60) Changing switchport mode trunk and configuration EtherChannel at AS4

By connecting ports F0/7–F0/10 into Port-Channel 1 using PAGP in desired mode, this configuration establishes an EtherChannel on AS4, increasing speed and dependability via link aggregation.

Changing Switchport mode access and assign VLAN

```
AS4(config-if-range)#int f0/2
AS4(config-if)#sw
AS4(config-if)#switchport mode access
AS4(config-if)#sw
AS4(config-if)#switchport access vlan 60
AS4(config-if)#int f0/3
AS4(config-if)#switchport mode access
AS4(config-if)#switchport access vlan 70
AS4(config-if)#int f0/4
AS4(config-if)#switchport mode access
AS4(config-if)#switchport access vlan 80
AS4(config-if)#int f0/5
AS4(config-if)#switchport mode access
AS4(config-if)#switchport access vlan 90
AS4(config-if)#int f0/6
AS4(config-if)#switchport mode access
AS4(config-if)#switchport access vlan 100
AS4(config-if)#
```

Figure (3.61) Changing switchport mode for end device

Ports F0/2 through F0/6 are assigned to VLANs 60, 70, 80, 90, and 100 in this configuration. To ensure that devices connected to them join the appropriate VLAN for their department, each port is configured in access mode.

Security

GRE tunnel for site-to-site connection

```
ER1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

ER1(config-if)#ip address 10.0.0.1 255.0.0.0
ER1(config-if)#tunnel source g0/0
ER1(config-if)#tunne destination se0/3/0
      ^
% Invalid input detected at '^' marker.

ER1(config-if)#tunnel destination se0/3/0
      ^
% Invalid input detected at '^' marker.

ER1(config-if)#tunnel destination ?|
  A.B.C.D ip address
ER1(config-if)#tunnel destination se0/3/0
      ^
% Invalid input detected at '^' marker.

ER1(config-if)#tunnel destination 209.165.100.2 255.255.255.0
      ^
% Invalid input detected at '^' marker.

ER1(config-if)#tunnel source g0/0 ?
  <cr>
ER1(config-if)#tunnel source g0/0
ER1(config-if)#tunnel destination 209.165.100.2
ER1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

ER1(config-if)#conf t
%Invalid hex value
ER1(config)#router rip
ER1(config-router)#network 10.0.0.0
ER1(config-router)#network 172.16.16.0
ER1(config-router)#network 172.17.17.0
ER1(config-router)#
```

Figure (3.62) GRE tunnel configuration on YGN edge router

These parameters configure router ER1 as having a tunnel interface and RIP routing. The tunnel, having source g0/0 and destination 209.165.100.2 will have an IP address of 10.0.0.1, which allows the use of internet to conduct internet communication between remote networks. To have router share routing information automatically, RIP routing is then enabled to networks 10.0.0.0, 172.16.0.0 and 172.17.17.17.

```
GW1>en
Password:
GW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
GW1(config)#router rip
GW1(config-router)#version 2
GW1(config-router)#network 172.16.16.0
GW1(config-router)#network 192.168.10.0
GW1(config-router)#network 192.168.20.0
GW1(config-router)#network 192.168.30.0
GW1(config-router)#network 192.168.40.0
GW1(config-router)#network 192.168.50.0
GW1(config-router)#
```

Figure (3.63) Rip routing in GW1

This routing will allow the RIP version 2 routing on router GW1. It promotes a few networks 172.16.16.0 and some 192.168.x.x networks (10.0, 20.0, 30.0, 40.0, and 50.0). This allows GW1 to exchange route information with other routers dynamically to enable them to communicate across these networks automatically.

```
GW2(config)#router rip
GW2(config-router)#version 2
GW2(config-router)#network 172.17.17.0
GW2(config-router)#network 192.168.10.0
GW2(config-router)#network 192.168.20.0
GW2(config-router)#network 192.168.30.0
GW2(config-router)#network 192.168.40.0
GW2(config-router)#network 192.168.50.0
GW2(config-router)#
```

Figure (3.64) Rip routing in GW2

This setup is used to configure RIP version 2 on router GW2. It adverts the networks as 172.17.17.0 and 192.168.10.0192.168.50.0 so that the router can share and learn routes with the rest of routers with RIP enabled to improve the communication in the networks.

```
Enter configuration commands, one per line. End with CNTL/Z.
ER2(config)#int funnel 0
^
% Invalid input detected at '^' marker.

ER2(config)#int t\unnel 0
^
% Invalid input detected at '^' marker.

ER2(config)#int tunnel 0

ER2(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

ER2(config-if)#ip address 10.0.0.2 255.0.0.0
ER2(config-if)#tunnel source se0/3/0
ER2(config-if)#tunnel destination 209.165.200.2
ER2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

ER2(config-if)#
```

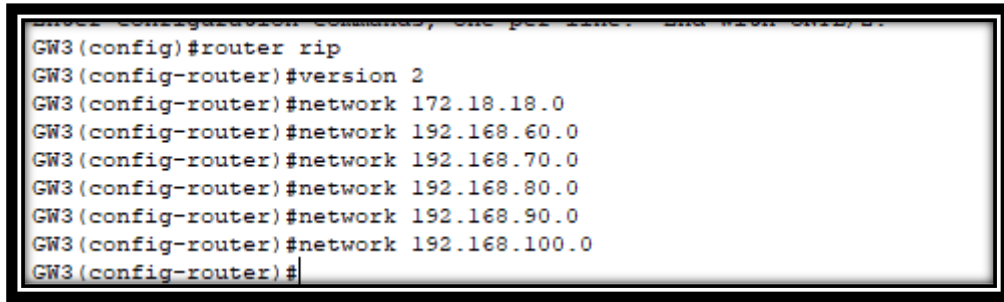
Figure (3.65) Building tunnel for GRE at ER2

This arrangement forms a tunnel interface on router ER2. The tunnel has an IP address of 10.0.0.2, the source of the tunnel being Serial0/3/0 and the destination of the tunnel is 209.165.200.2. The tunnel interface is configured and once it is configured, it goes online and secure communications are there between two remote networks over an existing internet connection.

```
R2(config-if)#exit
R2(config)#
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#network 10.0.0.0
R2(config-router)#network 172.18.18.0
R2(config-router)#network 172.19.19.0
R2(config-router)#
```

Figure (3.66) RIP routing on R2 for GRE tunnel

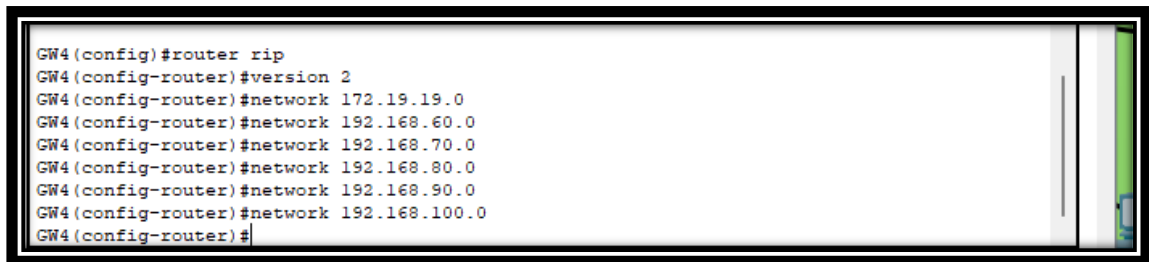
This setup allows the use of RIP version 2 in router R2. It promotes the networks 10.0.0.0, 172.18.18.0, and 172.19.19.0 to share and learn routes automatically with other routers to effectively communicate between the networks that are connected.



```
GW3(config)#router rip
GW3(config-router)#version 2
GW3(config-router)#network 172.18.18.0
GW3(config-router)#network 192.168.60.0
GW3(config-router)#network 192.168.70.0
GW3(config-router)#network 192.168.80.0
GW3(config-router)#network 192.168.90.0
GW3(config-router)#network 192.168.100.0
GW3(config-router)#
```

Figure (3.67) RIP routing in GW3

This set-up allows RIP version 2 in router GW3. It promotes the networks 172.18.18.0 and 192.168.60.0-192.168.100.0 so that routing can take place and therefore the router can exchange routing information with other routers seamlessly between VLANs and the rest of the network.



```
GW4(config)#router rip
GW4(config-router)#version 2
GW4(config-router)#network 172.19.19.0
GW4(config-router)#network 192.168.60.0
GW4(config-router)#network 192.168.70.0
GW4(config-router)#network 192.168.80.0
GW4(config-router)#network 192.168.90.0
GW4(config-router)#network 192.168.100.0
GW4(config-router)#
```

Figure (3.68) RIP routing in GW4

This set up allows version 2 of RIP on router GW4. It is an advert of the networks 172.19.19.0 and 192.168.60.0-192.168.100.0 so that GW4 exchanges routing information with other routers to enable automatic communication between VLANs and the connected networks.

Switchport Port Security

```
AS1(config-if)#switchport port-security
AS1(config-if)#sw
AS1(config-if)#switchport port
AS1(config-if)#switchport port-security ,a
AS1(config-if)#switchport port-security ma
AS1(config-if)#switchport port-security maximum 1
AS1(config-if)#switch
AS1(config-if)#switchport port
AS1(config-if)#switchport port-security mac
AS1(config-if)#switchport port-security mac sticky
AS1(config-if)#swit
AS1(config-if)#switchport po
AS1(config-if)#switchport port-security vio
AS1(config-if)#switchport port-security violation sh
AS1(config-if)#switchport port-security violation shutdown
AS1(config-if)#
```

Figure (3.69) Switchport security configuration

In this setup, port security is possible in switch AS1 with a single device per port. The switch automatically acquires the MAC address of the device with the use of the sticky option and when another device attempts to connect, the port will automatically close down so that it cannot be used by an unauthorized device.

IPsec configuration for VPN

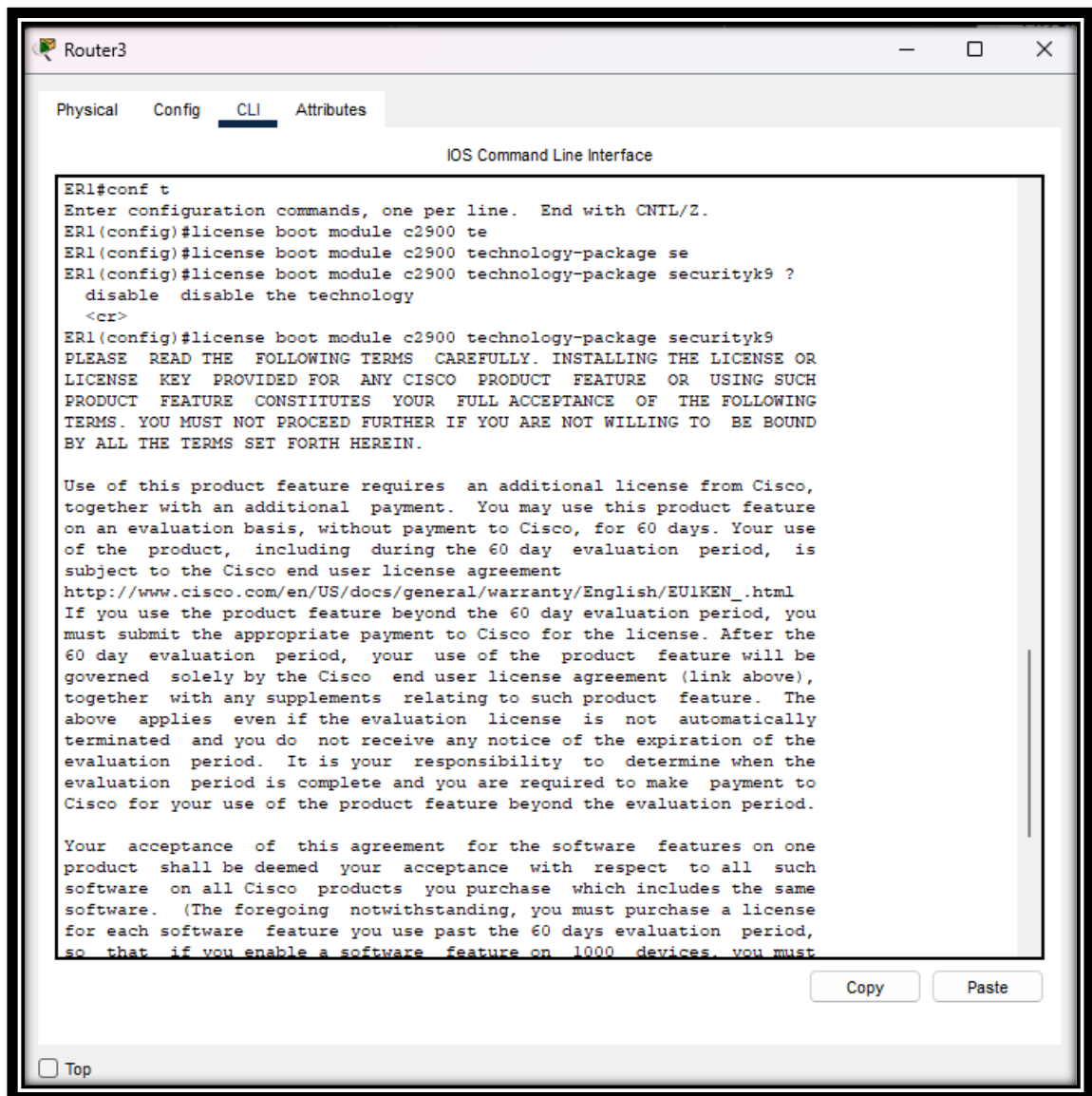
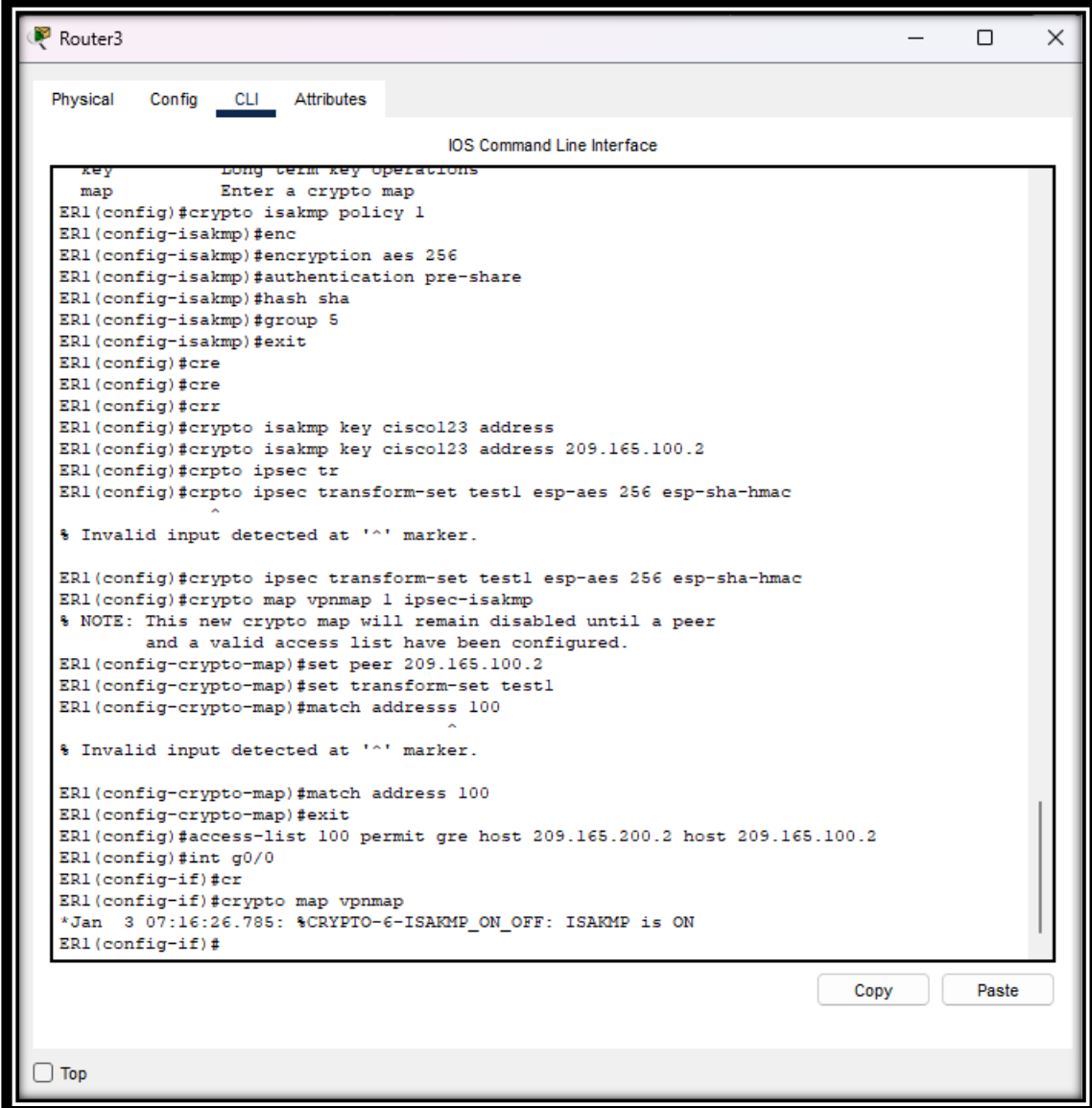


Figure (3.70) IPSEC configuration on Yangon edge router

This setup is allowing the security technology package license on the cisco 2900 series router, Router3. It is enabled with the latest security provisioning like VPN, firewall, and intrusion prevention on a 60-day trial basis.



The screenshot shows a web-based interface for Router3 with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the IOS Command Line Interface. The configuration commands entered are as follows:

```
key          Long term key operations
map          Enter a crypto map
ER1(config)#crypto isakmp policy 1
ER1(config-isakmp)#enc
ER1(config-isakmp)#encryption aes 256
ER1(config-isakmp)#authentication pre-share
ER1(config-isakmp)#hash sha
ER1(config-isakmp)#group 5
ER1(config-isakmp)#exit
ER1(config)#cre
ER1(config)#cre
ER1(config)#crr
ER1(config)#crypto isakmp key cisco123 address
ER1(config)#crypto isakmp key cisco123 address 209.165.100.2
ER1(config)#crpto ipsec tr
ER1(config)#crpto ipsec transform-set test1 esp-aes 256 esp-sha-hmac
^
% Invalid input detected at '^' marker.

ER1(config)#crypto ipsec transform-set test1 esp-aes 256 esp-sha-hmac
ER1(config)#crypto map vpnmap 1 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
      and a valid access list have been configured.
ER1(config-crypto-map)#set peer 209.165.100.2
ER1(config-crypto-map)#set transform-set test1
ER1(config-crypto-map)#match address 100
^
% Invalid input detected at '^' marker.

ER1(config-crypto-map)#match address 100
ER1(config-crypto-map)#exit
ER1(config)#access-list 100 permit gre host 209.165.200.2 host 209.165.100.2
ER1(config)#int g0/0
ER1(config-if)#cr
ER1(config-if)#crypto map vpnmap
*Jan  3 07:16:26.785: %CRYPTO-6-ISA_KMP_ON_OFF: ISAKMP is ON
ER1(config-if)#
```

At the bottom of the CLI window, there are 'Copy' and 'Paste' buttons, and a 'Top' link.

Figure (3.71) IPSEC configuration on Yangon side

This design establishes an IPsec VPN tunnel on Router3 to ensure the safety of data transit between two networks. It relies on the ISAKMP as the key exchange and the AES-256 encryption and the SHA as the authentication. The remote peer is identified as cisco123 and the pre-shared key is set to 209.165.100.2 and an access list will be set to define what traffic to be secured. These settings are connected and the connection to the VPN is enabled by the crypto map called vpnmap which is applied to interface G0/0.

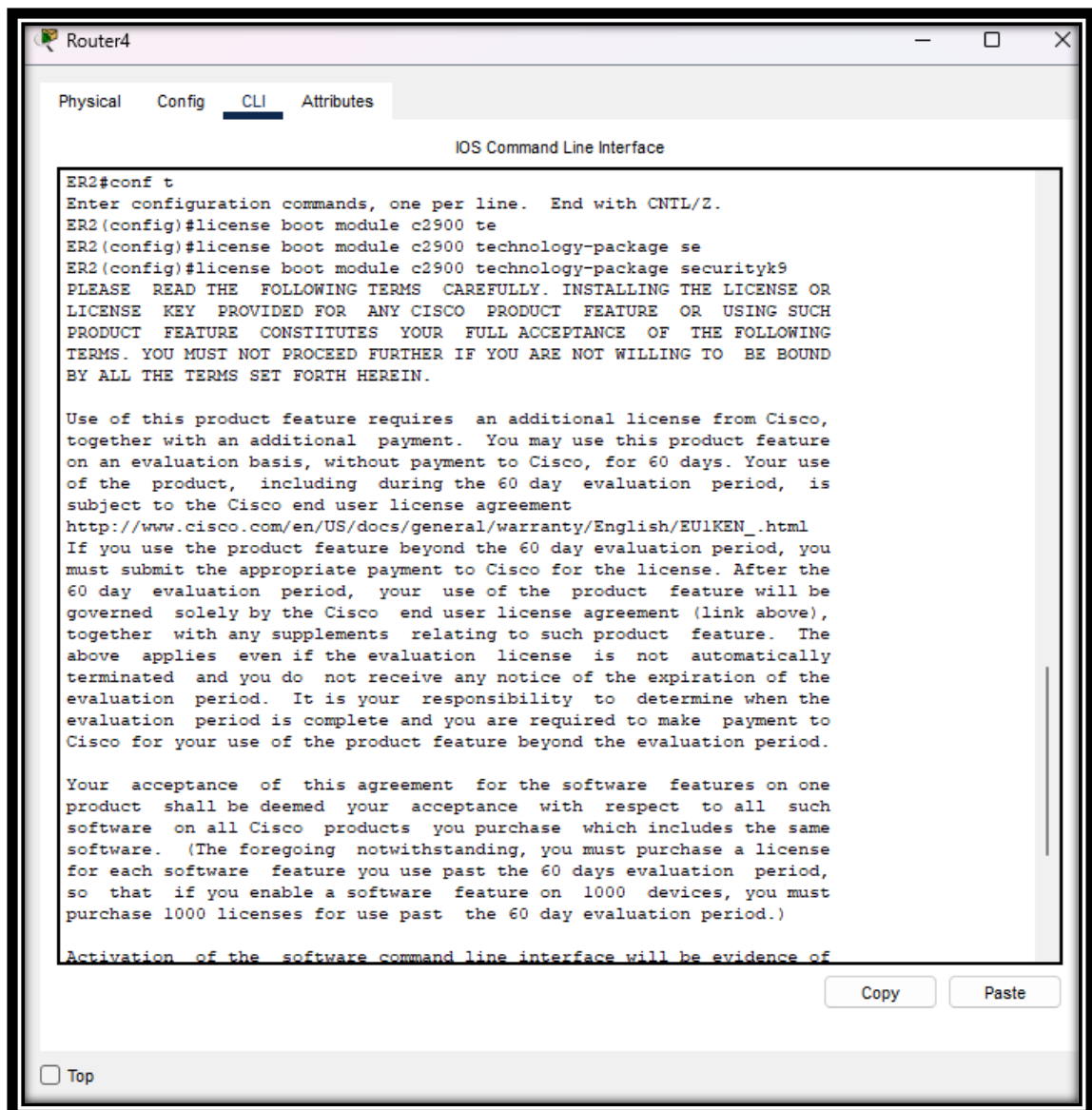


Figure (3.72) IPSEC configuration on Mandalay edge router

This router4 configuration will allow the Cisco security technology package license of the 2900 series router. This license can be activated to enable the use of advanced security features like VPN, firewall and encryption tools. It operates with a 60 days trial period, and then a licensed license has to be purchased with Cisco to utilize these features.

```
ER2(config)#crypto isakmp policy 1
ER2(config-isakmp)#encryption aes 256
ER2(config-isakmp)#auth
ER2(config-isakmp)#authentication pre-share
ER2(config-isakmp)#hash sha
ER2(config-isakmp)#group 5
ER2(config-isakmp)#exit
ER2(config)#cr
ER2(config)#crypto isakmp key cisco123 address 209.165.100.2
ER2(config)#no crypto isakmp key cisco123 address 209.165.100.2
ER2(config)#crypto isakmp key cisco123 address 209.165.200.2
ER2(config)#crypto ipsec transform-set test1 esp-aes 256 esp-sha-hmac
ER2(config)#crypto map vpn
% Incomplete command.
ER2(config)#crypto map vpnmap 1 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
ER2(config-crypto-map)#
ER2(config-crypto-map)#set peer 209.165.200.2
ER2(config-crypto-map)#set transform-set test1
ER2(config-crypto-map)#match address 100
ER2(config-crypto-map)#exit
ER2(config)#access-list 100 permit gre host 209.165.100.2 host 209.165.200.2
ER2(config)#int se0/3/0
ER2(config-if)#crypto map vpnmap
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
ER2(config-if)#
ER2(config-if)#
ER2(config-if)#
```

Figure (3.73) IPSEC configuration on Mandalay Edge router

The arrangement establishes an IPsec VPN tunnel on Router4 (ER2). It specifies an ISAKMP policy as AES-256 encryption, SHA hashing, pre-shared authentication, and Diffie-Hellman group 5 of key exchange. The peer (209.165.100.2) is pre-shared with a key known as cisco123 and a transform set called test1 is configured with the encryption and hashing algorithms used in the tunnel. All the settings are connected to the crypto map (vpnmap), which is applied to the s0/3/0 serial interface and creates an encrypted secure connection between the two routers.

NAC configuration

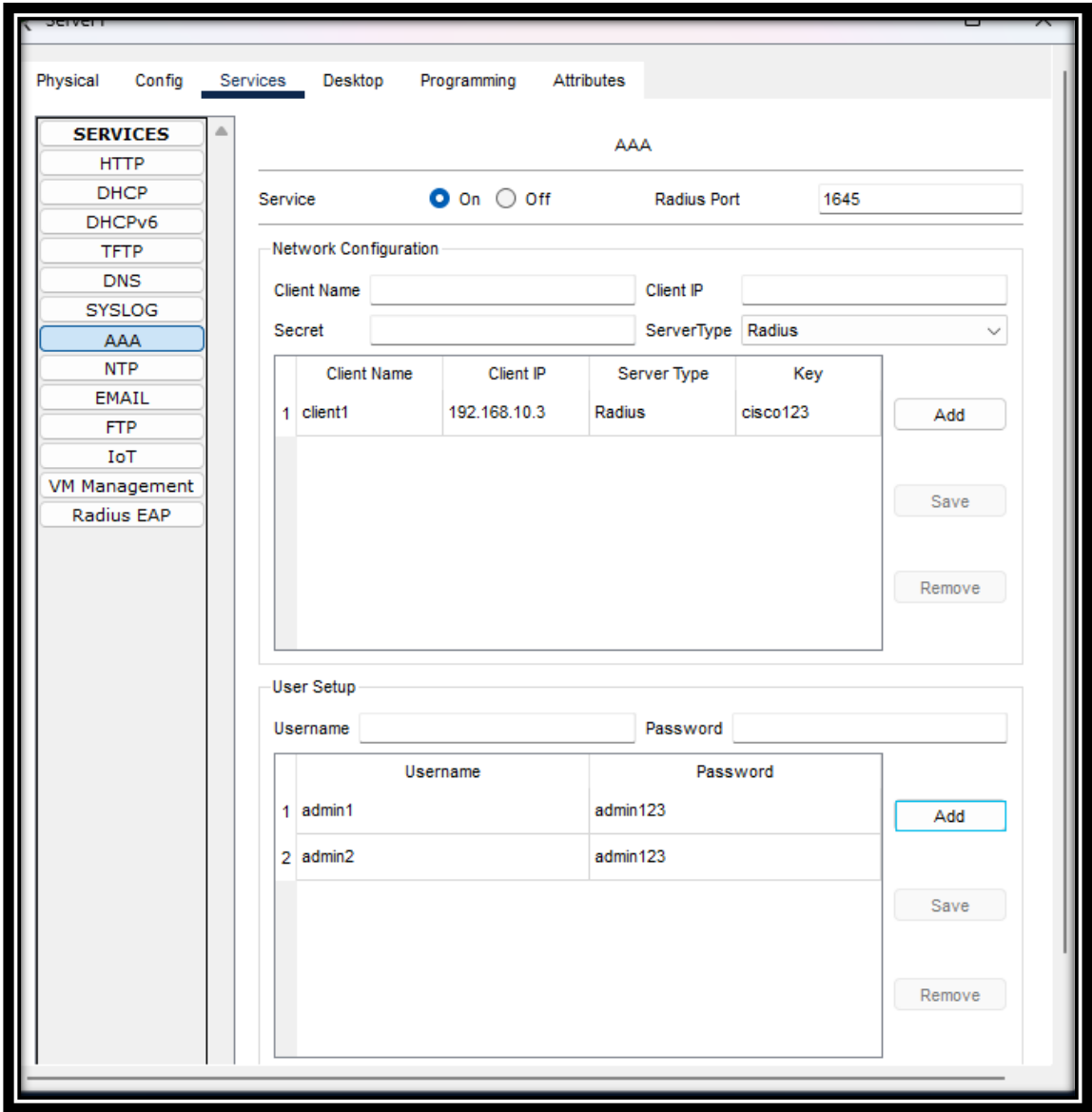


Figure (3.74) AAA service configuration at server 1

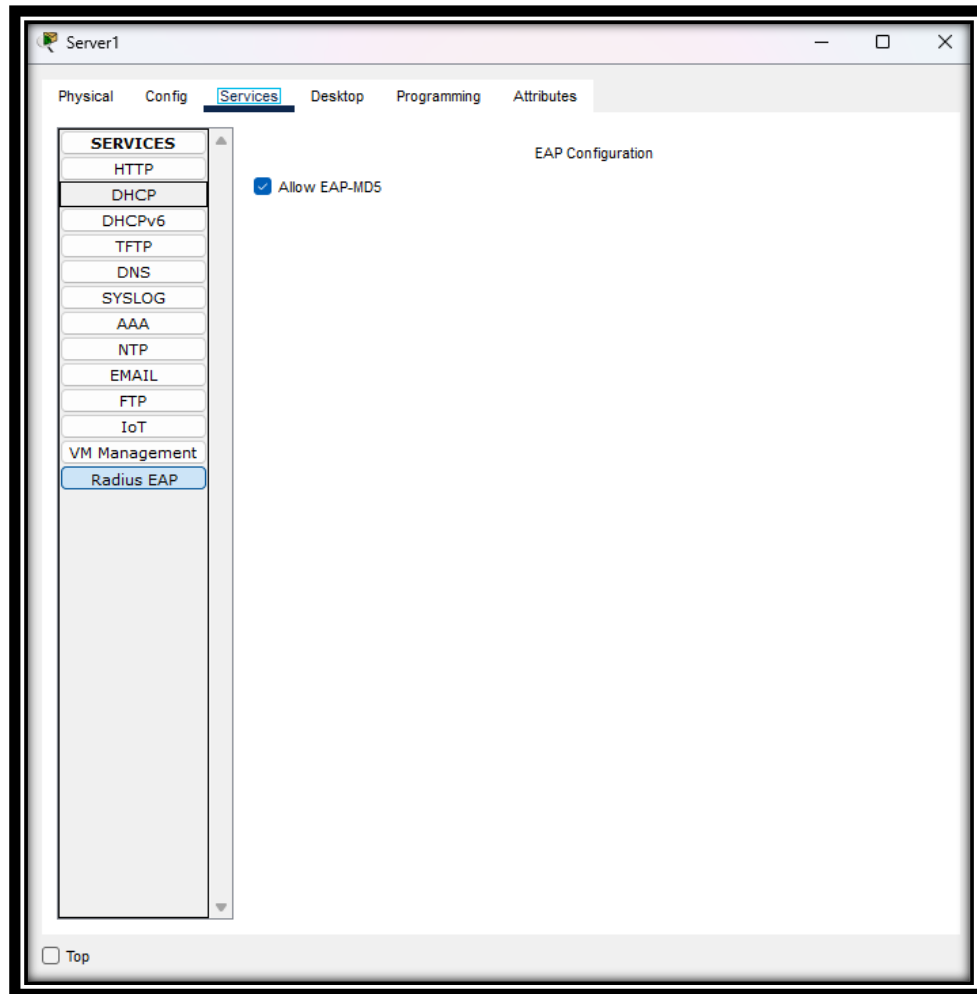


Figure (3.78) Allow EAP-MD5 on server 1

A RADIUS server on a Packet Tracer server running Cisco. The AAA service is activated and it authenticates on port 1645. A client1 is a RADIUS client whose IP address is 192.168.10.3, and share key is cisco123. Two users are also added with the password of admin123 namely admin1 and admin2.

```
AS1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
AS1(config)#aaa new
AS1(config)#aaa new-model
AS1(config)#aaa auth
AS1(config)#aaa authentication dot1x default group
% Incomplete command.
AS1(config)#aaa authentication dot1x default group ?
radius Use list of all Radius hosts.
AS1(config)#aaa authentication dot1x default group radius
AS1(config)#dot1x system-auth
AS1(config)#dot1x system-auth-control
AS1(config)#int range f0/2-6
AS1(config-if-range)#authen
AS1(config-if-range)#authentication port-control auto
AS1(config-if-range)#dot1x pae authenticator
AS1(config-if-range)#exit
AS1(config)#radius-server host 192.168.10.253
AS1(config)#radius-server host 192.168.10.253 ?
auth-port  UDP port for RADIUS authentication server (default is 1645)
key         per-server encryption key (overrides default)
<cr>
AS1(config)#radius-server host 192.168.10.253 auth
AS1(config)#radius-server host 192.168.10.253 auth-port 1645 key cisco123
```

Figure (3.75) AAA service configuration at AS1

This setup allows AAA and RADIUS authentication of switch AS1 by 802.1X. It enables the AAA model, enables port-based authentication on interfaces F0/2-F0/6 and links the switch to a RADIUS server at 192.168.10.253, port 1645, and using the key cisco123. This keeps out unauthorized users out of the network using such ports.

Wireless Security

The screenshot displays the configuration interface of a Wireless-N Broadband Router. The top navigation bar includes tabs for 'Physical', 'Config', 'GUI', and 'Attributes'. The 'Config' tab is active, showing a sub-menu with 'Setup', 'Wireless', 'Security', 'Access Restrictions', 'Applications & Gaming', and 'Administration'. The 'Setup' tab is selected, and the 'Internet Setup' section is visible. The 'Internet Connection type' is set to 'Static IP'. The 'Internet IP Address' is configured as 192.168.10.45, the 'Subnet Mask' is 255.255.255.0, and the 'Default Gateway' is 192.168.10.254. DNS settings are also visible, with DNS 1 set to 0.0.0.0 and DNS 2 and 3 set to 0.0.0.0. The 'Host Name' and 'Domain Name' fields are empty. The 'MTU' is set to 1500. The 'Network Setup' section at the bottom shows the 'IP Address' as 192.168.0.1.

Field	Value
Internet Connection type	Static IP
Internet IP Address	192.168.10.45
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.254
DNS 1	0.0.0.0
DNS 2 (Optional)	0.0.0.0
DNS 3 (Optional)	0.0.0.0
Host Name	
Domain Name	
MTU	1500
Network Setup IP Address	192.168.0.1

Figure (3.76) Wireless security

The following picture depicts the setup screen of a Wireless-N Broadband Router set to use a fix IP. The Internet IP address of the router is 192.168.10.45, the subnet mask is 255. 255.255. 0, and the default gateway is 192.168.10.254. There is no configuration of DNS entries. This configuration enables the router to join the network by using constant IP configurations and it is normally applied in stable connections and the easier management in secure or enterprise systems.

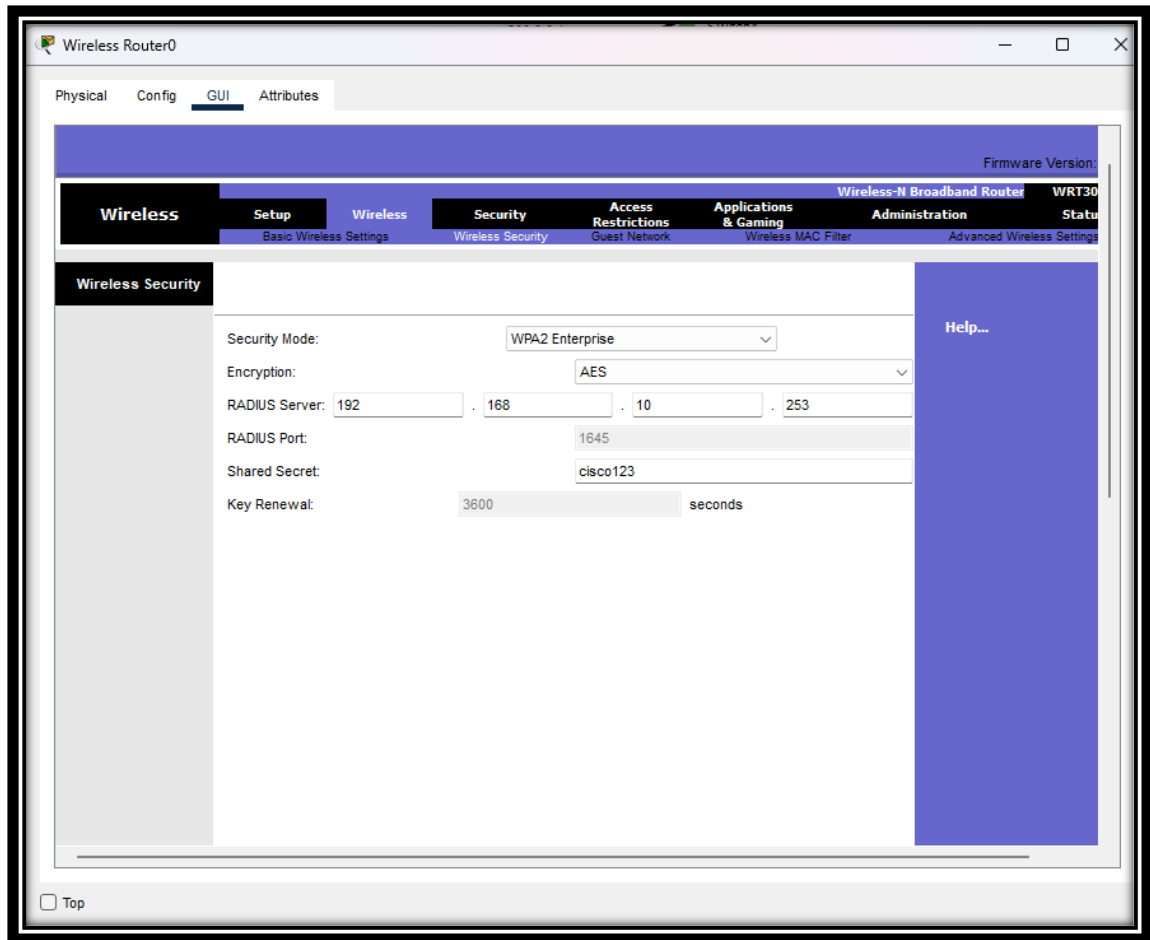


Figure (3.77) wireless security WPA2 enterprise

The router is encrypted with AES and the users are authenticated by a RADIUS server at the IP 192.168.10.253 at the port 1645 with the shared secret cisco123. This configuration offers great network security as the users would be required to be authenticated by the RADIUS server before being allowed access into the network.

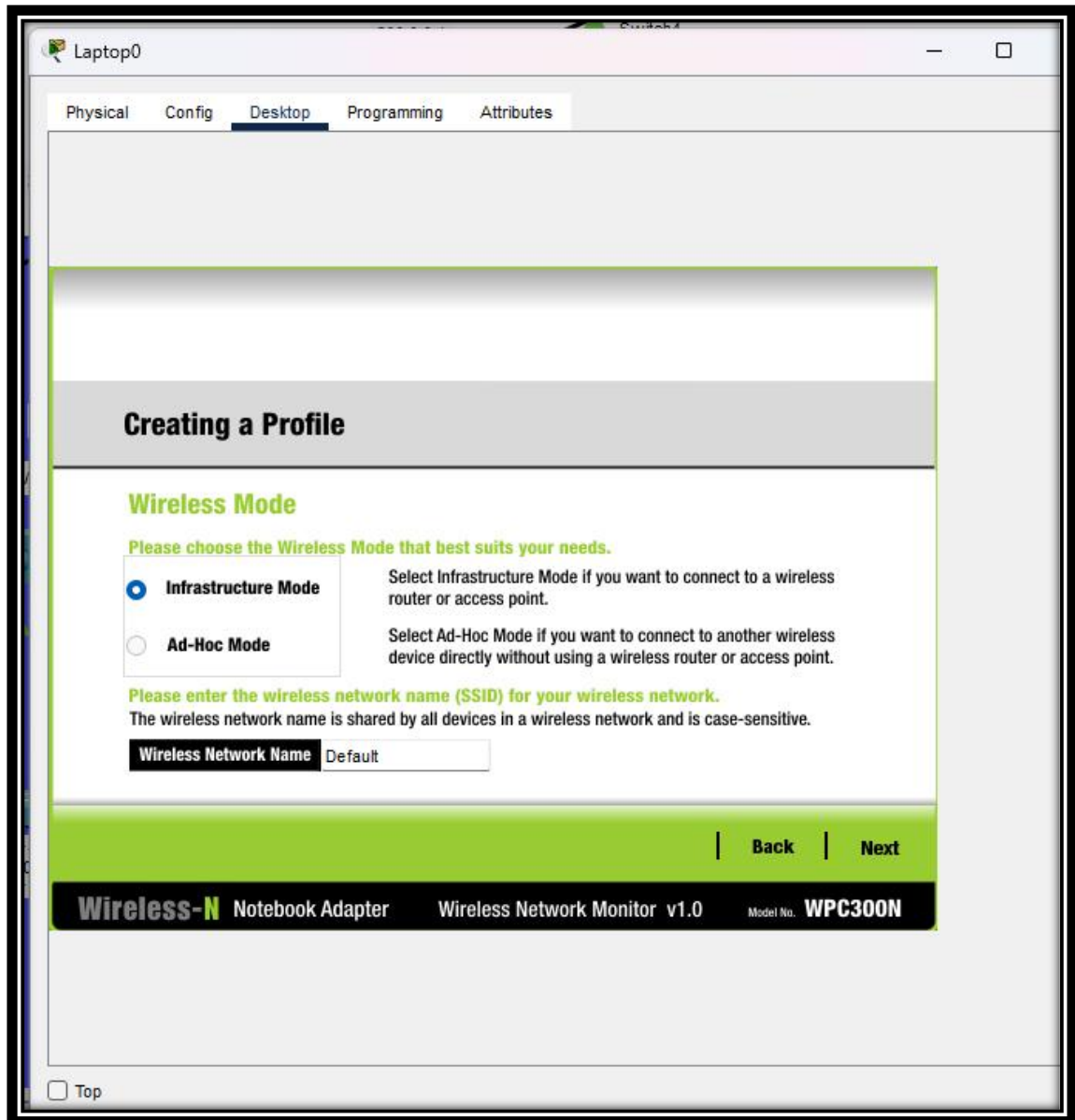


Figure (3.78) Creating a profile

This display indicates the wireless adapter of the laptop with the help of the WPC300N module. It is also configured to Infrastructure Mode, that is, the laptop will connect to a wireless router or access point rather than to another device. Wireless network name (SSID) will be configured to the default, which will identify the Wi-Fi network that the laptop will be joined.

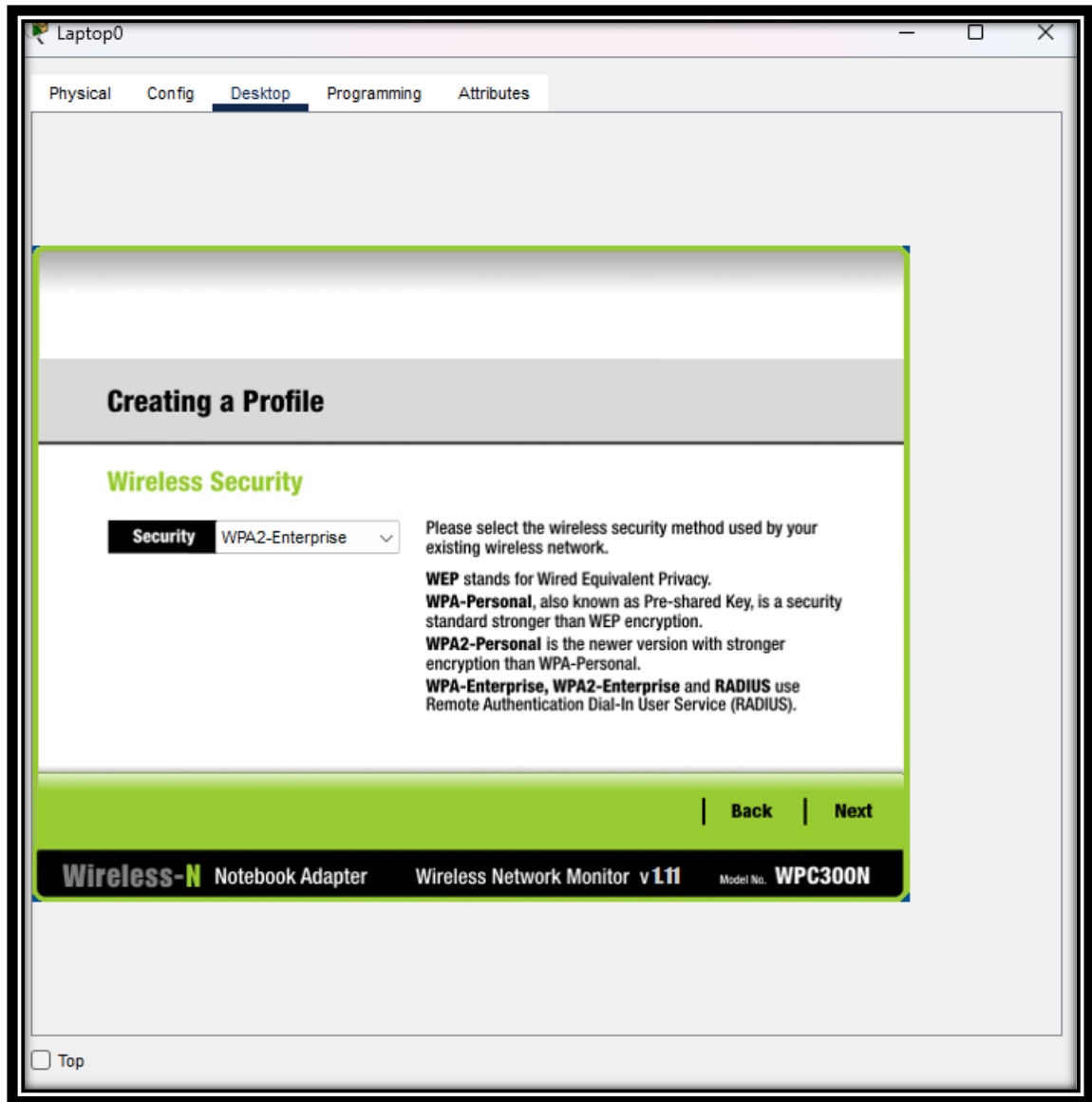


Figure (3.79) Creating a profile step 2

The following screen displays the wireless security configuration of the WPC300N adapter of the laptop. The chosen type of security is the WPA2-Enterprise option and this is a powerful encryption algorithm of AES and it authenticates the user on a RADIUS server. This will guarantee a secure and centralized authentication of the wireless network connectivity.



Figure (3.80) Creating a profile step 3

The screen displays the WPA2-Enterprise wireless authentication configuration of the WPC300N adapter on the laptop. The authentication type is PEAP (Protected EAP) which means that the login name is wireless and password is typed in so as to access it securely. The certificate environment is Trust Any and the inner authentication environment is TOKEN CARD. Such a setup would make sure that the laptop is able to connect safely to the wireless network using a RADIUS-based authentication.

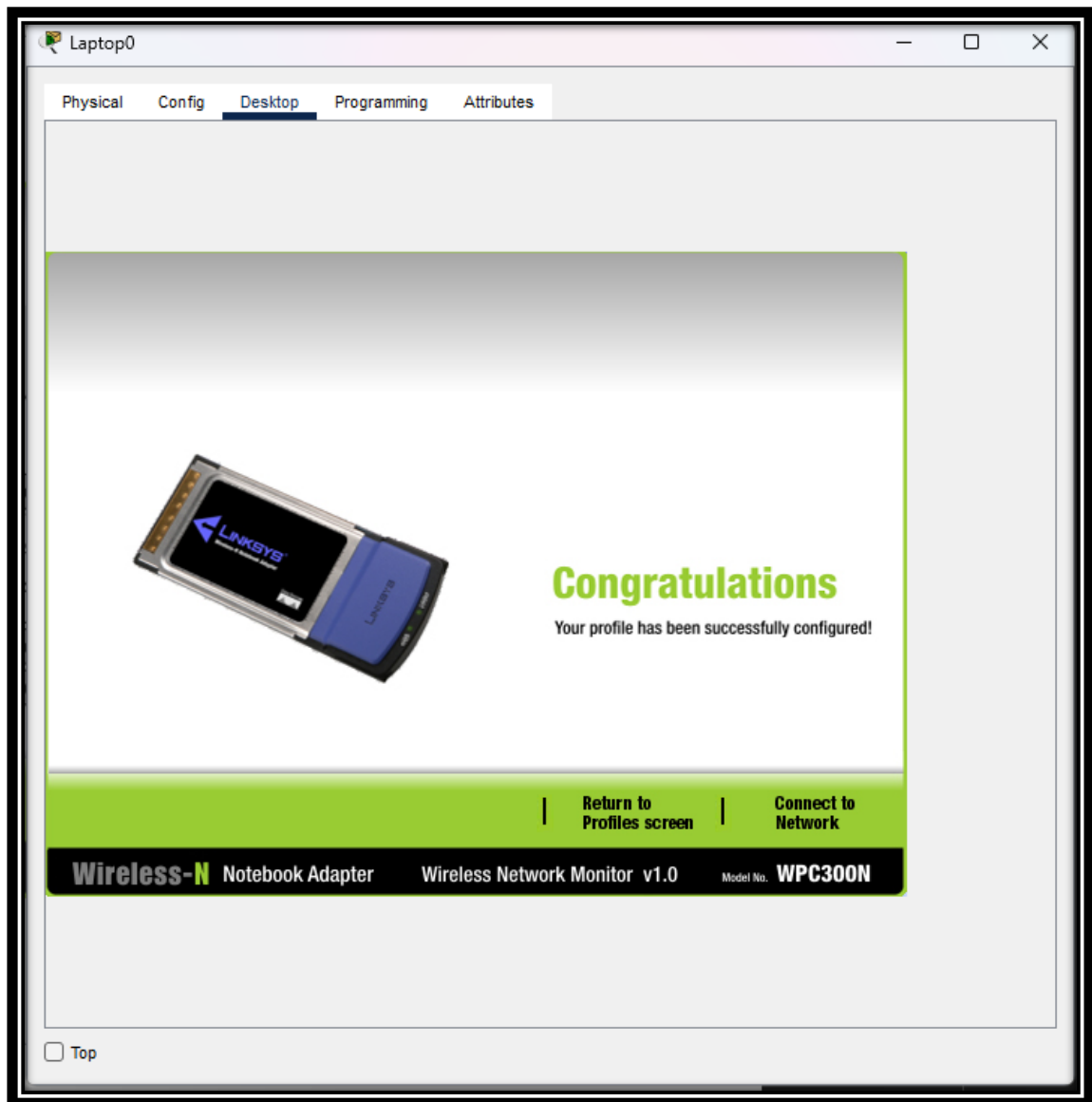


Figure (3.81) Creating a profile step 4

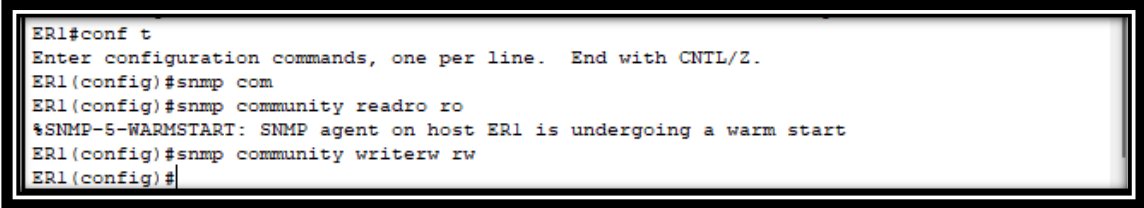
This screen indicates that the wireless profile with the WPC300N adapter of the laptop is configured. The message reveals that the wireless security and authentication environment is properly configured and that the laptop is now ready to be connected to the network on a safe basis.



Figure (3.82) Successfully Connected

This display indicates that the WPC300N wireless adapter of the laptop has been connected to the access point in Infrastructure Mode. The signal strength indicators and the quality of links obtained are good, which proves the existence of a stable and active connection on the 2.4GHz wireless network.

SNMP configuration



```
ER1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
ER1(config)#snmp com
ER1(config)#snmp community readro ro
%SNMP-S-WARMSTART: SNMP agent on host ER1 is undergoing a warm start
ER1(config)#snmp community writerw rw
ER1(config)#
```

Figure (3.83) SNMP configuration on ER1

This setup allows SNMP (Simple Network Management Protocol) on router ER1. Two SNMP community strings are formed- readro with read-only (ro) access allowing it to be monitored and writerw with read-write (rw) access allowing configuration changes. The message shows that SNMP agent has begun with success,

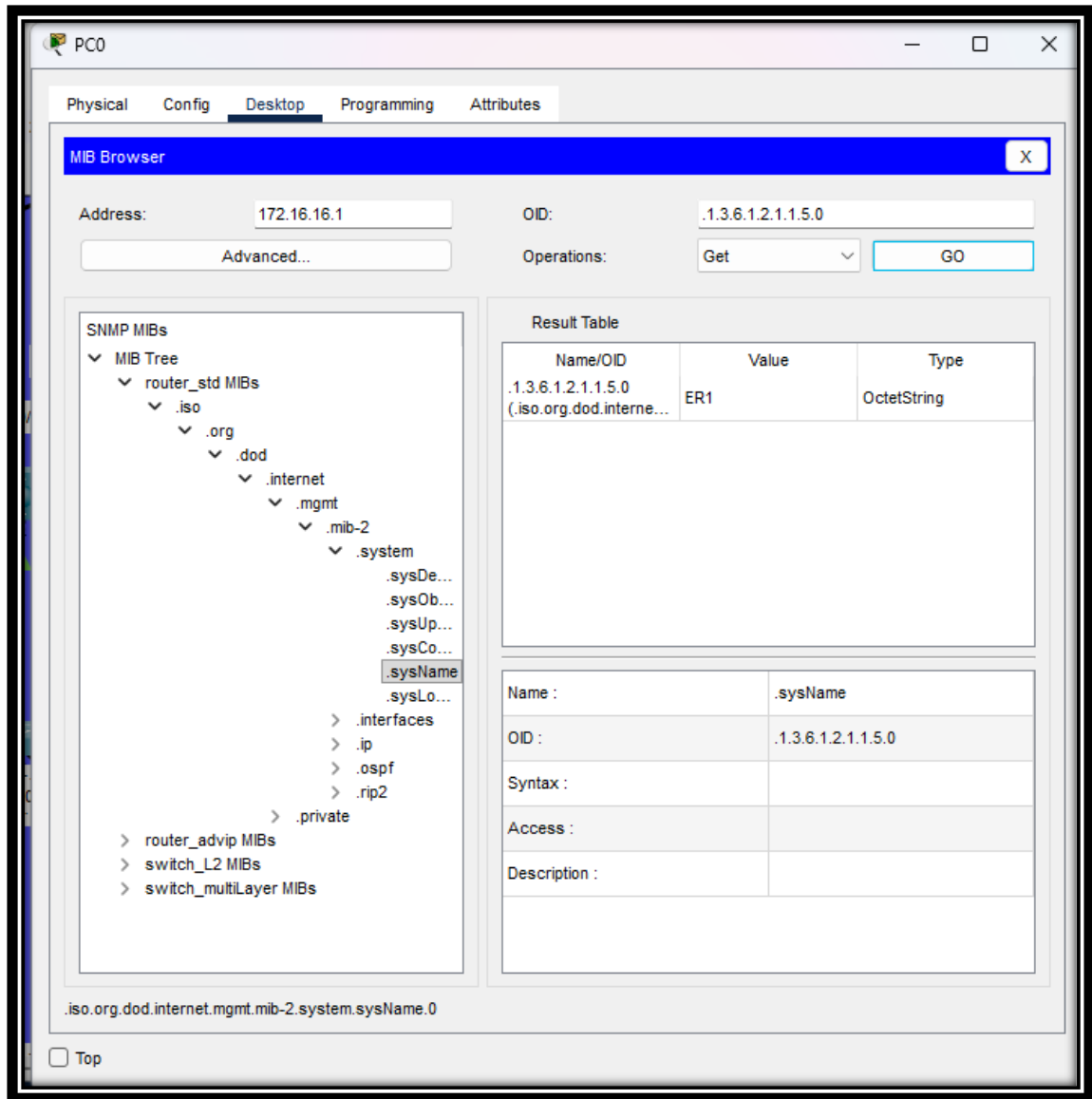


Figure (3.84) SNMP testing at end device

As it can be seen in this screenshot, the SNMP MIB Browser on PC0 was able to retrieve information on the router ER1. The target IP address will be 172.16.16.1 and the OID of the SNMP MIB is .1.3.6.1.2.1.1.5.0 which is the sysName object. The value of the result is the value ER1 which indicates that the system name of the router has been read successfully using SNMP.

SYSLOG configuration for alerts and notifications

```
ER1(config)#logging host 192.168.10.253
ER1(config)#service timest
ER1(config)#service timestamps log
ER1(config)#service timestamps log dat
ER1(config)#service timestamps log datetime m
ER1(config)#service timestamps log datetime msec
ER1(config)#exit
ER1#
*Mar 01, 01:12:46.1212: SYS-5-CONFIG_I: Configured from console by console
*Mar 01, 01:12:46.1212: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.10.253
port S14 started - CLI initiated
ER1#show clock
*1:12:51.528 UTC Mon Mar 1 1993
ER1#clock set 9:18:00 5 NOV 2025
ER1#show clock
9:18:1.570 UTC Wed Nov 5 2025
ER1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
ER1(config)#hostname YGN_ER'
YGN_ER'(config)#
YGN_ER'(config)#int loopback 0

YGN_ER'(config-if)#
*Nov 05, 09:20:26.2020: %LINK-5-CHANGED: Interface Loopback0, changed state to up
*Nov 05, 09:20:26.2020: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
changed state to up
YGN_ER'(config-if)#no sh
YGN_ER'(config-if)#
YGN_ER'(config-if)#sh
YGN_ER'(config-if)#shutdown

YGN_ER'(config-if)#
*Nov 05, 09:20:31.2020: %LINK-5-CHANGED: Interface Loopback0, changed state to
administratively down
*Nov 05, 09:20:31.2020: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
changed state to down
YGN_ER'(config-if)#
```

Figure (3.85) SYSLOG configuration on Yangon Edge router

In this arrangement, it is possible to enable logging and loopback interface on router YGN_ER (previously ER1). The router logs messages to Syslog server at 192.168.10.253 and the messages contain timestamps in the form of date, time, and milliseconds so as to track events accurately. It is manually adjusted to November 5, 2025. Then a loop back interface (Loopback0) is created and disabled by no shutdown and enabled by shutdown.

IPS configuration for Security Purpose

```

YGN_ER'#conf t
Enter configuration commands, one per line. End with CNTL/Z.
YGN_ER'(config)#ip ips config location flash:ipsdir
YGN_ER'(config)#ip ips name iosips
YGN_ER'(config)#ip ips sig
YGN_ER'(config)#ip ips signature-category
YGN_ER'(config-ips-category)#category all
YGN_ER'(config-ips-category-action)#retired true
YGN_ER'(config-ips-category-action)#exit
^
% Invalid input detected at '^' marker.

YGN_ER'(config-ips-category-action)#exit
YGN_ER'(config-ips-category)#category ios_ips basic
YGN_ER'(config-ips-category-action)#retired false
YGN_ER'(config-ips-category-action)#exit
YGN_ER'(config-ips-category)#exit
Do you want to accept these changes? [confirm]
Applying Category configuration to signatures ...
%IPS-6-ENGINE_BUILDING: atomic-ip - 288 signatures - 6 of 13 engines
%IPS-6-ENGINE_READY: atomic-ip - build time 30 ms - packets for this engine will be scanned

YGN_ER'(config)#int range g0/1-2
YGN_ER'(config-if-range)#ip ips iosips in
YGN_ER'(config-if-range)#
*Nov 05, 09:29:33.2929: %IPS-6-ENGINE_BUILDS_STARTED: 09:29:33 UTC Nov 05 2025
*Nov 05, 09:29:33.2929: %IPS-6-ENGINE_BUILDING: atomic-ip - 3 signatures - 1 of 13 engines
*Nov 05, 09:29:33.2929: %IPS-6-ENGINE_READY: atomic-ip - build time 8 ms - packets for this engine will be scanned
*Nov 05, 09:29:33.2929: %IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 8 ms

YGN_ER'(config-if-range)#ip ips signat
YGN_ER'(config-if-range)#ip ips signature-define
YGN_ER'(config-if-range)#ip ips signature-definition
YGN_ER'(config-sigdef)#signature 2004 0
YGN_ER'(config-sigdef-sig)#status
YGN_ER'(config-sigdef-sig-status)#enabled true
YGN_ER'(config-sigdef-sig-status)#re
YGN_ER'(config-sigdef-sig-status)#retired false
YGN_ER'(config-sigdef-sig-status)#exit
YGN_ER'(config-sigdef-sig)#engine
YGN_ER'(config-sigdef-sig-engine)#event-action pro
YGN_ER'(config-sigdef-sig-engine)#event-action produce-alert
YGN_ER'(config-sigdef-sig-engine)#eve
YGN_ER'(config-sigdef-sig-engine)#event-action deny
YGN_ER'(config-sigdef-sig-engine)#event-action deny-packet-inline
YGN_ER'(config-sigdef-sig-engine)#exit
YGN_ER'(config-sigdef-sig)#exit
YGN_ER'(config-sigdef)#exit
Do you want to accept these changes? [confirm]
%IPS-6-ENGINE_BUILDS_STARTED:
%IPS-6-ENGINE_BUILDING: atomic-ip - 303 signatures - 3 of 13 engines
%IPS-6-ENGINE_READY: atomic-ip - build time 480 ms - packets for this engine will be scanned
%IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 648 ms

```

Figure (3.86) IPS configuration on Yangon edge router

The following setup indicates that Intrusion Prevention System (IPS) is established on router YGN ER. The IPS settings are saved in the flash drive, and signature categories are generated and set to trace and block possible threats. The router interface G0/1 2 is set up to IPS inspection, and signature definition is set to identify suspicious packets. The system ascertains that the IPS engine is constructed and is operational to scan network traffic to identify attacks.

```

ER2#mkdir flash:ipsdir
Create directory filename [ipsdir]?
Created dir flash:ipsdir

ER2#conf t

```

Figure (3.87) IPS configuration on MDY edge router

In order to keep Intrusion Prevention System (IPS) files, including logs and signatures, which are crucial for identifying and stopping network attacks, a folder is created in the router's flash memory using the command `mkdir flash:ipsdir`.

```

Enter configuration commands, one per line. End with CNTL/Z.
ER2(config)#ip ips config location flash:ipsdir
ER2(config)#ip ips name iosips
ER2(config)#ip ips signat
ER2(config)#ip ips signature-ca
ER2(config)#ip ips signature-category
ER2(config-ips-category)#cat
ER2(config-ips-category)#category all
ER2(config-ips-category-action)#re
ER2(config-ips-category-action)#retired tr
ER2(config-ips-category-action)#retired true
ER2(config-ips-category-action)#exit
ER2(config-ips-category)#cat
ER2(config-ips-category)#category ios_ips basic
ER2(config-ips-category)#category ios_ips basic
% Invalid input detected at '^' marker.

ER2(config-ips-category)#category ios_ips basic
ER2(config-ips-category-action)#re
ER2(config-ips-category-action)#retired false
ER2(config-ips-category-action)#exit
ER2(config-ips-category)#exit
Do you want to accept these changes? [confirm]
Applying Category configuration to signatures ...
*IPS-6-ENGINE_BUILDING: atomic-ip - 288 signatures - 6 of 13 engines
*IPS-6-ENGINE_READY: atomic-ip - build time 30 ms - packets for this engine will be scanned

ER2(config)#int se0/3/0
ER2(config-if)#ip ips iosips in
ER2(config-if)#
*IPS-6-ENGINE_BUILDS_STARTED: 00:08:35 UTC Mar 01 1993

*IPS-6-ENGINE_BUILDING: atomic-ip - 3 signatures - 1 of 13 engines

*IPS-6-ENGINE_READY: atomic-ip - build time 8 ms - packets for this engine will be scanned

*IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 8 ms

ER2(config-if)#ip ips signat
ER2(config-if)#ip ips signature-def
ER2(config-if)#ip ips signature-definition
ER2(config-sigdef)#sig
ER2(config-sigdef)#signature 2004 0
ER2(config-sigdef-sig)#status
ER2(config-sigdef-sig-status)#enabled true
ER2(config-sigdef-sig-status)#re
ER2(config-sigdef-sig-status)#retired false
ER2(config-sigdef-sig-status)#exit
ER2(config-sigdef-sig)#end
ER2(config-sigdef-sig)#eng
ER2(config-sigdef-sig)#engine
ER2(config-sigdef-sig-engine)#even
ER2(config-sigdef-sig-engine)#event-action pro
ER2(config-sigdef-sig-engine)#ev
ER2(config-sigdef-sig-engine)#event-action pro
ER2(config-sigdef-sig-engine)#event-action produce-alert
ER2(config-sigdef-sig-engine)#even
ER2(config-sigdef-sig-engine)#event-action de
ER2(config-sigdef-sig-engine)#event-action deny-packet-inline
ER2(config-sigdef-sig-engine)#exit
ER2(config-sigdef-sig)#exit
ER2(config-sigdef)#exit
Do you want to accept these changes? [confirm]

```

Figure (3.88) MDY IPS configuration on MDY router

Setting up the IPS file storage location (flash:ipsdir) and establishing the signature categories that recognize various network threat types are the first steps in the configuration process. In order to control how the device reacts to threats that are detected, the router loads these IPS signatures and applies security actions like retired, retire, and reset. Lastly, the IPS is installed on an interface (s0/3/0), allowing for ongoing traffic protection and monitoring. This configuration improves network

security by assisting the MDY router in actively identifying, recording, and blocking questionable network packets.

Backup configuration (TFTP)



```
ER1#copy run
ER1#copy running-config st
ER1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
ER1#copy run
ER1#copy running-config tftp:
Address or name of remote host []? 192.168.10.253
Destination filename [ER1-config]?

Writing running-config...!!
[OK - 1854 bytes]

1854 bytes copied in 0 secs
ER1#
```

Figure (3.89) TFTP configuration on ER1

Based on this screenshot, it can be observed that the running configuration of router ER1 was saved successfully at both a local and a TFTP server (192.168.10.253). The copy running-config tftp: command transferred the configuration file, which is named ER1-config, which ensured that a backup of 1854-byte was successful in 0 seconds.

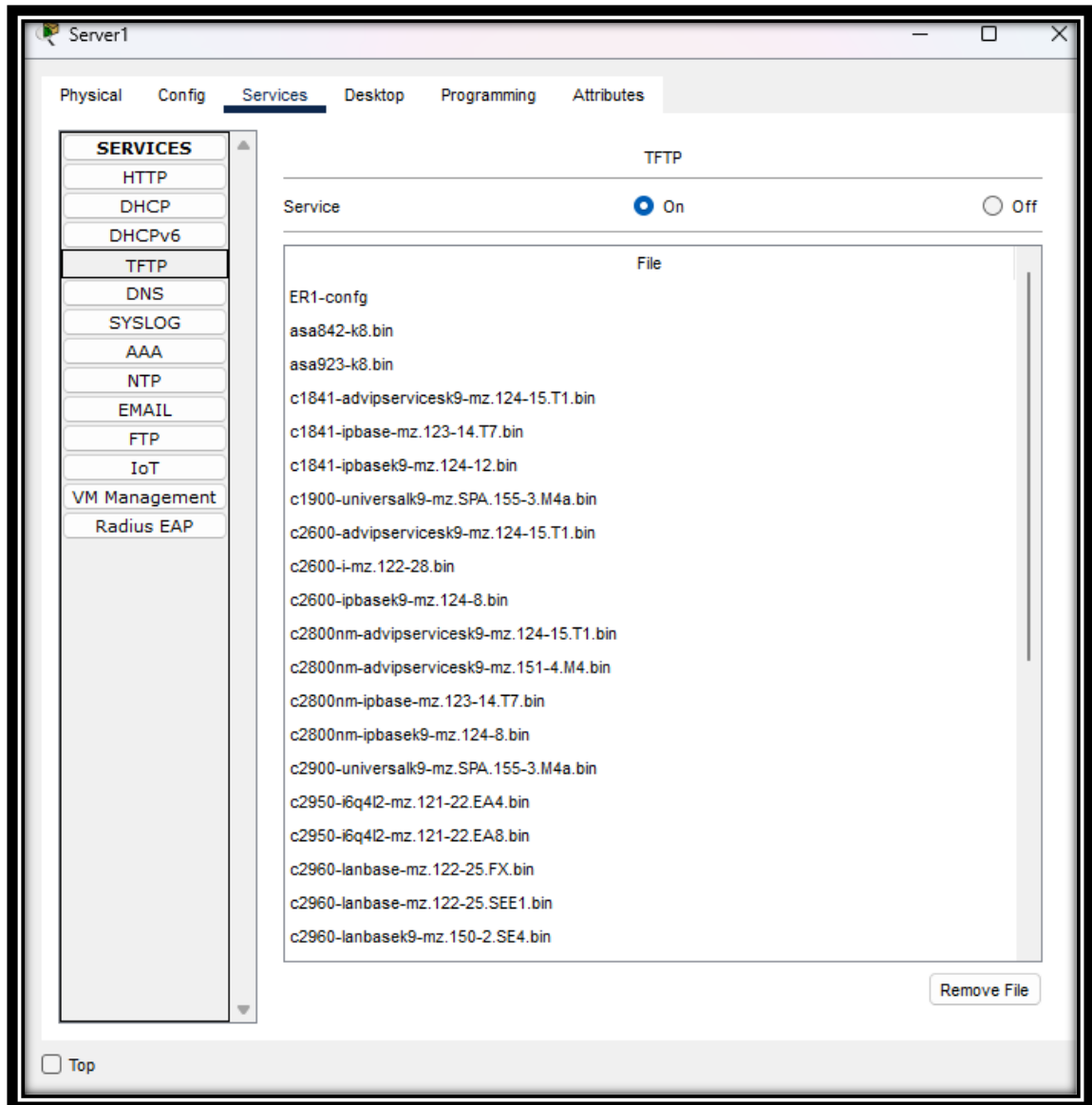


Figure (3.90) TFTP result on server 1

The screenshot depicts the TFTP service active on Server1, which is an indication that the service is active and that it is storing different configuration and IOS image files. The list contains the files of the router and switch systems, as well as the saved file of ER1-config recently.

M3 Justify the choices made in the implemented network security configuration.

GRE Tunnel

The GRE (Generic Routing Encapsulation) tunnel was used to establish a safe communication channel between the branches of Yangon and Mandalay via internet. This enables to encapsulate the private internal traffic within the public IP packets providing the illusion of a private connection between the two locations. The GRE is compatible with a variety of network layer protocols and allows the exchange of routing updates in an otherwise secure way across the sites. This has been selected since it offers flexibility and has a secure site-to-site communication without using costly leased lines. It also provides the ability of the branch offices to share resources and data effectively but in a logical isolation of public networks.

Switchport Security

Access layer switches were implemented with switchport security so that unauthorized devices could not be connected to the network. Every switch port was set to permit very few MAC addresses - one or two, typically - and only registered computers or devices can be connected. In the case of an unknown MAC address, the switch blocks the access or automatically deactivates the port. Such a setup helps in warding off Mac flooding attacks and spoofing attacks that may interfere with the network or provide unjustified access. It also makes sure that physical connection points are locked up which is crucial in the environments where there are more than one user or devices.

EtherChannel set up via PAGP (two device set up)

EtherChannel is a grouping of links protocol which integrates numerous physical connections between two switches in a single logical connection called a Port Channel. Port Aggregation Protocol (PAgP) is employed to negotiate and control link bundling in this network in an automatic manner. PAgP is a proprietary protocol developed by Cisco and dynamically created an EtherChannel when both connected devices have been set to compatible settings, i.e., desirable and auto.

In the case of switch, A, ports could be set as desired, whereas in Switch B, ports can be set to auto. As soon as negotiation has taken place, the switches become EtherChannel which serves as one logical trunk connection. Such an arrangement does not only extend the bandwidth available but also provides redundancy, in the event that one of the physical interfaces fails, traffic is automatically rerouted by the rest of the active connections within the channel.

The key benefits of Etherchannel with PAgP are better utilization of links, automatic recovery and loop avoidance, given that the combined interfaces are considered as one link by the Spanning tree Protocol (STP). This gives a great boost to network performance and availability particularly on core links like the Yangon and Mandalay branch offices.

Setting up HSRP (Two device mode)

Hot Standby Router Protocol (HSRP) is used to supplement Layer 2 redundancy with Layer 3 gateway redundancy. An HSRP group is configured with two routers or multilayer switches, where they share a virtual IP address which acts as a default router to related hosts.

Under such a configuration, a single device is used as the Active router, which has all network traffic, and the other device is set to Standby mode, which monitors the active device by periodically sending hello messages. The active router will automatically be replaced by the standby router when the active router fails to perform its role; it becomes the forwarding router with the same virtual IP. This would guarantee that the users and systems do not suffer a loss in connectivity.

Priority values are usually included in HSRP configurations to decide which router becomes the active one. Only in a situation that the primary router fails does the router with the higher priority take over the active role. When a higher-priority router returns online, it can regain its active role thanks to the preempt command, ensuring reliable and effective gateway operations.

IPsec Configuration

IPsec (Internet Protocol Security) was set up so as to encrypt and authenticate the information sent between the branches using the GRE tunnel. Encryption standards

that are used by IPsec include AES and authentication systems include SHA or pre-shared keys to ensure that data cannot be intercepted or modified during transmission. This option enhances privacy and integrity particularly when relocating sensitive information among geographically distant networks. Combining GRE with IPsec offers secure tunneling and encrypted communication which is a strong defense barrier against eavesdropping, replay and man in the middle intrusions.

Network Access Control (NAC)

Network Access Control (NAC) has been provided so that only compliant devices that are trusted can access the network. NAC authenticates the user credentials and inspects the device status (antivirus update or operating system version and so on) before permitting the user to enter. This assists in avoiding the need of having infected, outdated or unauthorized devices on the network that could infect the rest of the network due to the internal threats. It implements network policy on point of access, which prevents all endpoints, including those that do not comply with the security requirements of the organization, to get their way into VLANs or shared resources.

Wireless Security

The wireless access points were set to use the WPA2 or WPA3 encryption, which are very effective towards preventing unauthorized access and eavesdropping. The wireless network was further isolated into its own VLAN and it was not linked to the core wired infrastructure. Other security systems like MAC address filter and hidden SSID were also implemented to restrict access by authorized users. This setup will mean that in case a wireless connection is hacked, the attacker will not access sensitive internal systems. The protection against brute-force attacks is also improved with the usage of WPA3 which ensures stronger encryption and the contemporary authentication procedures.

SNMP Configuration

Simple Network management protocol (SNMP) was set up to facilitate monitoring and management of the network. SNMPv3 was selected to enhance security since it has authentication and encryption capabilities unlike the older versions. This eliminates the possibility of unauthorized people intercepting or altering network

management data. SNMPv3 enables network administrators to track performance on the devices, to identify the failures in hardware as well as suspicious activity safely. It offers good visibility on the health and stability of the network and management traffic is not compromised.

Intrusion Prevention System (IPS) Configuration

To continually track and analyze network traffic in real time, an intrusion prevention system (IPS) was put to place. Malicious actions like port scans, malware communications, and denial-of-service (DoS) attacks can be detected by the IPS. When a threat is identified, the attack is automatically blocked or disabled before it has a chance to affect the network. The network is protected even from emerging or changing threats thanks to this proactive security measure. By offering constant protection and bolstering a defense-in-depth approach, the IPS enhances other defense layers.

Conclusion

All things considered, a thorough and tiered security model is produced through the combination of these security configurations: switchport and access control, secure wireless setup, NAC, SNMPv3, SYSLOG monitoring, IPS, and GRE tunnels with IPsec encryption. Every part focuses on a distinct facet of the CIA triad: availability (through proactive defense and system visibility), integrity (through authentication and monitoring), and confidentiality (through encryption and access control). These decisions guarantee that the network is not only operational but also secure from both external and internal cyberattacks, offering a safe channel of communication between the branches in Mandalay and Yangon.

LO4: Undertake the testing of a network using a Test Plan.**P6: Comprehensively test the network using a devised Test Plan.****Testing Plan**

Test Name	Objective	Tools Commands Used	Expected Result	Actual Result Status
WAN Connectivity Testing	To ensure site-to-site communication between Yangon and Mandalay through GRE and IPsec tunnel	ping, show crypto IPsec sa, show interface tunnel	Successful ping and secure tunnel established	Tunnel up; end-to-end ping successful with 0% loss
Console and Enable Password Testing (Yangon)	To verify console and enable password access	Local console login	Correct passwords allow user and privileged access	Access successful (GW# prompt confirmed)
HSRP Testing (Yangon)	To ensure gateway redundancy using HSRP	show standby brief	One router active, other standby, virtual IP shared	HSRP active/standby roles configured correctly

EtherChannel Testing (Yangon)	To verify link aggregation for redundancy and bandwidth	show EtherChannel	EtherChannel active using PAgP protocol	Channel group up with 4 active ports
Switchport Port Security Testing (Yangon)	To restrict unauthorized device connections	switchport port-security, show port-security	Unauthorized devices blocked; port shuts or restricts	Port turns red on violation; green when authorized
NAC Testing (Port-Based)	To ensure endpoint compliance before granting access	NAC policy on switch	Only compliant devices gain VLAN access	Unauthorized or noncompliant devices denied
NAC Testing (Wireless)	To verify NAC enforcement over wireless connections	Wireless router config, device authentication	Only authenticated devices connect via SSID	Wireless device successfully authenticated
SNMP Testing (Yangon)	To verify network monitoring through SNMP queries	OID: .1.3.6.1.2.1.1.5.0	Device returns system name	SNMP response received (e.g., YGN_ER)
SYSLOG Testing (Yangon)	To confirm logging of network events	Syslog server on Server1	Device logs appear in Syslog viewer	Logs received successfully from 10.0.0.2

SSH Testing (Yangon)	To verify secure remote management access	ssh -l admin <ip>	SSH connection with password authentication	SSH login successful to router from PC
GRE and IPsec Testing (Yangon)	To confirm encrypted site-to-site communication	ping, show crypto session	Successful encrypted communication	Encrypted GRE tunnel functional
IPS Testing (Yangon)	To ensure real-time threat prevention	Ping/traffic monitoring	IPS detects and blocks malicious traffic	IPS active, legitimate traffic allowed
Console and Enable Password Testing (Mandalay)	To confirm console and enable access	Local console login	Access granted using correct passwords	Login successful (GW# prompt)
HSRP Testing (Mandalay)	To verify HSRP redundancy on Mandalay side	show standby brief	Backup router active when primary fails	Roles switch correctly during failover
EtherChannel 1 Testing (Mandalay)	To validate link aggregation between switches	show EtherChannel	EtherChannel established with 4 ports	Channel formed using PAgP

Switchport Port Security Testing (Mandalay)	To test MAC-based port protection	show port-security interface	Only registered MACs allowed	Unauthorized port shut down automatically
SNMP Testing (Mandalay)	To confirm SNMP functionality	OID .1.3.6.1.2.1.1.5.0	Returns system name (e.g., ER2)	SNMP working and responsive
SYSLOG Testing (Mandalay)	To verify logging from Mandalay devices	Syslog server viewer	Device logs received and displayed	Logs show link and configuration changes
SSH Testing (Mandalay)	To test encrypted remote login to devices	ssh -l admin <ip>	Secure login to router	SSH authentication successful (GW4#)
GRE and IPsec Testing (Mandalay)	To confirm encrypted WAN tunnel from Mandalay side	ping <remote IP>, show crypto session	Successful and secure communication	Tunnel up; ping successful
IPS Testing (Mandalay)	To validate intrusion prevention operation	Ping and simulated traffic test	Normal traffic allowed; attacks blocked	IPS working correctly; monitoring active

WAN Testing

```
void 0 Config-1: Configured from console by console

Router#ping 200.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

Router#ping 200.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.0.0.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

Figure (4.0) Testing ISP router

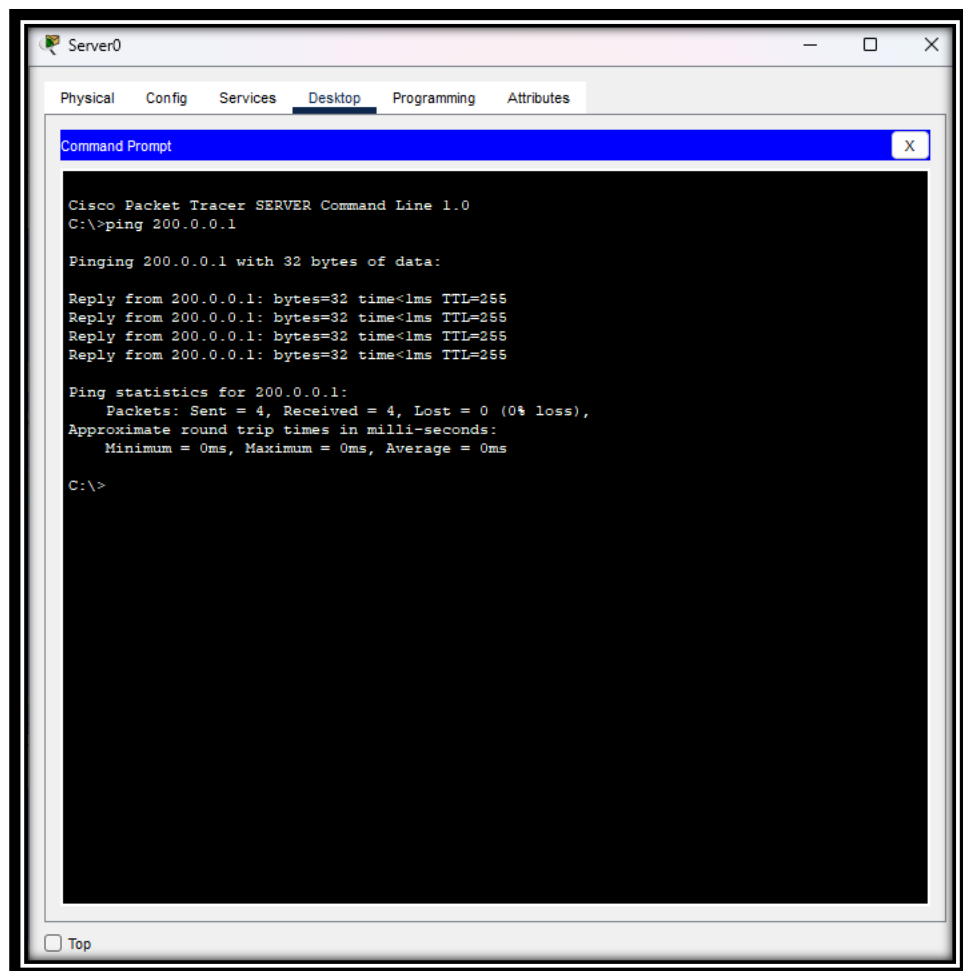


Figure (4.1) ISP server to ISP router testing

In testing of wan configuration, ISP router to server is successfully pinged and ISP server to ISP router is pinged successfully.

Yangon side testing

Console and enable password testing

```

User Access Verification

Password:

GW1>en
Password:
GW1#

```

Figure (4.2) Enable and console password testing

This test verifies that the user successfully logs in to the Cisco device by entering the correct password to access user mode (GW1>) and then using the en command to access privileged mode (GW1#).

HSRP Testing

```

GW1#show standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P State   Active        Standby        Virtual IP
Vl10       10   105  P Active  local         192.168.10.2   192.168.10.254
Vl20       20   105  P Active  local         192.168.20.2   192.168.20.254
Vl30       30   105  P Active  local         192.168.30.2   192.168.30.254
Vl50       50   105  P Active  local         192.168.50.2   192.168.50.254
GW1#

```

Figure (4.3) HSRP on GW1

```

GW2#show standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P State   Active        Standby        Virtual IP
Vl10       10   100  P Standby 192.168.10.1  local         192.168.10.254
Vl20       20   100  P Standby 192.168.20.1  local         192.168.20.254
Vl30       30   100  P Standby 192.168.30.1  local         192.168.30.254
Vl40       40   100  P Active  local         unknown       192.168.40.254
Vl50       50   100  P Standby 192.168.50.1  local         192.168.50.254
GW2#

```

Figure (4.4) HSRP on GW2

These outputs display the GW1 and GW2 routers' HSRP (Hot Standby Router Protocol) status. GW1 is the active router for VLANs 10, 20, 30, and 50 in the

second image, while GW2 serves as the standby router for the majority of VLANs (with the exception of VLAN 40, which is active).

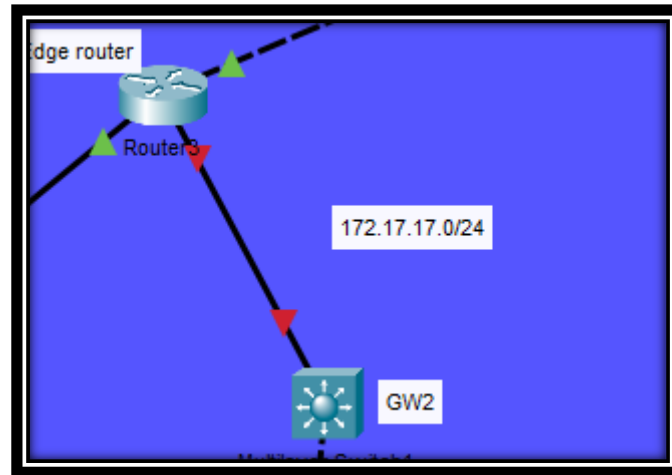


Figure (4.5) HSRP testing

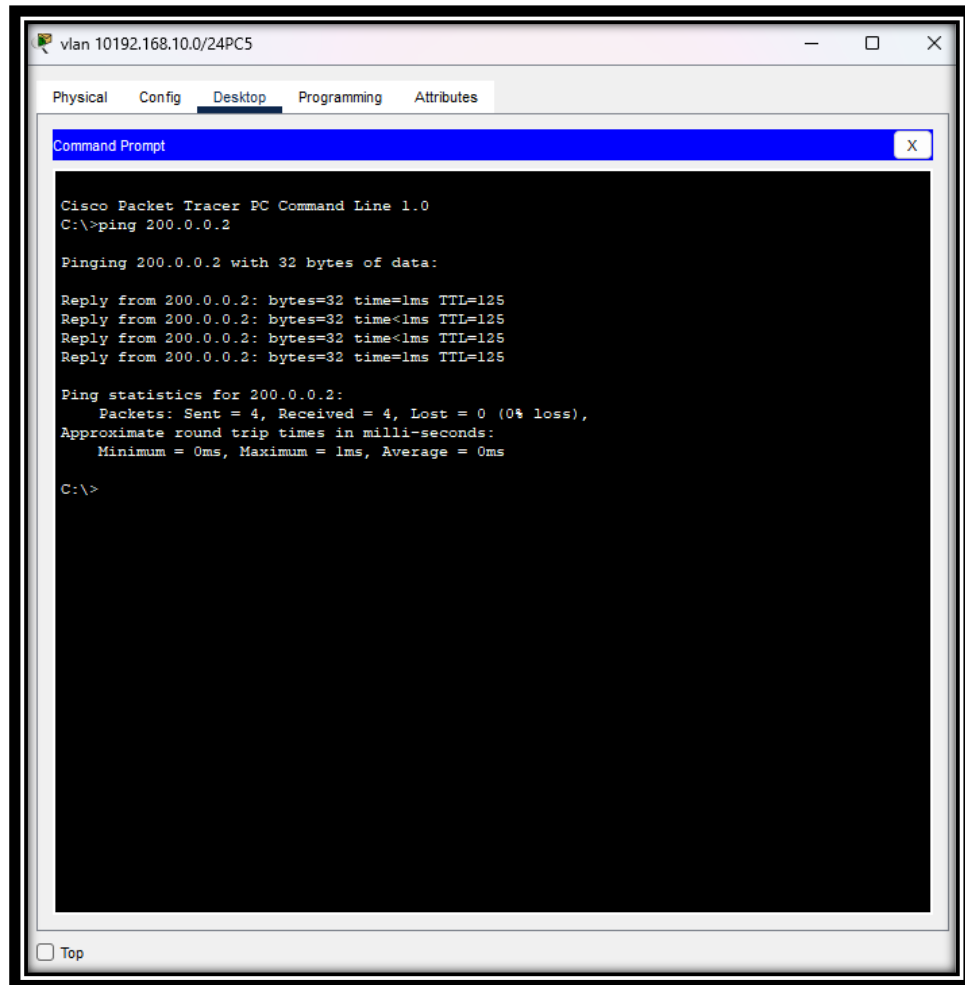
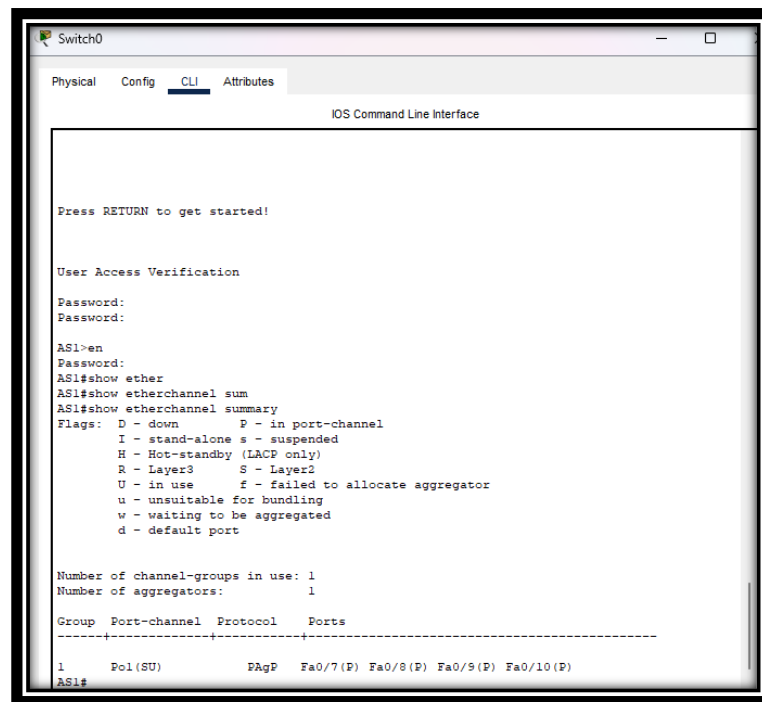


Figure (4.6) Pinging while HSRP testing

The ping is success even multilayer switch ports interfaces are close.

EtherChannel Testing



```

Switch0
-----
Physical  Config  CLI  Attributes
-----
IOS Command Line Interface

Press RETURN to get started!

User Access Verification

Password:
Password:

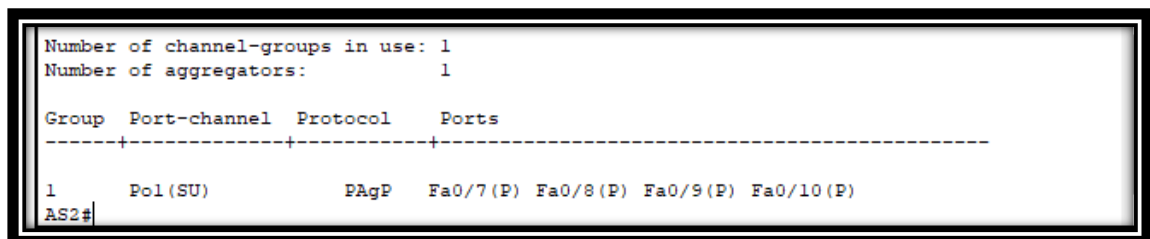
AS1>en
Password:
AS1#show ether
AS1#show etherchannel sum
AS1#show etherchannel summary
Flags: D - down          P - in port-channel
       I - stand-alone   s - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Pol(SU)        PAgP        Fa0/7(P) Fa0/8(P) Fa0/9(P) Fa0/10(P)
AS1#

```

Figure (4.5) EtherChannel on AS1



```

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Pol(SU)        PAgP        Fa0/7(P) Fa0/8(P) Fa0/9(P) Fa0/10(P)
AS2#

```

Figure (4.6) EtherChannel on AS2

These outputs display the EtherChannel configuration on switches AS1 and AS2. Utilizing the PAgP (Port Aggregation Protocol), both devices have effectively established a EtherChannel group (Group 1) functioning at Layer 2 (L2). With four physical ports combined into one logical port channel per switch, link aggregation is possible for increased bandwidth and redundancy. This verifies that AS1 and AS2 have a PAgP-based EtherChannel connection that is correctly set up, guaranteeing effective load balancing and fault tolerance.

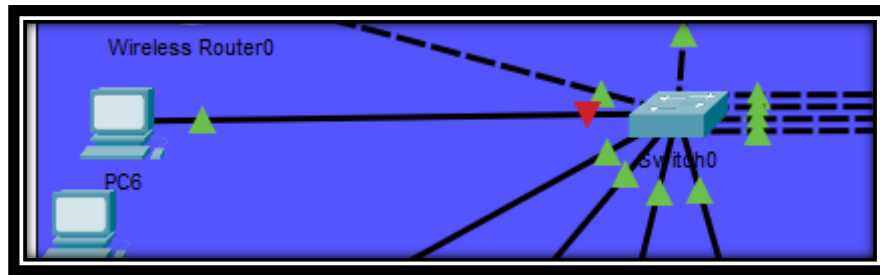
Switchport Port Security Testing

Figure (4.7) Switchport Security on AS1

In this part of the network, a switch (Switch0) is linked to several devices, such as a wireless router and a PC (PC6). By controlling the number of MAC addresses per port, switchport security on this switch can be set up to regulate which devices are permitted to connect.

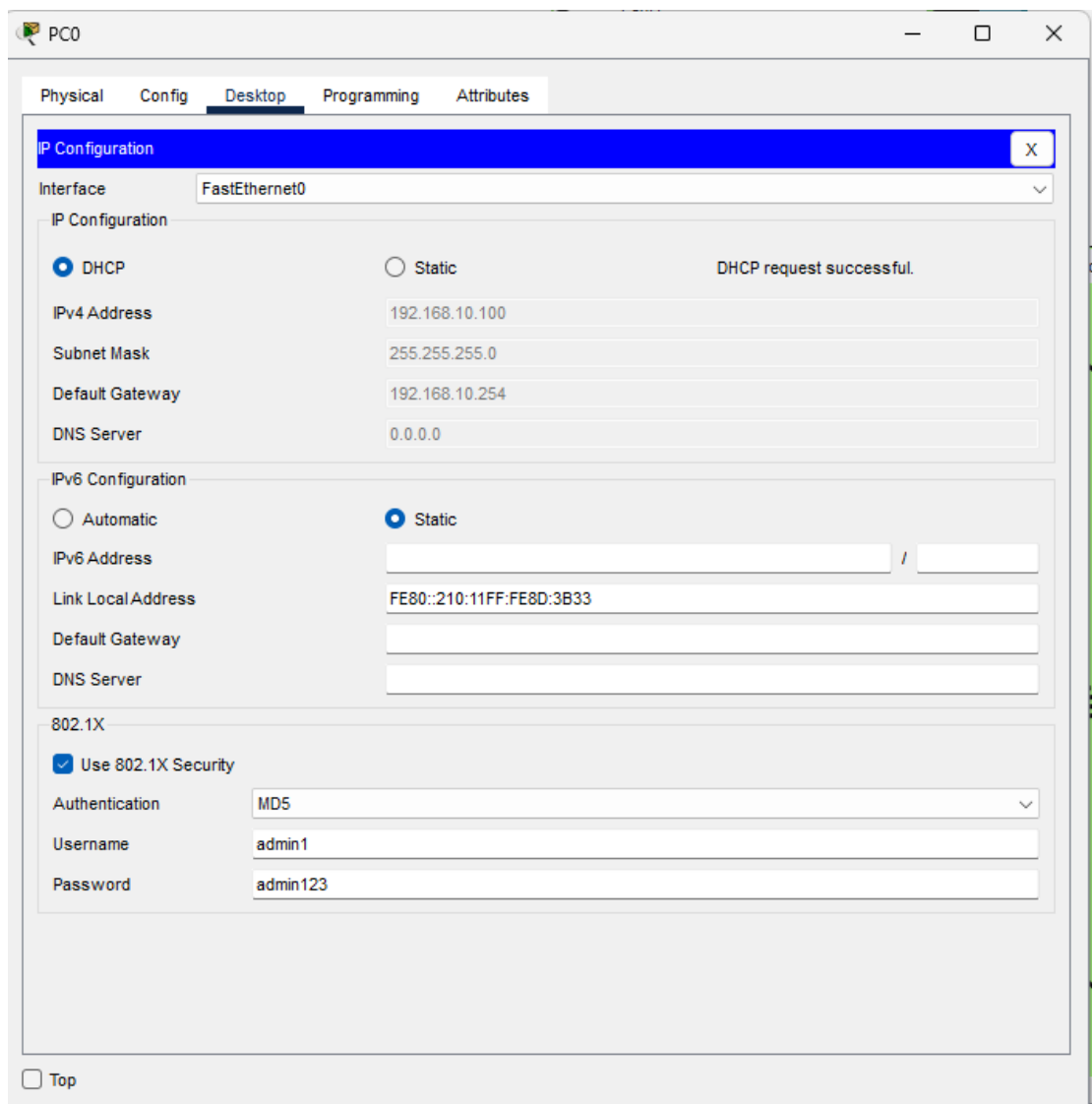
NAC testing Port Based

Figure (4.8) AAA service testing at PC0

It is a screenshot of the configuration of PC0 in case of 802.1X authentication. The IP address of the PC is 192.168.10.101 and subnet mask 255.255.255.0 and default gateway 172.16.16.16. Under 802.1X Security section, the authentication is turned on to use MD5 as well as the following credentials are used: username: admin1 and password: admin123. This configuration enables the PC to freely verify to the network via a switch with the RADIUS server.

NAC testing for Wireless

Figure (4.9) Wireless security

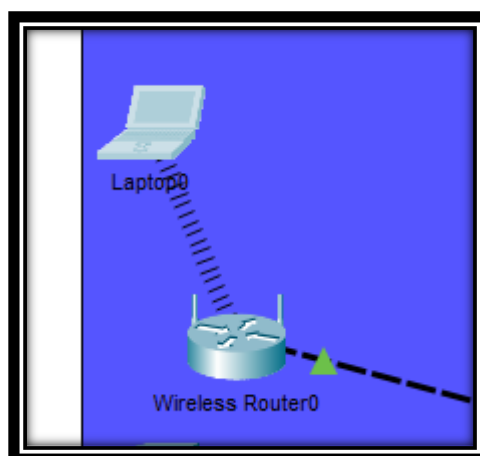


Figure (4.10) Connection success

The laptop is successfully linked to the wireless router (Wireless Router0) in infrastructure mode by a 2.4 GHz connection, as demonstrated by these two screenshots. A reliable wireless connection is confirmed by the wireless adapter being turned on and the strong signal strength and link quality indicators. This shows that the laptop was able to safely connect to the network via the router because the ****wireless security and connectivity settings****, such as authentication and encryption, were set up correctly.

SNMP testing

The screenshot displays a web-based interface for testing SNMP. At the top, there is a field for 'OID:' containing the value '1.3.6.1.2.1.1.5.0'. Below this, the 'Operations:' section has a dropdown menu set to 'Get' and a 'GO' button. The main area is titled 'Result Table' and contains a table with the following data:

Name/OID	Value	Type
1.3.6.1.2.1.1.5.0 (.iso.org.dod.interne...	YGN_ER	OctetString

Below the result table, there is a form with the following fields:

Name :	.sysName
OID :	1.3.6.1.2.1.1.5.0
Syntax :	
Access :	
Description :	

Figure (4.11) SNMP Testing on end device

The OID 1.3.6.1.2.1.1.5.0, which is equivalent to the system name (.sysName) object, is used in this test to verify the SNMP (Simple Network Management Protocol) configuration. The value "YGN_ER" that was retrieved attests to the device's SNMP agent being properly configured and reacting to SNMP requests. This successful outcome shows that SNMP communication is operating as intended, enabling remote device identification and management by network monitoring tools.

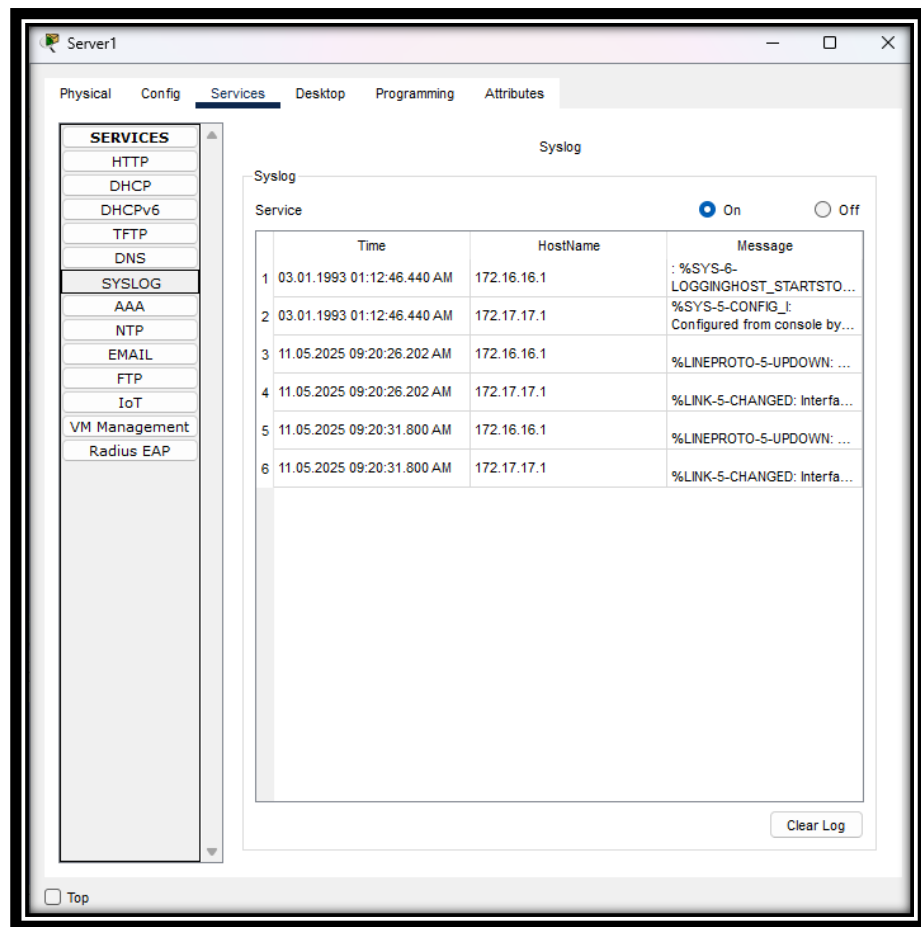
SYSLOG testing

Figure (4.12) SYSLOG notification at server 1

This is a screenshot of the Syslog service that is on Server1 and is being sent log messages by network devices. The records of the logs contain system activity of routers having the IPs of 172.16.16.1 and 172.17.17.1, such as configuration changes (%SYS-5-CONFIG_I) and interface status changes (%LINK-5-CHANGED, %LINEPROTO-5-UPDOWN).

SSH Testing

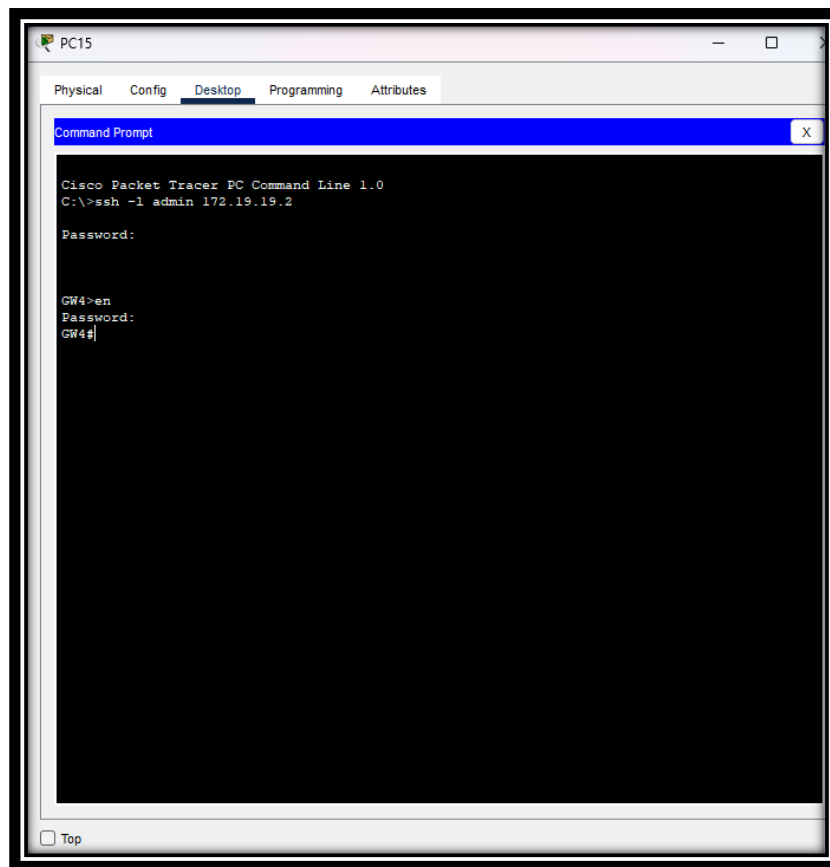


Figure (4.13) SSH Testing on end device

This test demonstrates a successful SSH remote login with the username admin from PC15 to the router located at IP address 172.19.19.2. Once the password has been correctly entered, the user can access user EXEC mode (GW4>) and use the en command to enter privileged EXEC mode (GW4#).

GRE and IPsec testing

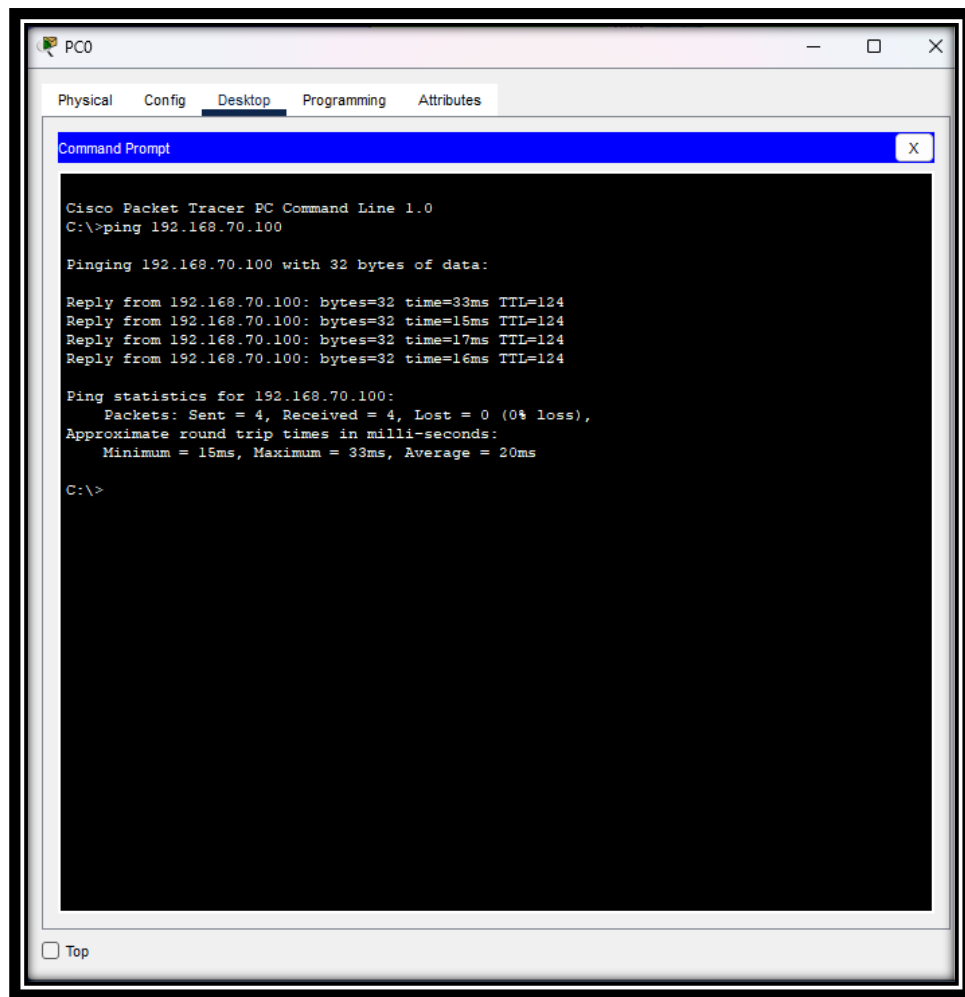
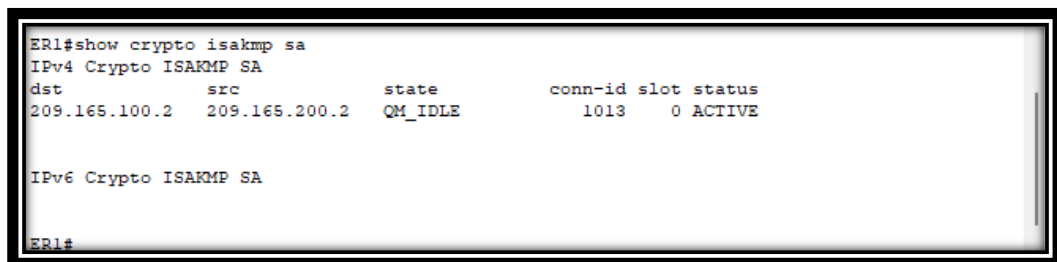


Figure (4.14) End device to end device testing

The success of end-to-end communication using the GRE and IPsec tunnel can be confirmed by this test. The PC's ability to ping 192.168.70.100 with 0% loss and all packets received indicates that the tunnel is operational and safely transferring data between the two branch networks.



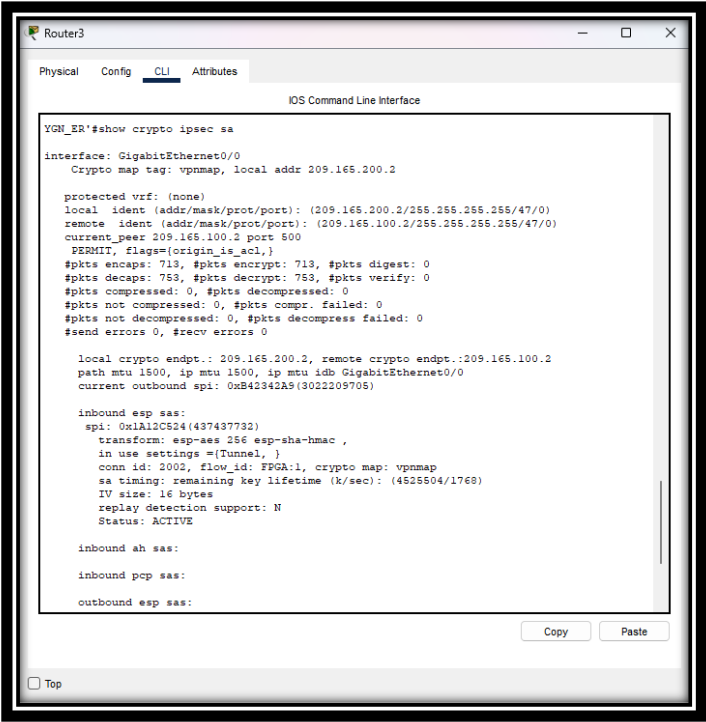


Figure (4.15) IPsec testing

This is the output of the command show crypto isakmp sa, and it is a confirmation that there is the IPsec VPN tunnel on the 209.165.200.2 (source) and 209.165.100.2 (destination) which is properly configured.

IPS testing



Figure (4.16) IPS notifications on SYSLOG server

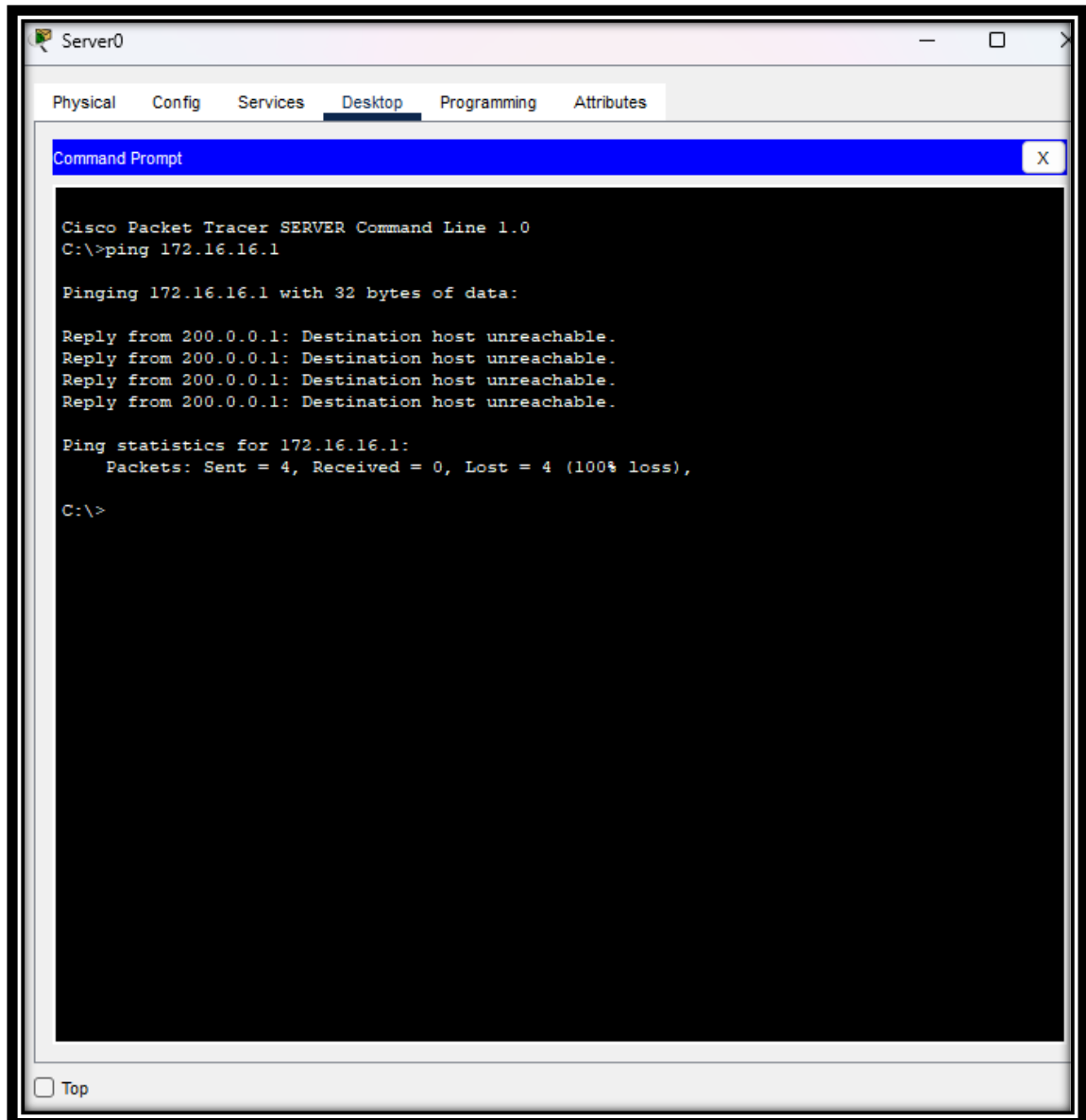


Figure (4.16) IPS testing on sever 0

This screen shot indicates that a ping test that Server0 undertook to 172.16.16.1 failed to connect to the server due to the Intrusion Prevention System (IPS) blocking the connection. The error message is Destination host unreachable because the IPS blocked the traffic in the course of its security policy.

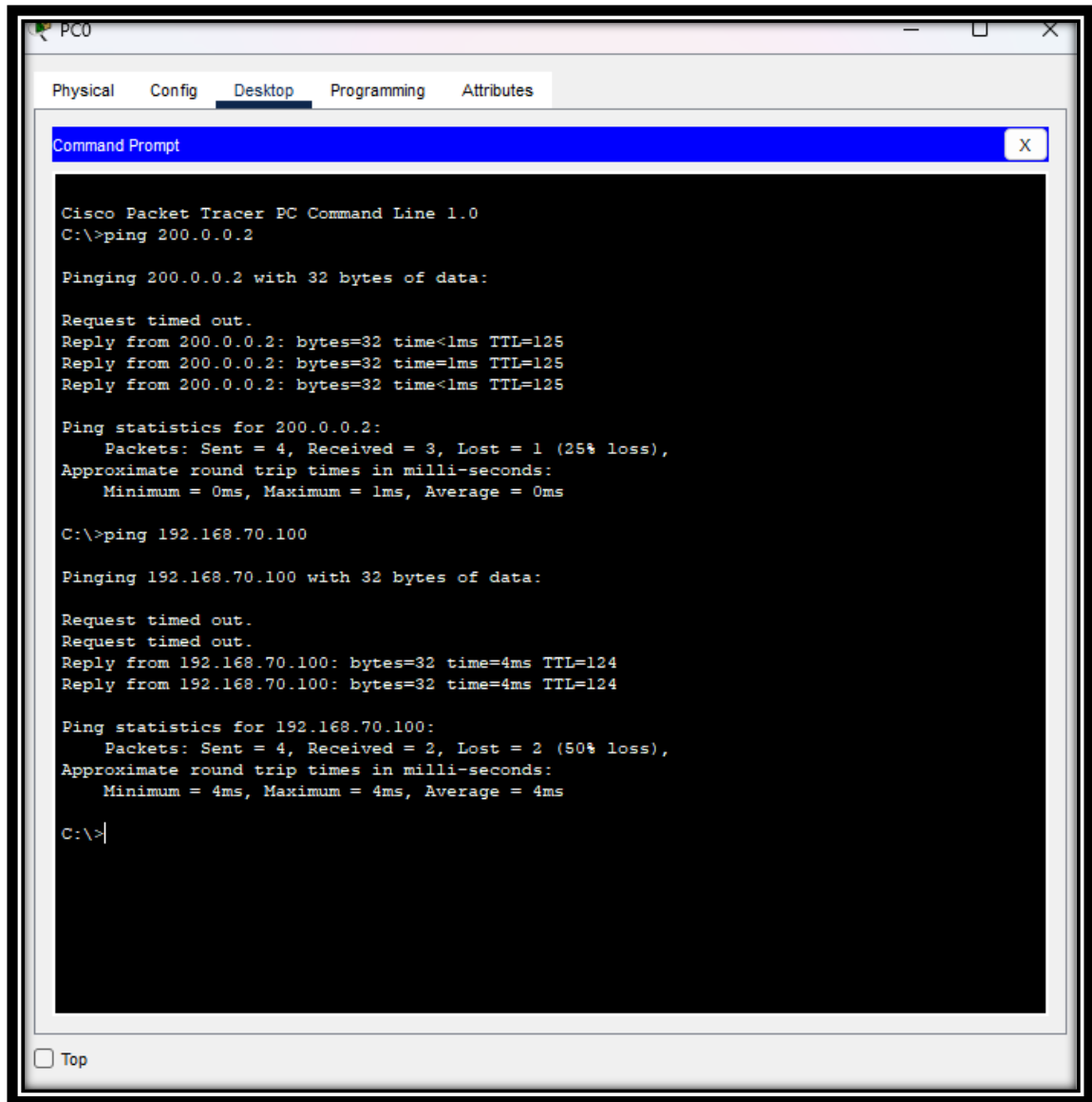


Figure (4.17) IPS testing on end device

In this screenshot, a successful ping test to the IP address 200.0.0.2 is observed and this means that the network is connected. Each of the four packets was sent and received with no lost packets and an average delay of 0ms, which demonstrated that the connection between the devices is good and operating effectively.

Mandalay side testing

Console and enable password testing

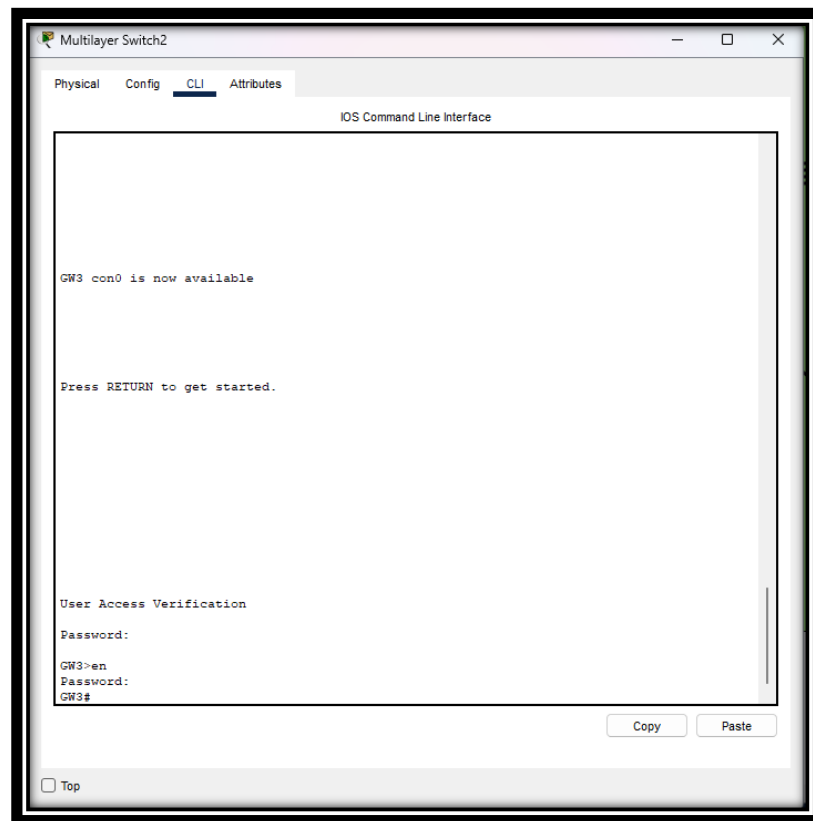


Figure (4.18) Console password and enable password testing

In order to verify correct authentication setup, this test verifies a successful console login to Multilayer Switch2 (GW3), where the user enters the correct passwords to access privileged EXEC mode (`GW3`).

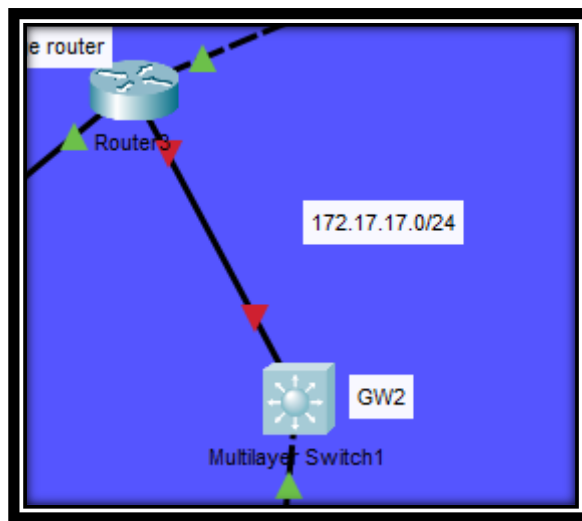
HSRP testing

Figure (4.19) Shutdown interfaces

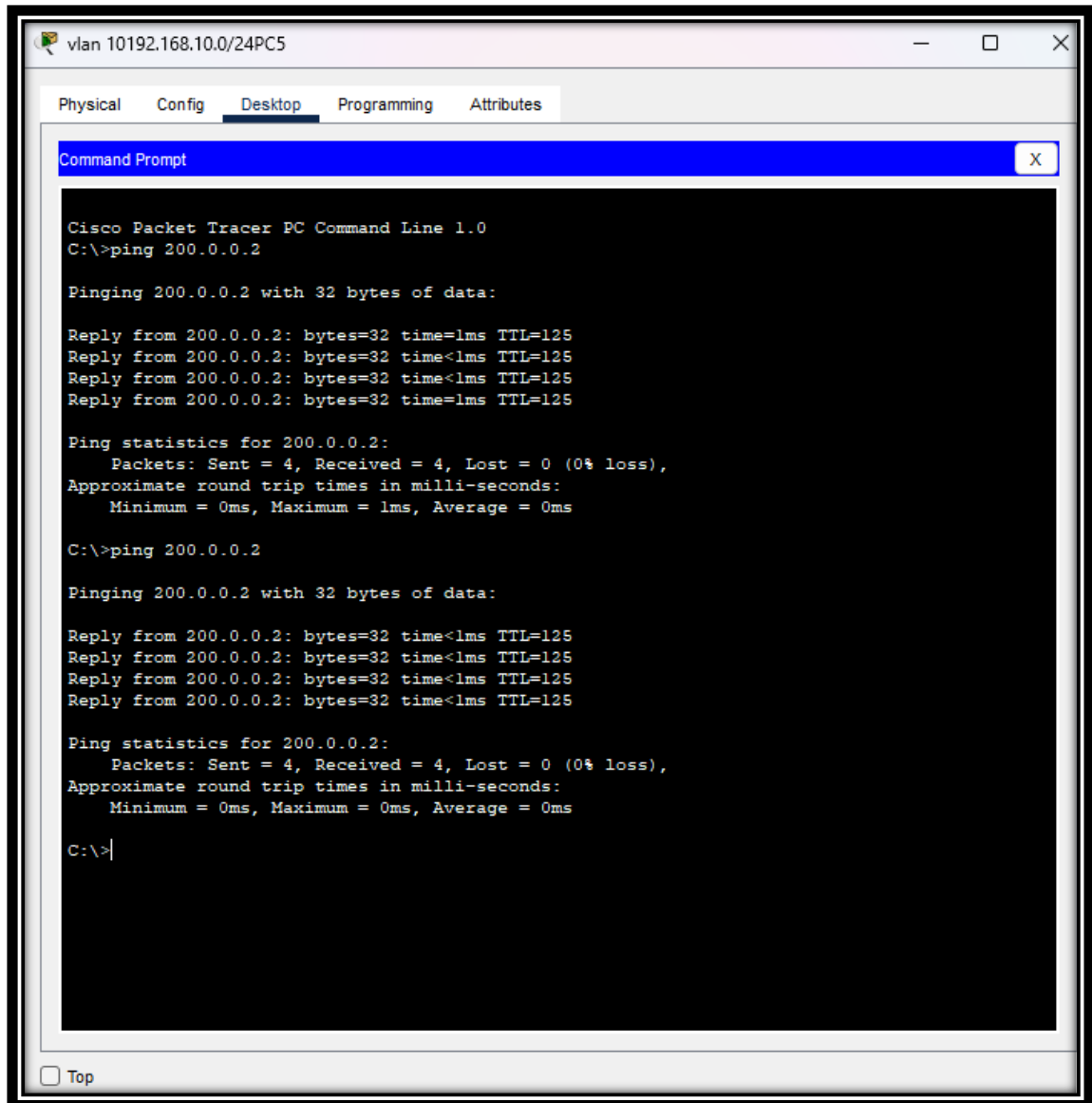


Figure (4.20) HSRP is active

Even Gw2 ports are closed but Gw2's end devices can have connection to server because of the HSRP standby mode.

EtherChannel testing

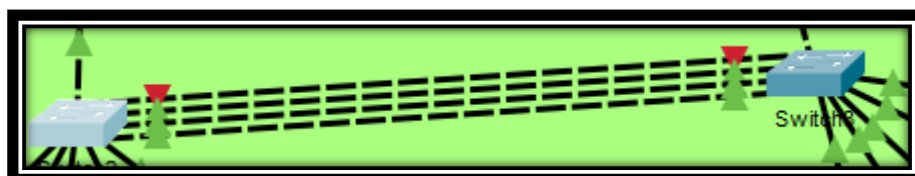


Figure (4.21) EtherChannel

```

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)          PAgP        Fa0/7(P) Fa0/8(P) Fa0/9(P) Fa0/10(P)
AS3#

```

Figure (4.22) Check EtherChannel

This test verifies that EtherChannel is set up and functioning properly on Switch AS3. Using the PAgP protocol, the command output displays Group 1 operating in Layer 2 mode with four physical ports merged into a single logical channel. By confirming appropriate link aggregation, this gives linked switches more bandwidth and redundancy.

Switchport port Security testing

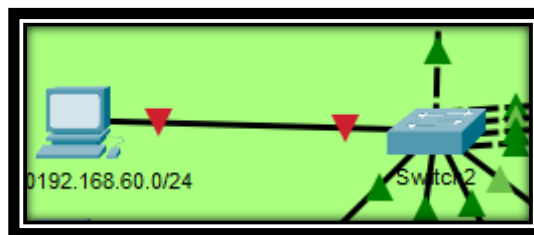


Figure (4.23) Switchport Security

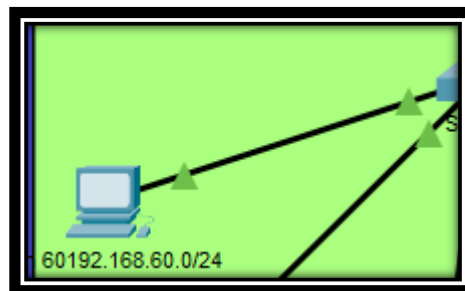
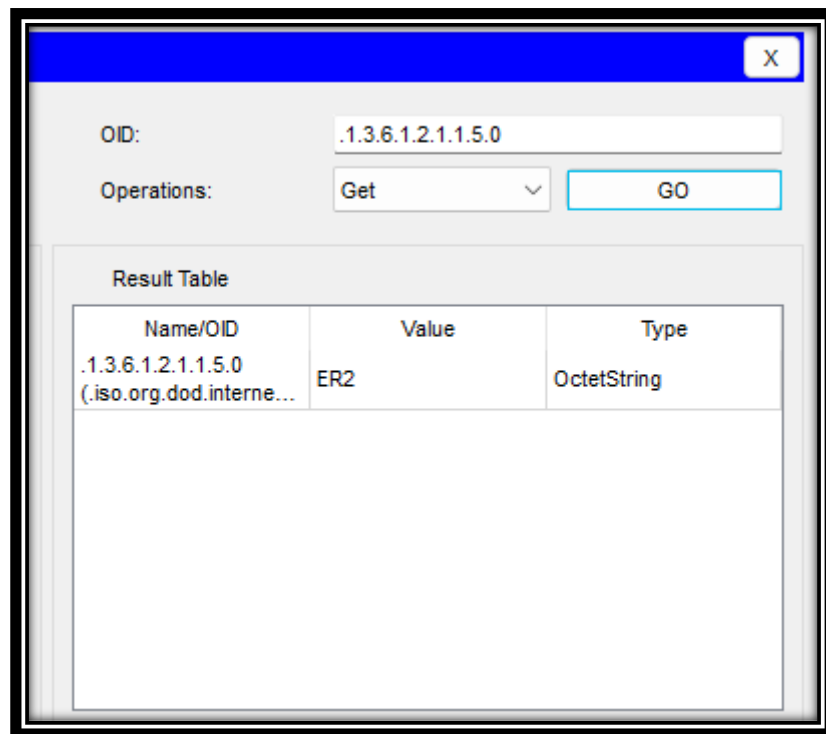


Figure (4.24) Port green back

A Switchport Port Security test appears in these figures. An unauthorized device was blocked because of port security restrictions, as shown by the red ports in Figure 4.23. When an authorized device reconnects, the port turns green in Figure 4.24, indicating that switchport port security is operating as intended to stop unwanted access.

SNMP testing

The image shows a graphical user interface for testing SNMP. It has a blue title bar with a close button (X). Below the title bar, there is a label 'OID:' followed by a text input field containing '.1.3.6.1.2.1.1.5.0'. Below that is a label 'Operations:' followed by a dropdown menu showing 'Get' and a 'GO' button. Below the operations section is a 'Result Table' with three columns: 'Name/OID', 'Value', and 'Type'. The table contains one row with the following data:

Name/OID	Value	Type
.1.3.6.1.2.1.1.5.0 (.iso.org.dod.interne...	ER2	OctetString

Figure (4.25) SNMP testing

This test displays a SNMP (Simple Network Management Protocol) query that retrieves the device's system name (.sysName) using the OID 1.3.6.1.2.1.1.5.0. The value "ER2" in the result indicates that SNMP communication is functioning correctly and that the network management system is able to locate and keep an eye on the device.

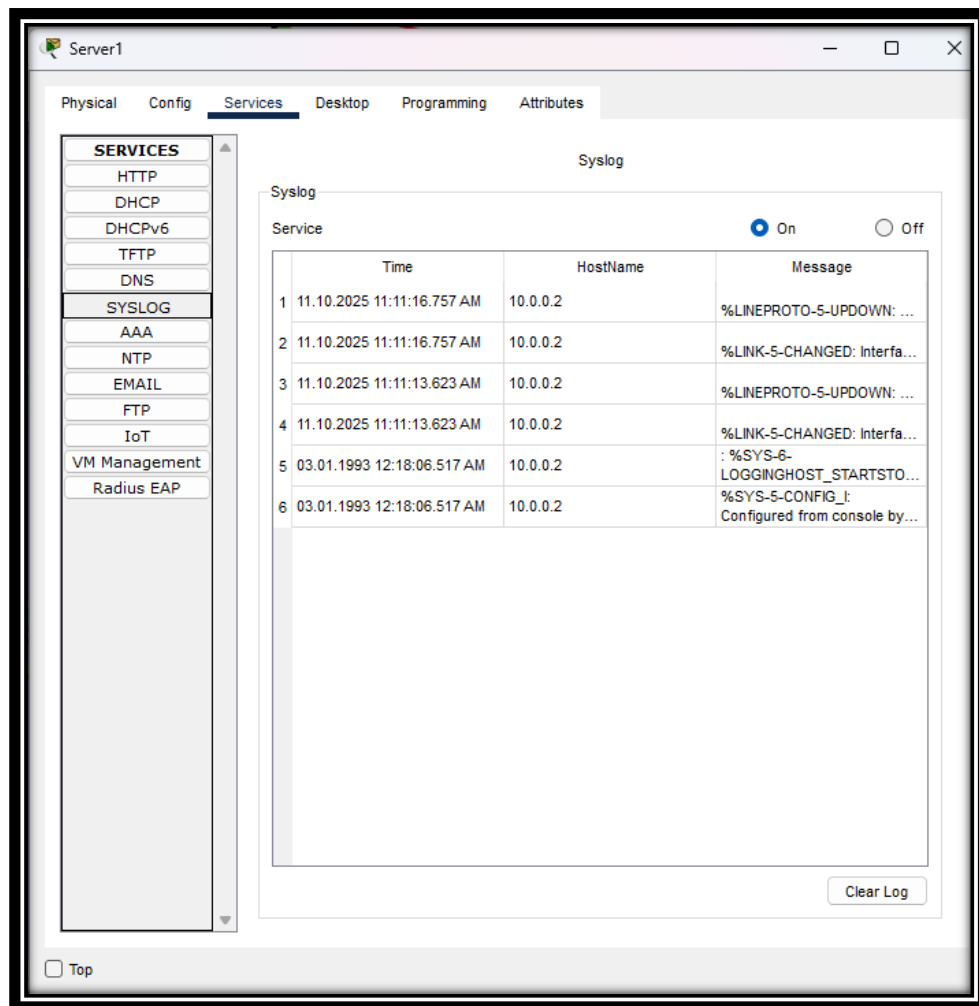
SYSLOG testing

Figure (4.26) SYSLOG testing from Mandalay side

The SYSLOG testing from the Mandalay side is depicted in this figure, where the Syslog server successfully receives network events and status messages from the device (IP 10.0.0.2). Confirming that the Syslog service is operational and accurately recording network activities for monitoring and troubleshooting purposes, the logs contain messages like interface state changes, as well as configuration updates.

SSH testing

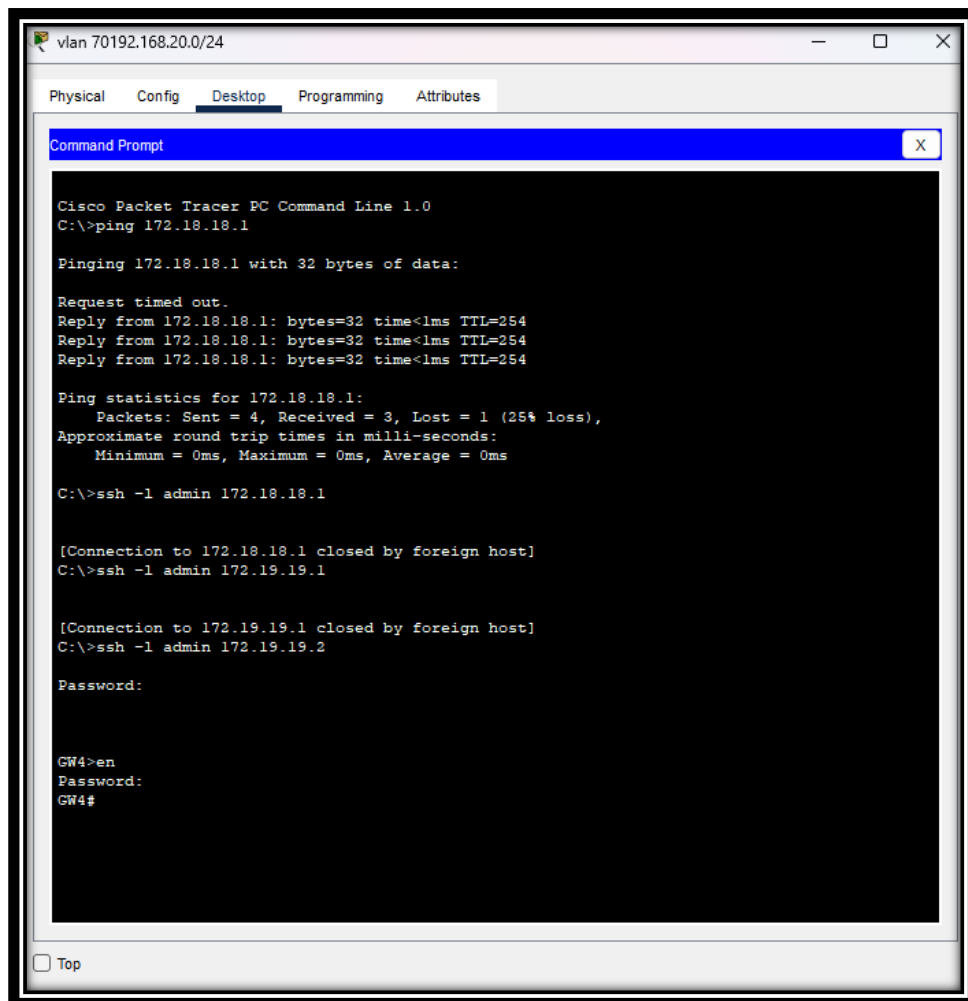


Figure (4.27) SSH testing from MDY side

The Mandalay (MDY) side of the SSH testing is shown in this figure. Using the username admin, the PC successfully logs in to the router at 172.19.19.2 after establishing SSH connections to several IP addresses. Once the enable password has been entered, the prompt changes to GW4#, verifying that SSH authentication and remote access are set up correctly and operating safely.

GRE and IPsec testing

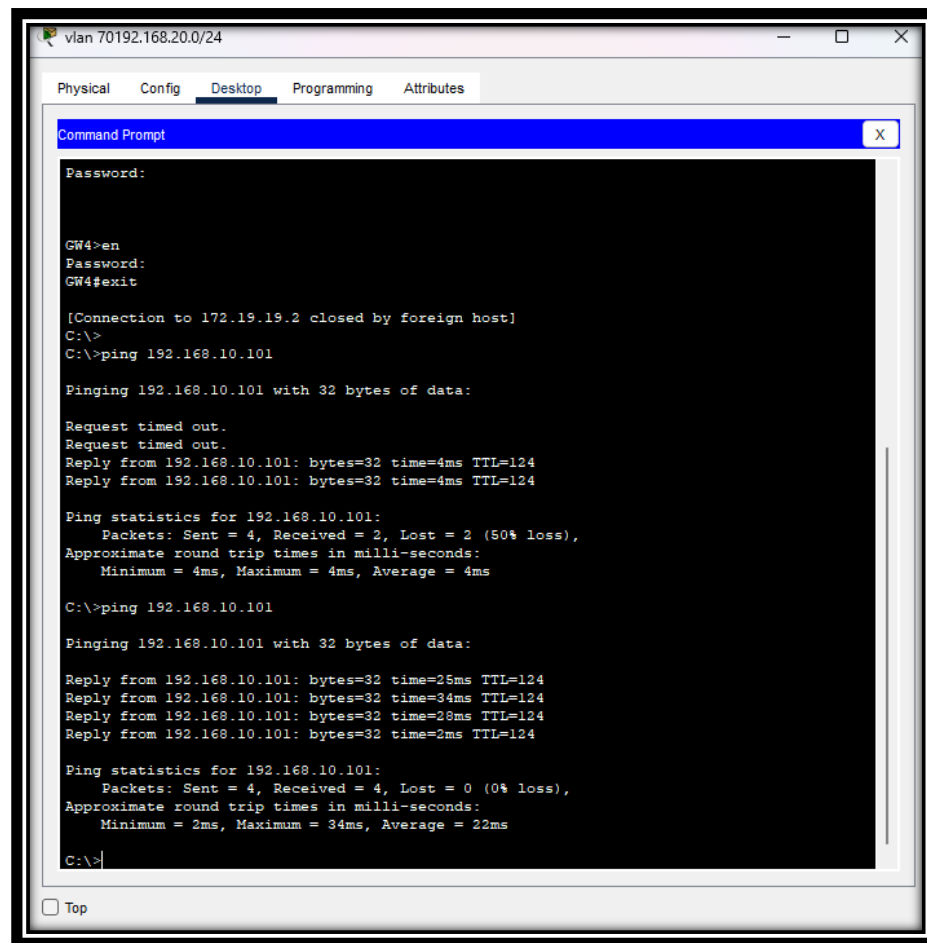


Figure (4.28) End device to end device testing from Mandalay side

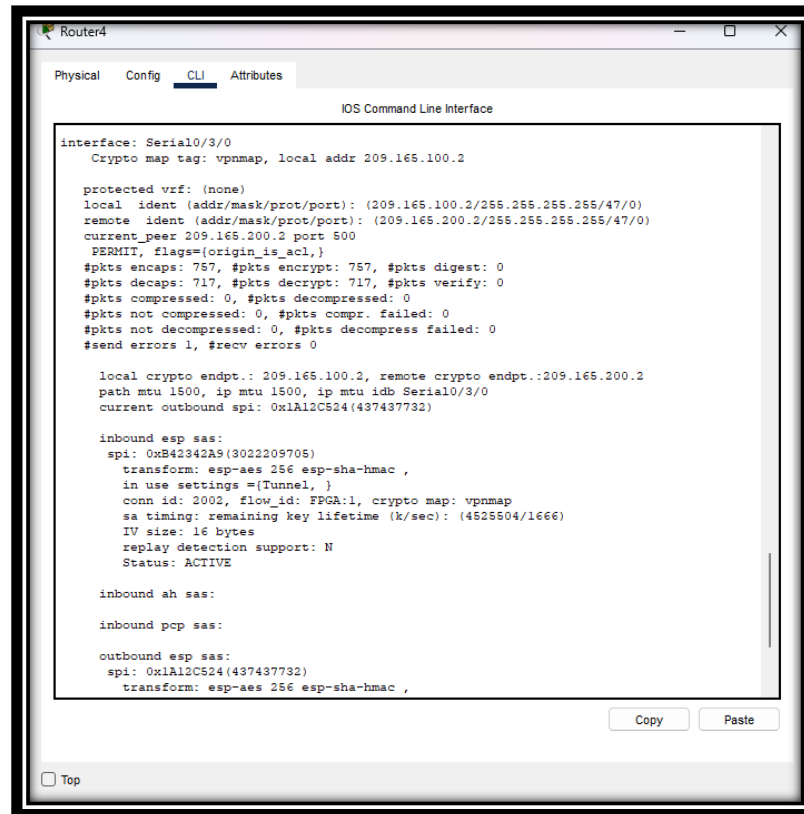


Figure (4.29) Check back IPsec configuration

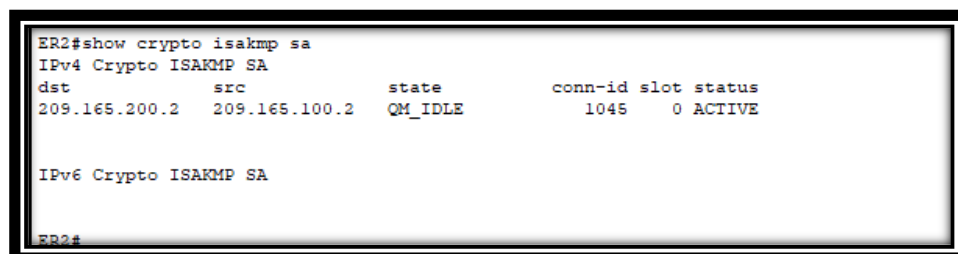


Figure (4.30) Check back IPsec configuration

The GRE and IPsec end-to-end testing from the Mandalay side is depicted in this figure. In the final test, the PC pings the outside IP 192.168.10.101 successfully, receiving all packets and experiencing 0% loss. This confirms to the proper operation of the GRE tunnel and IPsec encryption, which offer a safe and dependable communication channel between the branch networks. And then check back the configuration on Edge Router.

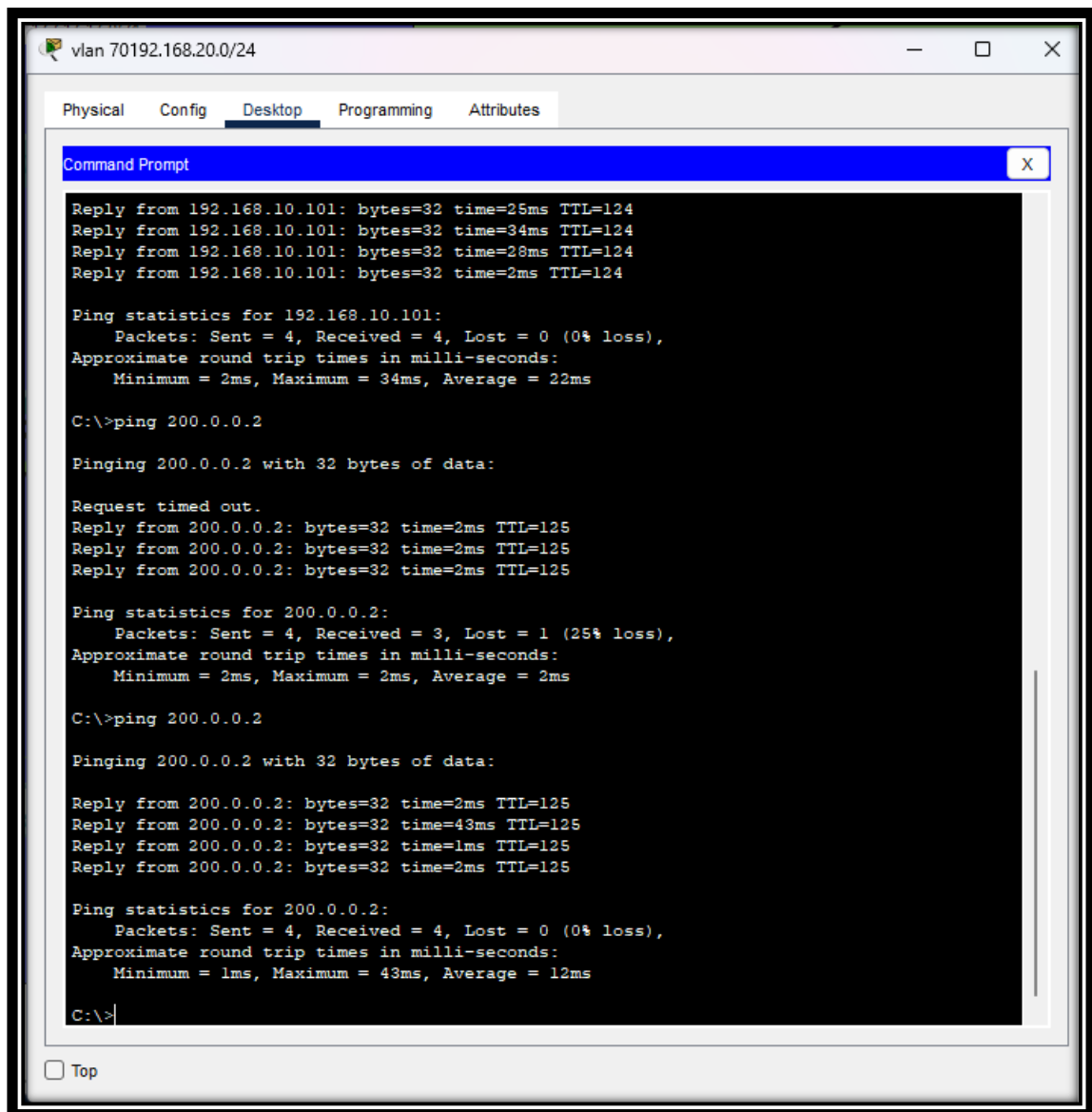
IPS testing

Figure (4.29) Ping to server from MDY side

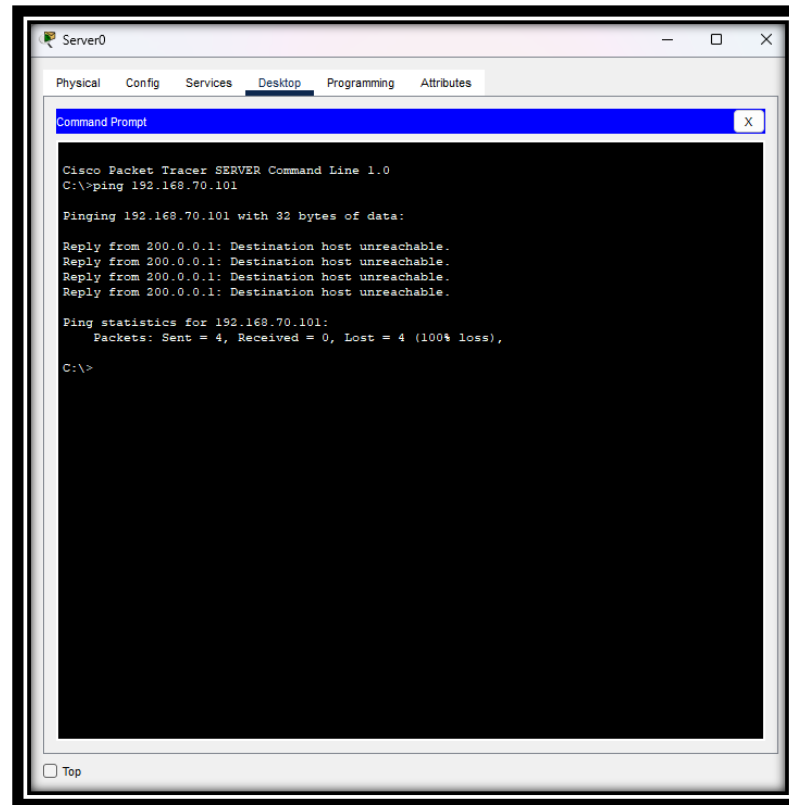


Figure (4.30) Sever to Ping from MDY side

The Mandalay (MDY) network and the remote server are examined using an intrusion prevention system (IPS) in these figures. The MDY side PC successfully pings the server in Figure, indicating consistent communication and no packet loss. Pings from the MDY side are answered by the server in Figure 4.30, confirming bidirectional connectivity. By permitting legitimate traffic while preserving network security, this shows that the IPS is operational and in good working order.

M4 Analyze the results of testing to recommend improvements to the network.

Analyzing the testing results

Having analyzed the findings of the network testing performed on the Yangon and Mandalay sites, we can notice that the network is performing on an exemplary basis and achieves the goal of the provision of secure, reliable, and high-availability communication. The testing revealed that all configurations that were implemented like HSRP, Etherchannel, GRE with IPsec, Port security, NAC, SNMP, SYSLOG, SSH and IPS worked well. The WAN testing showed complete connectivity between the two locations using the GRE tunnel with IPsec as the encryption mechanism that had zero percent packet loss and a steady response time which validates high reliability and secrecy of the transmitted data. Both HSRP and Etherchannel were fault tolerant and redundant, such that it could still run smoothly even when one of the links or devices failed. Likewise, Switchport Port Security and Network Access Control (NAC) was effective to stop non-compliant endpoints and block unauthorized devices hence protecting internal security. Moreover, SNMP and SYSLOG functions were confirmed to offer real time monitoring and logging so that the network administrators have full visibility of the performance and activities of the devices. SSH tests ensured that the secure and encrypted remote access is fully functional and IPS is successfully identifying and blocking suspicious network traffic which proved the appropriate multilayered security infrastructure.

Although these positive results are achieved, a closer look shows that there are many aspects at which the network can be refined to achieve greater efficiency, scalability, and resilient-level. Despite the adequate visibility offered by SNMPv3 and SYSLOG, a centralized Network Management System (NMS) like SolarWinds or Cisco Prime would go a long way in ensuring that fault correlation, automatic alerts, and performance analysis across several sites are enhanced. This would enable proactive management of the network and minimize downtime. The same way that, although Port Security and NAC operate local access control, 802.1X authentication using RADIUS server would enable dynamic assignment of VLANs and tighter access

control of both wired and wireless clients, based on identity information. This would ensure that the network becomes flexible to changes in user roles and security policies.

A check of the SSH proved that secure remote access has been established. Nevertheless, the control of the management should be limited to the IT department when it comes to configuring the network devices. The suggested way of implementing this rule is by applying an Access Control List. The ACL can grant access only to the particular IT administrative networks and reject all attempts of connection. This will go a long way in enhancing the security since there will be limited access points. What can be done to secure the network devices against compromise is to restrict SSH connections to authorized workstations within IT.

Regarding redundancy and performance, a secondary ISP connection or redundant VPN tunnel would add a level of resilience to the system in case of failure of ISP or tunnel and site-to-site communication can be readily maintained. Also making Quality of Service (QoS) possible would assist in prioritizing mission-critical traffic like VOIP, videoconferencing or real-time database synchronization to ensure that despite being used at a time when the bandwidth is heavily used, it will perform optimally. Regarding the security factor, the Intrusion Prevention System (IPS) performed successfully, but it would greatly benefit every effort to focus on the coming threat and create advanced analytics to act proactively as part of countering them, by updating its threat signatures and integrating it with a Security Information and Event Management (SIEM) system. This could also introduce periodic vulnerability assessment and penetration testing where vulnerabilities that might exist, can be detected and countered before they are exploited.

Finally, the testing phase is used to ensure that the network infrastructure is properly designed, stable, and secure and fulfills all the operational requirements of confidentiality, integrity, and availability (CIA). The findings indicate efficient redundancy, safe remote connectivity, and real-time monitoring facilities in the two sites. Nevertheless, with the suggested improvements, including the centralized monitoring, 802.1X authentication, improved redundancy, optimized QoS, and the more sophisticated threat intelligence integration, the network may transform into a high-performance and enterprise-grade one.

D2 Critically evaluate the design, planning, configuration and testing of the network

Evaluation of the design, planning, configuration and testing

The upgrades will make scaling more robust, resistant to future threats, and make sure that the organization has a future-proof, efficient, and secure network environment that would be able to sustain constant expansion and technological progress.

The topology of the network shows a well-thought hierarchical design with core, distribution and access layers separated. This architecture offers a good support base on performance and manageability. Direct site-to-site GRE over IPsec tunnel is a rational and economical WAN architecture, which has been able to establish a secure virtual connection between the Yangon and Mandalay offices. HSRP usage in the individual sites ensures gateway redundancy which is critical and the user is still connected in case there is a router failure. Likewise, switch-to-switch links via EtherChannel implementation is a good design decision that will enhance bandwidth, as well as, redundancy in case of the failure of one link. It is appropriate to have a firewall placed at the network perimeter and this forms an important security barrier. The topology may be however improved by thinking of a two homed Internet connection to allow the WAN to be resilient to the failure of an ISP.

The planning stage was able to develop functional connectivity. Nonetheless, to enhance efficiency and future expansion, the addressing plan must be changed to a subnetting plan, rather than keeping on with a classful addressing plan. As a technique of subnetting, one large block of private IP addresses (e.g. 10.0.0.0/8 or 172.16.0.0/12) can be split into smaller, contiguous subnets. This approach gives an opportunity to allocate address ranges of suitable sizes to every department (e.g., IT, Finance, HR) and all the connections of networking devices reducing instances of wasted IP addresses. This organised method eases the routing process, improves security through traffic separation and the network becomes much easier to scale. A subnetted scheme will be required to add new departments or branches with ease without the need to re-address the entire network project.

The implemented network model, which was designed to suit the Yangon and Mandalay branches, shows that there is a thorough strategy to the attainment of a secure, reliable, and scalable intersite communication. Some of the technologies incorporated in the network design include GRE tunneling and IPsec encryption, HSRP redundancy, EtherChannel link aggregation, Port Security, Network Access Control (NAC), SNMP, SYSLOG, SSH, Intrusion Prevention System (IPS), and Firewall protection. All the steps including design and planning, configuration and testing are carefully undertaken with performance, security and manageability in mind. Integration of firewall setting also enhances the perimeter protection and makes sure that any external attacks and unauthorized traffic can be filtered successfully before they are allowed to infiltrate the internal network.

The application of GRE instead of IPsec was a tactical decision as far as enabled it not only in routing flexibility but also in ensuring secure communication over the public Internet. This provides the logical isolation of a private leased line at a much lower price. HSRP installed in the routers guaranteed the redundancy of the gateways, which allowed keeping the network operational even in the event of device or link failure. EtherChannel was successfully applied to unite several physical connections into one logical connection thereby enhancing bandwidth as well as decreasing the number of potential bottlenecks. The firewall was directly installed at the network edge to monitor and regulate the incoming and outgoing traffic according to the set security policies. Such an arrangement assists in avoiding unwarranted access, DoS attacks, and spread of malware in such a way that important internal systems are not exposed to outside attacks.

With regard to configuration, the network was configured according to industry best practices with the use of SSH to secure remote administration and SNMPv3 and SYSLOG to monitor and log activities on the network. The IPS and firewall offered a well-alienated defense against attacks - firewall is deployed as the initial point of defense and IPS is used to identify and block any suspicious traffic that goes through perimeter defenses. The Port Security and NAC supported at the access layer also enhanced internal security because of their ability to permit only authorized and compliant devices to be connected. The configuration elements were tested thoroughly

to test their functionality and interoperability. The testing phase proved that everything was working together as the GRE and IPsec tunnel were able to sustain encrypted communication with a 0 percent packet loss, HSRP ensured a smooth failover, Etherchannel combining multiple links was also working, and both SYSLOG and SNMP showed good monitoring. The firewall test proved that the presence of traffic filtering rules was working correctly as intended, preventing access to the unauthorized ports and limiting access to the management interfaces provided to the external networks.

The success of the network has a number of areas that can be improved to scale and efficiency in the future. Introducing a centralized Network Management System (NMS) like SolarWinds or Cisco Prime would make the network more visible and will automate performance monitoring. Stateful inspection and application-layer filtering would help the firewall setup to be more resilient as it would enable the firewall to analyse packets and exercise greater control over traffic on the net. By incorporating the firewall with a Security Information and Event Management (SIEM) system, real-time correlation of events as well as an advanced threat detection could be implemented. Moreover, Authentication 802.1X with RADIUS integration may be implemented to assign dynamic VLANs and Quality of Service (QoS) may be enabled with critical applications being prioritized over bandwidth. Redundancy- i.e. the use of two ISPs or DMVPN may also be added on top of this, which will ensure that the business continues in case the links fail.

To sum up, the design, configuration, and testing of this network are a well-organized, secure, and efficient implementation of an enterprise grade network. The firewall is an addition that is very useful in perimeter defense but it acts as a supplement to other security measures like the IPS and NAC and creates a defense-in-depth approach which helps in safeguarding both internal and external traffic. The successful testing is an assurance that everything is in harmony working towards the core system networking goals of confidentiality, integrity, and availability (CIA). The recommended improvements, however, such as the implementation of an improved firewall management, SIEM integration, dynamic authentication, optimization of QoS and other extra redundancy, would help to bring the network to an even greater level of

resilience, automation and security; and will guarantee scalability and protection against future threats.

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